

## **CHAPTER I**

### **THE PROBLEM AND ITS BACKGROUND**

This chapter includes an introduction, background of the project, the project's objectives, including the general and specific objectives, purpose and description, scope and limitations, theoretical framework, conceptual framework, and significance of the project.

#### **Introduction**

Government-issued documents such as PSA birth certificates, marriage certificates, death certificates, and certificates of no marriage are essential for personal, legal, and official use. These documents are part of many daily life transactions and are often required by various agencies.

In the Philippines, these records are verified and issued nationally by the Philippine Statistics Authority (PSA). However, the process usually starts at the local level, particularly in offices such as the Municipal Civil Registrar. This office handles the recording and issuance of these civil documents. In the Municipality of Calauan, the current process of requesting documents is mostly manual, which can cause delays, long queues, and additional workload for both staff and residents.

To address these challenges, this project introduces an automated solution designed to streamline the document request process. Through automation, residents can easily request documents online, track their status, and receive

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|  | <div> <p>notification updates without needing to go back and forth to the municipal office. This makes the entire process faster, more efficient, and more organized.</p> <p>Moreover, the project integrates data analysis techniques to further enhance operational efficiency and decision-making. Aside from automating the system, this project also applies K-Means Clustering to group document requests based on factors such as request type, frequency, and time of submission. This approach helps the office identify trends, such as which documents are most in demand and when requests peak, allowing administrators to allocate resources more efficiently.</p> <p>The title of this capstone project is <i>“Development of a Document Request System with K-Means Clustering for the Municipal Registrar of the Municipality of Calauan.”</i> It aims to streamline existing procedures, resulting in a process that is not only more efficient and accurate for the staff but also more convenient and accessible for the residents of Calauan.</p> <p><b>Background of the Study</b></p> <p>Calauan is a second-class municipality in the province of Laguna, Philippines, with a population of over seventy-four thousand (74,000) based on the 2010 census. The town, named after the word “kalawang” (rust), is known for its Pineapple Festival held every May 15 and its patron saint, San Isidro Labrador. Calauan is also notable for being a resettlement site for informal settlers from Metro Manila through the Bayan ni Juan and Kapit-Bisig para sa Ilog Pasig</p> </div> |   |
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|  | <p>projects. With its continuous population growth and development, the demand for efficient public services in Calauan has become increasingly evident.</p> <p>One key area where this need for improvement is most apparent is the Municipal Civil Registrar’s Office. This office manages and issues important records, such as birth, marriage, and death certificates. These documents are required for legal, financial, and personal purposes, including applying for government services, obtaining loans, and verifying one’s identity. However, the office currently uses a manual process for handling requests and processing records.</p> <p>Based on a survey of thirty (30) respondents, it was found that all participants (100%) currently request documents through walk-in transactions. This indicates that the municipality does not offer alternative methods, such as phone or online requests, which limits accessibility and convenience for residents. The most commonly reported challenge was the long processing time, identified by twenty-two (22) respondents (73.3%). In addition, sixteen (16) respondents (53.3%) cited inconvenient office hours. The survey also revealed that thirteen (13) respondents (43.3%) experienced waiting times of four to seven days, while twelve (12) respondents (40%) waited one to three days. Only seven (7) respondents (23.3%) reported receiving their documents on the same day, whereas six (6) respondents (20%) experienced delays of more than a week. These findings point to inefficiencies in the current document request system. The absence of digital</p> |   |
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|  | <p>alternatives requires residents to visit the office in person, often within limited operating hours, which contributes to delays and user dissatisfaction.</p> <p>In response, by implementing a Document Request System, the Municipality of Calauan can enhance efficiency, reduce processing time, and improve the overall experience for both citizens and government employees. This system aims to increase the efficiency of document request processes to ensure that municipal services are more responsive to community needs.</p> <p><b>Objectives of the Study</b></p> <p><b>General Objective</b></p> <p>To develop a Document Request System for the Municipal Civil Registrar of Calauan that enhances the efficiency, accuracy, and accessibility of civil registry document requests, ultimately improving public service delivery.</p> <p><b>Specific Objectives</b></p> <ol style="list-style-type: none"> <li>1. To develop a Document Request Management System that streamlines and automates the processes of requesting and approving documents.</li> <li>2. To develop a digital dashboard that saves time and facilitates staff work by allowing them to quickly view, approve, and manage document requests.</li> </ol> |             |
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|  | <div> <div> 3. To integrate data analytics and K-Means clustering to automatically group similar request types, enabling staff to easily identify patterns and organize records more efficiently. </div> <div> 4. To evaluate the Document Request System using ISO 25010 software quality standards by determining whether the system meets the requirements for functionality, reliability, usability, and performance. </div> <div> 5. To prepare an implementation plan that guides how the development of the Document Request System will be set up and used in the Municipal Civil Registrar’s Office. </div> </div> <div> <b>Purpose and Description</b> </div> <div> <p> The system will include essential features that facilitate a smoother transaction process for document requests. It will be designed to modernize the process of requesting civil registry documents, such as birth certificates, marriage certificates, and death certificates. The system will enhance efficiency by digitizing request submissions and notifying residents, thereby reducing long waiting times and manual requests. With its user-friendly interface, the system will allow residents to submit requests online, check the status of their applications, and receive automated notifications regarding document status, specifically when a request is ready for release and when it has been officially released. </p> <p> The system will use K-Means clustering to group requests by age, document type, and barangay. This will allow staff to easily organize and track pending records, identify request patterns, and improve decision-making and </p> </div> |             |
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|  | <p>service allocation. The system will also improve transparency and accountability by maintaining a clear archive of transaction records and reducing the risk of lost or misplaced documents. Furthermore, it will enhance staff efficiency and help avoid delays.</p> <p>The system will minimize paper-based transactions and reduce administrative costs. It will be accessible both online and in-office, ensuring convenience for residents while maintaining data security and compliance with municipal regulations.</p> <p><b>Scope and Limitation</b></p> <p>This capstone project will focus on developing a Document Request System for the residents of Calauan, which will allow them to request and process civil registry documents from the Municipal Civil Registrar’s Office. The system will be designed for use by registered residents of Calauan, ensuring that it remains exclusive to the local population.</p> <p>The system will consist of two main parts: the Admin Side and the User Side. The User Side will allow residents to register accounts, submit requests for various civil documents, receive notifications once their account registration is approved, receive real-time updates on the status of their document requests, leave reviews based on their experiences, and receive notifications when their requested documents are ready for pickup. These features aim to improve transparency, user experience, and access to civil registry services.</p> | 6           |
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|  | <p>On the Admin Side, administrators will be responsible for managing and validating system activities. They will be able to approve or reject resident account registrations and document request submissions. K-Means clustering and data analytics will be integrated to help organize and analyze resident data, providing reports and summaries that support performance monitoring and service improvement. Additionally, administrators will be able to manually fill out request forms for walk-in clients to ensure that all transactions, whether online or in person, are properly recorded in the system.</p> <p>The system will support the digital processing of the following civil documents: Local Birth Certificate, Death Certificate, and Marriage Certificate. These documents represent the most frequently requested records at the Civil Registrar’s Office and are essential for various legal and administrative purposes.</p> <p>The system will have certain limitations. It will only be available to residents of Calauan and will be web-based, requiring users to have internet access and basic digital skills. It will not integrate with other government systems, such as direct communication with the PSA or online payment platforms, due to budgetary and security constraints. Additionally, the Municipal Civil Registrar will only provide physical copies of the requested documents and will not have access to any digital copies stored in the system, as the platform will function solely as a request management system.</p> | 7           |
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|  | <div>Theoretical Framework</div> <div> <p>This study is anchored in Connectionism Theory by Thorndike (1910), which explains how people learn through connections between actions and their results. According to this theory, learning occurs through trying different actions, observing what works, and repeating those that lead to success. This idea can be applied to the development of intelligent and user-friendly systems.</p> <p>One of Thorndike’s key ideas is the Law of Effect, which states that people are more likely to repeat actions that lead to positive results. This principle aligns with how the proposed system operates. The system will improve by learning from actions that produce successful outcomes. By applying the K-Means clustering algorithm, the system can categorize user requests more intelligently by grouping requesters with similar characteristics. In the document request system, each effective retrieval and accurate categorization of user inquiries constitutes a “positive outcome.” The system identifies patterns in successful interactions and strengthens these patterns through a grouping method known as a clustering algorithm. As it processes increasing amounts of user data, the system gradually improves its ability to group information. This enhancement makes document sorting and searching faster and more effective. This process is comparable to how individuals improve their actions over time based on feedback. The more often the system succeeds, the better it becomes at predicting and organizing new requests independently. As a result, user satisfaction increases, and the system operates more efficiently.</p> </div> |   |
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|  | <p>Another important principle is the Law of Exercise, which suggests that the more frequently an action is repeated, the stronger the connection between the action and the outcome becomes. This implies that as users interact with the system more frequently, it will become more effective at responding accurately to common requests. The Law of Readiness indicates that learning occurs most effectively when users are prepared and able to use the system, thereby making their experience faster and easier.</p> <p>By incorporating these principles into the system's design, the document request process will become more user-centered, data-driven, and responsive. The clustering method demonstrates how repeated successful actions can contribute to continuous system improvement.</p> <p><b>Conceptual Framework</b></p> <p>This conceptual framework outlines the development of a Document Request System with K-Means Clustering for the Municipal Civil Registrar of Calauan. It focuses on how the system will organize and process document requests by grouping similar requests using clustering techniques. The goal is to improve efficiency, streamline workflows, and provide faster and more accurate services to citizens.</p> |             |
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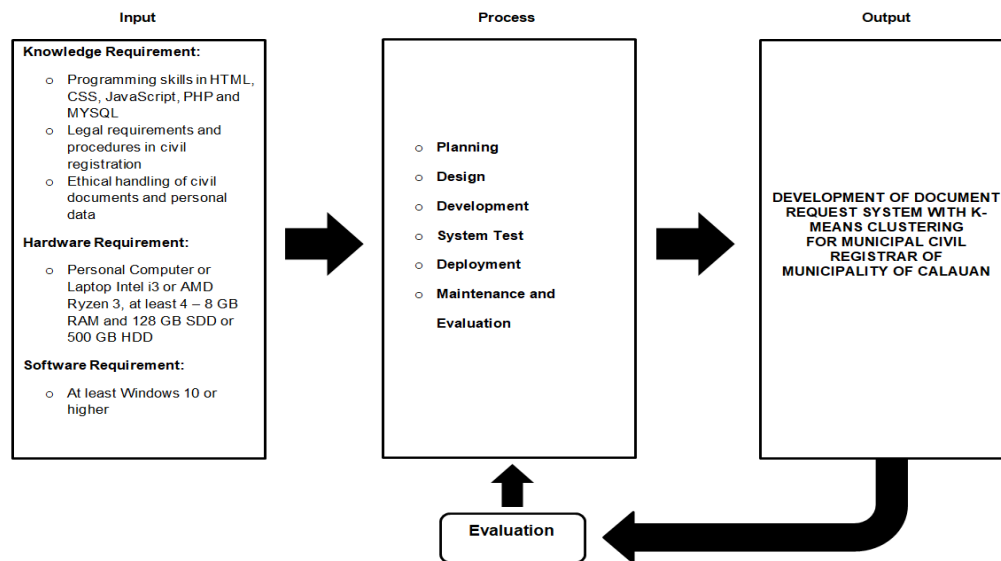
**Figure 1***Conceptual Framework (Input-Process-Output)*

Figure 1 shows the framework for developing the “*Document Request System with K-Means Clustering*” for the Municipal Civil Registrar of Calauan. Before development begins, it is important to have the appropriate knowledge and tools. This includes an understanding of front-end and back-end technologies such as HTML, CSS, JavaScript, PHP, and MySQL, as well as familiarity with civil registration procedures and the proper handling of personal data. The system should be developed on a Windows 10 (or later) machine with an Intel i3 or AMD Ryzen 3 processor, 4–8 GB of RAM, and at least a 128 GB SSD or a 500 GB HDD.

The project follows the standard System Development Life Cycle. In the planning stage, the goals, requirements, and success criteria, such as clustering accuracy, are defined. During the design phase, the database structure is planned, and the user interface and system architecture are outlined. In the development

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|  | <p>phase, the front-end and back-end components are implemented, and the K-Means clustering algorithm is integrated to group similar request characteristics. The system testing phase evaluates whether all functions operate correctly, including clustering accuracy and overall system performance. Once testing is completed, the system is deployed at the registrar’s office with appropriate security settings, backup mechanisms, and access controls.</p> <p>After deployment, the project enters the maintenance and evaluation phase, which feeds back into the development process. This stage involves monitoring system performance, collecting user feedback, and refining the clustering algorithm and other system components. Continuous evaluation helps improve the system to meet real-world needs, making the document request process more efficient and user-friendly.</p> <p><b>Significance of the Study</b></p> <p><b>Municipal Civil Registrar of Calauan, Laguna:</b> The new system will significantly improve the way documents are processed and managed. It will reduce the need for manual work, which will help minimize delays in fulfilling document requests. This improvement will allow the Municipal Civil Registrar to operate more efficiently, ensuring that residents receive the necessary civil registry documents faster and with fewer errors.</p> <p><b>Residents of Calauan:</b> For the residents of Calauan, the system will provide a more convenient and accessible way to request civil registry documents such as</p> |    |
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|  | <p>birth certificates, marriage licenses, and death certificates. Traditionally, requesting these documents involves long lines and extended waiting periods. With the implementation of the system, the process will become much faster, allowing residents to submit requests online or through a more streamlined procedure that reduces prolonged waiting times.</p> <p><b>Municipal Staff:</b> The system will assist municipal staff in several ways. It will organize and simplify the process of approving requests and tracking document statuses. The approval process will become faster and more systematic. Staff members will be able to easily monitor requests, identify those that require immediate attention, and provide improved service to residents. This will make their workload more manageable and allow them to focus on tasks that require greater attention.</p> <p><b>Researchers:</b> This study will contribute valuable knowledge to the field of digital government services. It will provide insights into how digital systems can enhance municipal services, particularly in document request processes. Researchers may use this study as a reference to explore similar systems in other areas or to build upon the findings to improve existing models.</p> <p><b>Future Researchers:</b> For individuals who plan to research and develop similar systems in the future, this study will serve as a foundational reference. It will highlight the benefits and challenges of implementing digital systems in municipal civil registries. By examining this research, future developers can gain a better understanding of how to design and improve digital services for local governments,</p> | 12          |
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|  | <div> <p>ensuring that such systems are effective and user-friendly. This may lead to the development of similar systems in other municipalities, contributing to the modernization and efficiency of public services.</p> <p><b>Definition of Terms</b></p> <p><i>Conceptual Definition of Terms</i></p> <p><b>Automation:</b> The use of technology to perform tasks with minimal human involvement (Bovens, 2020).</p> <p><b>Customer Satisfaction:</b> The level of contentment users feels after using a service (Kotler, Keller, &amp; Chernev, 2021).</p> <p><b>Data Analytics:</b> The process of examining data to find meaningful patterns, trends, and insights. (Provost &amp; Fawcett, 2020)</p> <p><b>Data-Driven Governance:</b> The practice of using analyzed information to guide government decisions (World Bank, 2021).</p> <p><b>Digital Governance:</b> The use of information technology to improve public service delivery (United Nations, 2022).</p> <p><b>Document Request System:</b> A system designed to make it easier for people to request official documents from a government office (Laudon &amp; Laudon, 2022).</p> <p><b>Efficiency:</b> The ability to accomplish tasks with minimal waste of time and resources (OECD, 2020).</p> </div> |    |
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|  | <p><b>Information Transparency:</b> The ability of users to see and understand transaction progress (Florini &amp; Pauli, 2020).</p> <p><b>K-Means Clustering:</b> A machine learning technique that groups similar data points into clusters for easier analysis (Aggarwal, 2020).</p> <p><b>Operational Effectiveness:</b> The ability of an organization to perform its functions accurately and promptly (Porter &amp; Kramer, 2020).</p> <p><b>Public Service Quality:</b> The overall performance of government services from the user’s perspective (OECD, 2021).</p> <p><b>Service Accessibility:</b> The ease with which residents can obtain services (United Nations, 2021).</p> <p><b>Service Delivery:</b> The act of providing services to users (World Bank, 2020).</p> <p><b>System Suitability:</b> The ability of a system to remain useful over time (ISO/IEC, 2020).</p> <p><b>User Experience:</b> How users feel when interacting with a system (ISO, 2020).</p> | 14          |
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*Technical Definition of Terms*

**Admin Account:** The system profile used by staff to manage users and document requests.

**Admin Dashboard:** The interface showing request statistics and management tools.

**Algorithm:** A set of programmed instructions used to process and organize data.

**Authentication:** The feature that verifies user identity during login.

**Clustering Analytics:** The grouping of document requests using K-Means for analysis.

**Database:** The electronic storage of all system data.

**Data Visualization:** Charts and graphs that show request patterns.

**Document Request Form:** The online form is used to submit document applications.

**Email Notification System:** The feature that sends alerts to users.

**K-Means Clustering:** The algorithm used to group requests based on age, barangay, and document type.

**Login System:** The module that allows secure system access.

**Request Management Module:** The system tool for approving and rejecting requests.

**Request Tracking System:** The feature that shows request status.

**Resident Account:** The user profile created by a resident.

**System Database:** The storage of user and request information.

**Walk-In Module:** The feature for recording in-person requests.

**Web-Based Application:** The online system is accessed through browsers.

**Web Server:** The computer that hosts the system.

# **PAMANTASAN NG LUNGSOD NG SAN PABLO**

## **CHAPTER II**

### **REVIEW OF RELATED LITERATURE AND SYSTEMS**

This chapter reviews previous studies and existing systems related to the project. It discusses key ideas and findings that support the development of the proposed system.

#### **Technical Background**

The technical background section provides an overview of the tools, technologies, and frameworks used in the development of the project. These include Visual Studio Code (VS Code), PHP, HTML, and CSS.

#### **Visual Studio Code**

Visual Studio Code, commonly known as VS Code, is a popular and versatile code editor developed by Microsoft. Its widespread use is attributed to its extensive collection of extensions, which allow developers to customize the editor for various programming tasks. VS Code includes features such as code suggestions, debugging tools, and built-in support for Git, all of which help developers work more efficiently. The editor is lightweight and compatible with multiple operating systems, making it appealing to a broad audience. Being open source, the community can contribute to its development and improvement.

#### **HyperText Markup Language (HTML)**

HTML is the basic language used to structure web pages. It uses tags to define elements such as headings, paragraphs, images, and links, forming the foundation

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|  | <p>of web content. The structure of HTML organizes information in a way that is understandable to users. The language is easy to learn, making it accessible to beginners in web development. HTML also serves as the framework for other web technologies, such as CSS and JavaScript, which add style and functionality to web pages.</p> <p><b>Cascading Style Sheets (CSS)</b></p> <p>CSS is a language used to style HTML elements and control their appearance on a webpage. It allows developers to define styles such as colors, fonts, layouts, and how a website adapts to different devices. The “cascading” feature of CSS means that styles can be inherited or overridden, making it easier to manage the appearance of elements. CSS provides various methods to select elements for styling.</p> <p><b>Hypertext Preprocessor (PHP)</b></p> <p>PHP is a widely used server-side scripting language that is often embedded within HTML. It allows developers to create dynamic websites that interact with databases and display personalized content. PHP includes numerous libraries and frameworks that simplify tasks such as handling forms, managing databases, and managing user sessions. It works well with databases like MySQL for efficient data management. Although PHP has a large community and abundant support resources, developers must pay careful attention to security to avoid vulnerabilities.</p> |             |
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|  | <div>Related Literature</div> <p>This section reviews books, articles, and studies relevant to the topic of this research. It focuses on key ideas, findings, and theories that help explain the problem being studied. The purpose is to understand what has already been done and to identify how this research can contribute to the existing body of knowledge.</p> <div>Document Request System</div> <p>According to Alade (2023), the development of a Web-Based Electronic Document Management System (WBEDMS) using the Object-Oriented Hypermedia Design Method represents an important step in simplifying document access and management in offices. The system was created to address the challenges of manual processes and scattered storage. The study employed in-depth interviews, use case analysis, and careful examination of existing processes to gather data. Testing revealed strong results, including high user satisfaction (96.60%), accuracy (95%), and ease of use (99.20%). These outcomes demonstrate the system's positive impact on work efficiency and workflow. The study also emphasizes the importance of such systems in managing organizational knowledge, which supports decision-making and planning. Additionally, the research highlights the potential for these systems to scale and address larger organizational challenges.</p> <p>Abacı and Medeni (2022) examined the effectiveness of Electronic Document Management Systems (EDMS) in Turkey's Ministry of Science, Industry, and Technology. They evaluated the system's impact on managerial</p> |    |
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|  | <div> <div> <p>control, workflow efficiency, job satisfaction, and data security. Despite challenges such as weak technical infrastructure and resistance to change, the study found that EDMS improved key operational areas by enhancing oversight, accelerating processes, and strengthening data protection. The study further suggests that addressing identified problems, providing adequate training, and implementing gradual system rollout can reduce resistance and maximize system adoption.</p> <p>Dante et al. (2023) developed a web-based Document Management System (DMS) for ISPSC, Tagudin Campus, to address slow paper-based processes. By reducing reliance on physical files and streamlining document handling, the system significantly improved workflow. User satisfaction, measured using the WAMMI instrument, indicated that the system was effective and user-friendly. The study highlights the importance of user-centered design and demonstrates how digital technology can improve organizational efficiency in educational institutions. The findings also suggest opportunities for integrating cloud-based tools to enhance accessibility and collaboration.</p> <p>Velasco et al. (2024) developed an online document tracking and request system for a Philippine university's records office using the System Development Life Cycle (SDLC) methodology. Multiple programming languages were used to ensure flexibility and adaptability. The system complied with ISO/IEC 25010 standards, achieving a high average score of 4.85 in areas including functionality, performance, interoperability, usability, reliability, security, maintainability, and portability. The study demonstrates that digital systems can improve school record</p> </div> </div> |    |
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|  | <div> <p>management, streamline document handling, and serve as a reference for similar implementations. The research also notes the benefits of incorporating features such as automated notifications and data analytics to optimize administrative processes.</p> <p>Similarly, Fagar (2024) developed a Record Management System with Document Control for Northern Iloilo Polytechnic State College. The study, published in the <i>International Journal of Management Research and Analysis (IJMRA)</i>, evaluated system usability and functionality. Users rated the system as "Very Good" in terms of functionality, usability, and performance, reinforcing the positive impact of digital systems on organizational efficiency and user satisfaction (Velasco et al., 2024).</p> <p>Abang et al. (2022) implemented an online paper request system for City College of Tagaytay using Agile methodology to improve manual processes. The system proved particularly useful during the COVID-19 pandemic by providing a safer, faster, and more organized method of handling documents. User evaluations confirmed that the system was effective and reliable, indicating its potential applicability in other schools. The study emphasizes the importance of online systems in enhancing workflow efficiency and ensuring continuity of services during challenging circumstances. Recommendations include integrating real-time tracking and linking with other databases to expand functionality.</p> <p>Casicas et al. (2025) developed an Online Document Request System (ODRS) for PHINMA College of Cagayan, improving inefficient and risky manual</p> </div> |    |
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|  | <p>processes. The system achieved a high usability score of 79.4 in user tests. The authors also suggested incorporating machine learning techniques to identify usage patterns and further optimize system performance.</p> <p>Bartosova et al. (2022) explored the technical aspects of document retrieval systems using a Hopfield neural network to enhance search rankings. Their research focused on hybrid approaches for relevance verification and sorting, demonstrating the potential of artificial intelligence to improve digital document management. While this study differs from conventional EDMS applications, it contributes to the broader field of intelligent document handling and aligns with the integration of machine learning (Casicas et al., 2025).</p> <p>Frese (2023) examined document management within the European Commission under Articles 101 and 102 of the Treaty on the Functioning of the European Union (TFEU). The study highlights the critical role of document oversight in regulatory compliance and legal accountability. It also underscores the challenges organizations face in balancing regulatory requirements with the protection of sensitive data. Firms must navigate legal frameworks carefully to comply with regulations while maintaining data privacy, illustrating the complex interaction between legal obligations and organizational information practices. Understanding these legal frameworks equips firms to better respond to document requests.</p> <p>Requinto et al. (2019) studied record and document management in Barangay Paligui, Apalit, Pampanga, where inefficiencies were impacting service</p> |    |
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|  | <div> <p>quality. They developed a central Document Management System (DMS) using Kanban Agile methodology and OwnCloud for secure file storage. The system, accessible via computer and mobile devices, improved functionality, usability, and met the needs of barangay officials. This study demonstrates the importance of technology in streamlining administrative processes, enhancing service delivery, and ensuring secure access to critical records. It also highlights the potential for scaling similar systems in other barangays with comparable challenges and recommends further research to customize tools for local needs.</p> <p>Overall, these studies indicate that digital document request and management systems improve workflow efficiency, enhance service delivery, and enable organizations to operate more effectively. By leveraging modern technologies, user-friendly designs, and well-planned methodologies, these systems address common challenges in both government and private sectors. To maximize their effectiveness, adequate user training, infrastructure improvements, and system scalability are essential.</p> <p><b>K-means Clustering</b></p> <p>Clustering is a key technique in unsupervised learning, helping to identify hidden patterns or groups in data. Traditional clustering methods, such as K-means, provide only a single way to group data, which may be insufficient for complex or high-dimensional datasets. To address this limitation, multiple clustering methods have been introduced, offering several alternative groupings of the same data. This approach helps researchers uncover more meaningful</p> </div> |    |
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|  | <p>patterns that might be missed by a single clustering solution. As data becomes increasingly complex, employing multiple clustering techniques is becoming more critical in data analysis (Yu &amp; Wang, 2024).</p> <p>K-means clustering is a widely used unsupervised learning method for grouping similar data points without pre-assigned labels. Unlike supervised learning, it does not rely on known outcomes. K-means is considered a “hard” clustering method, meaning each data point belongs exclusively to one cluster. It is applied in diverse areas, including market segmentation, document organization, image partitioning, and image compression. Its popularity stems from its simplicity, speed, and effectiveness across various applications (Kavlakoglu &amp; Winland, 2024).</p> <p>Despite its usefulness, K-means faces several challenges. These include sensitivity to data errors, poor initialization, high-dimensional data, and datasets that do not conform to spherical clusters (Kumar &amp; Patel). The algorithm also requires specifying the number of clusters in advance, assumes clusters are roughly equal in size, and may overfit or fail to capture complex data relationships. To address these issues, researchers have developed more advanced algorithms that enhance accuracy and robustness.</p> <p>To improve interpretability, some studies propose using small decision trees to explain clustering results. This approach has been evaluated with K-means and K-medians, but limitations exist: the top-down construction of trees can degrade results, and using trees with a fixed number of leaves may prevent optimal</p> |    |
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|  | <div> <div> <p>clustering. While decision trees enhance interpretability, they do not always yield the highest-quality clusters (Moshkovitz et al., 2020).</p> <p>Several improvements to K-means have been proposed. The U-K-means algorithm automatically determines the optimal number of clusters and requires minimal parameter tuning, making it fully unsupervised. Experimental results indicate that U-K-means outperforms traditional K-means in accuracy and reliability (Sinaga &amp; Yang, 2020). Nie et al. (2023) proposed a modified K-means that optimizes an alternative objective function, eliminating the need to update cluster centers at every iteration. This method runs as fast as traditional K-means while improving convergence speed and performance on real-world datasets.</p> <p>Other hybrid approaches combine K-means with optimization techniques. For example, integrating K-means with Particle Swarm Optimization (PSO) improves cluster initialization, enhancing accuracy and stability, particularly for analyzing public opinion data (Qi et al., 2024). Lk-means, which incorporates Lévy flight, enables broader exploration during clustering, improving results on large or uneven datasets without increasing complexity. Experiments on ten public datasets showed superior performance in cluster quality and distribution compared with other methods (Hu et al., 2023).</p> <p>Another improved K-means variant addresses challenges such as poor initialization, slow convergence, and unknown cluster numbers. By employing the elbow method to determine the optimal number of clusters and selecting better initial centroids, this approach improves speed, accuracy, and clustering quality,</p> </div> </div> |    |
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|  | <div> <p>outperforming traditional K-means and Canopy-KMeans on benchmark datasets (Zhao, 2022).</p> <p>Despite its limitations, K-means remains one of the most widely used clustering algorithms due to its simplicity, interpretability, and computational efficiency. Its major drawbacks include the need to predefine the number of clusters, sensitivity to initial centroid placement, and limited ability to handle complex or non-spherical data. Researchers continue to develop enhancements, including better initialization strategies, alternative similarity measures, and hybrid methods combining K-means with other techniques. These improvements ensure that K-means remains a powerful and practical tool for real-world data analysis (Ikotun et al., 2022).</p> <p>In summary, K-means is a fast and effective clustering algorithm that continues to evolve through research addressing its weaknesses. Advances such as U-K-means, hybrid optimization methods, and Lévy flight-based approaches improve clustering accuracy, stability, and applicability to complex datasets, maintaining K-means' relevance in modern data analysis.</p> <p><b>Municipal Civil Registrar</b></p> <p>Celeste and Tinagan (2020) conducted a study on civil registration awareness and compliance in a municipality in Negros Occidental, Philippines. The researchers surveyed 376 residents and found high awareness of registration requirements for births, marriages, and deaths. However, while compliance rates</p> </div> |    |
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|  | <p>were high for births and marriages, they were significantly lower for death registrations. The study identified challenges such as the lack of valid identification cards, transportation costs, and time constraints, which hindered compliance, particularly for death registrations. Women and more educated residents demonstrated higher awareness and compliance. The study concluded that improving access to identification cards and reducing logistical barriers could enhance registration compliance, especially for deaths.</p> <p>Becedas et al. (2022) reviewed 18 studies to examine how civil registration systems contribute to achieving the United Nations’ Sustainable Development Goal of leaving no one behind by 2030. The findings emphasize that proper civil registration, including births, deaths, and marriages, is crucial for individuals to access their rights and improve outcomes in health, education, and economic opportunities. Well-functioning civil registration systems have long-term positive effects, benefiting not only individuals but also future generations. The study underscores the importance of investing in robust civil registration systems, as they provide essential data for tracking development progress and empowering citizens to claim services and rights.</p> <p>Okoth (2023) analyzed security challenges in digital civil registration systems managing records of births, deaths, marriages, and divorces. The study identified three major security threats: cybersecurity risks, identity fraud, and privacy breaches. To address these issues, the author recommended implementing encryption, biometric verification, and strict access controls.</p> |    |
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|  | <p>Additionally, the digital divide may exclude marginalized groups, so mobile registration solutions and community outreach were suggested to ensure inclusive access. The study concluded that while digital transformation improves efficiency, comprehensive security frameworks are necessary to protect data and maintain public trust.</p> <p>Carreon (2024) examined service quality and client satisfaction at the Calamba City Civil Registry Office in the Philippines. The study surveyed 22 employees, 71 applicants, and 52 clients, finding significant differences between employee and client perceptions of service quality. High levels of reliability, responsiveness, empathy, and timeliness were linked to greater customer satisfaction. The study further recommended staff training, system upgrades, and the implementation of a one-stop-shop service model to enhance service delivery. Overall, maintaining high-quality service was shown to be essential to meeting and exceeding customer expectations.</p> <p>De Guzman (2024) explored civil registration services in Calamba City, focusing on birth, marriage, and death certificates. The study found overall satisfaction to be high, especially for death certificates, followed by marriage and birth certificates. However, some issues were noted, such as long waiting times for marriage certificates and inaccuracies in birth certificate data. The research highlighted areas for improvement, including expediting death registrations and providing clearer instructions for marriage registration, providing valuable guidance for improving civil registration services.</p> |    |
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|  | <p>Latupeirissa et al. (2024) studied the impact of digital transformation on public services, including civil registration systems. The study identified three primary benefits of digital transformation: improved operational efficiency, increased citizen engagement, and enhanced accountability. However, challenges were also noted, including ensuring accessibility, managing costs, and protecting data privacy. The research concluded that digital transformation must be inclusive, ensuring access for all, including marginalized populations, to be successful in civil registration systems.</p> <p>Trofimets (2021) discussed the transition of civil registration systems from paper-based to digital formats. The study emphasized that as governments modernize these systems, legal concepts such as “digital landscape” and “digital hygiene” must evolve. Civil registration systems must update laws and technical standards while ensuring proper legal oversight to manage vital records efficiently in today’s information society.</p> <p>Kumar et al. (2022) examined challenges faced by the Civil Registration System in Bihar, India, focusing on birth and death registration. While residents were aware of registration procedures, death registrations were often neglected, particularly for women and children, due to a lack of immediate benefits. Barriers such as corruption, inadequate service delivery, and limited staffing were identified. The study recommended linking health institutions with registration centers, introducing mobile registration in remote areas, and conducting</p> | 29          |
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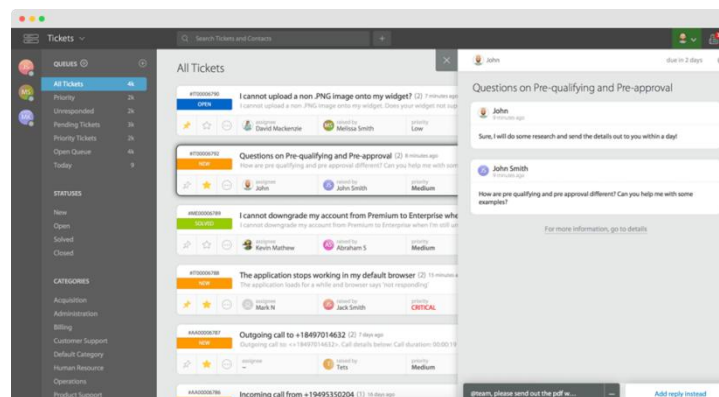
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|  | <p>awareness campaigns to improve registration rates. These measures are essential for ensuring accurate vital statistics and guiding healthcare interventions.</p> <p>Ndiyun and Mukonza (2025) investigated innovative transformations in Cameroon’s civil registration system following legislative reforms in 2024. The study highlighted improvements including electronic declarations, the creation of a national database, and the transition to electronic certificates. These changes are expected to enhance governance efficiency, transparency, and accessibility. However, challenges such as inadequate infrastructure, low digital literacy, and limited internet access must be addressed to ensure successful implementation. A balanced approach is required to guarantee that digitalization enhances service delivery without compromising citizens’ rights.</p> <p>Lascuña and Junsay (2023) studied how service quality affects customer satisfaction at the Civil Registry of the Island Garden City of Samal, Philippines. The study found that reliability, responsiveness, empathy, and timeliness were strongly correlated with high customer satisfaction. Service quality accounted for 77.6% of the variation in satisfaction, indicating that maintaining high service standards is crucial for exceeding customer expectations. The study recommended further research in other sectors to examine similar service quality indicators.</p> |    |
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## Related Systems

This section examines systems and tools that are similar to the one being developed in this research. It analyzes how these systems function, their strengths, and their limitations. This evaluation helps identify potential improvements and new features that can be incorporated to create a more effective and efficient system.

### Figure 2

#### *HappyFox Request Management System*



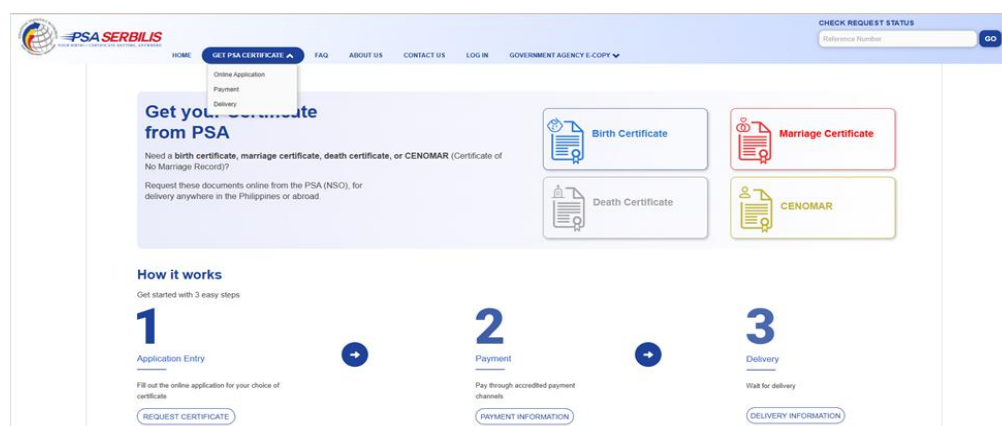
<https://www.happyfox.com/software/request-management-system/>

HappyFox's Request Management System enables businesses to manage customer requests via email, live chat, and voice calls on a single platform, offering features such as SLA management and workflow automation to improve efficiency. In comparison, the proposed document request system will allow residents to register, submit requests for civil documents, receive real-time updates, leave feedback, and receive notifications when their documents are ready, with the goal of enhancing user experience and transparency.

On the administrative side, the document request system will support the management of registrations and document requests, while incorporating Walk-In Request Form processing, K-means clustering, and data analytics to improve organization, service quality, and decision-making. The system will also include color-coded duration alerts to indicate when the status of a document request has not changed for more than three days.

**Figure 3**

*PSA Serbilis*



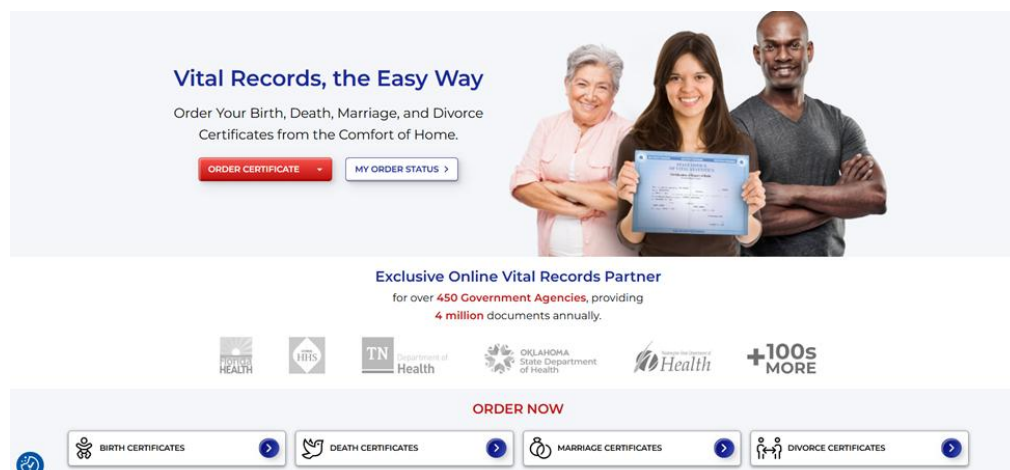
[https://www.psaserbilis.com.ph/?fbclid=IwY2xjawJglplleHRuA2FibQlXMAABHjLD7wOm2YyLkNzDfPDk-0vbJ3WC8PQqLs1zW\\_8pXgF-hywNnTwVx7eCdF6w\\_aem\\_s3x9a2FgUCi5jFmFHGa2OQ#](https://www.psaserbilis.com.ph/?fbclid=IwY2xjawJglplleHRuA2FibQlXMAABHjLD7wOm2YyLkNzDfPDk-0vbJ3WC8PQqLs1zW_8pXgF-hywNnTwVx7eCdF6w_aem_s3x9a2FgUCi5jFmFHGa2OQ#)

The PSA Serbilis website enables users to request civil registry documents, including birth, marriage, and death certificates, as well as CENOMAR, by completing an online form, making a payment, and awaiting delivery. The platform provides both local and international shipping options and ensures secure transactions to protect user data. In comparison, the proposed Document Request

System will allow residents to request civil documents while offering enhanced features such as real-time status updates, account registration, feedback submission, and notifications for document pickup, thereby improving transparency. The system will also include a Walk-In Request Form for applicants who prefer to process their documents onsite, ensuring that all requests—whether online or in person—are accurately recorded.

**Figure 4**

*VitalCheck*



[https://www.vitalchek.com/v/?fbclid=IwY2xjawJgIyVleHRuA2FibQIxMAABHsFEHZQYKy4NmPdruZTSuDb\\_miXCWL5ZpfATwQePEQVnnfDzIP\\_aP7WDsNAw\\_aem\\_bAZ9nWg7pPK1S2nRs8X3U](https://www.vitalchek.com/v/?fbclid=IwY2xjawJgIyVleHRuA2FibQIxMAABHsFEHZQYKy4NmPdruZTSuDb_miXCWL5ZpfATwQePEQVnnfDzIP_aP7WDsNAw_aem_bAZ9nWg7pPK1S2nRs8X3U)

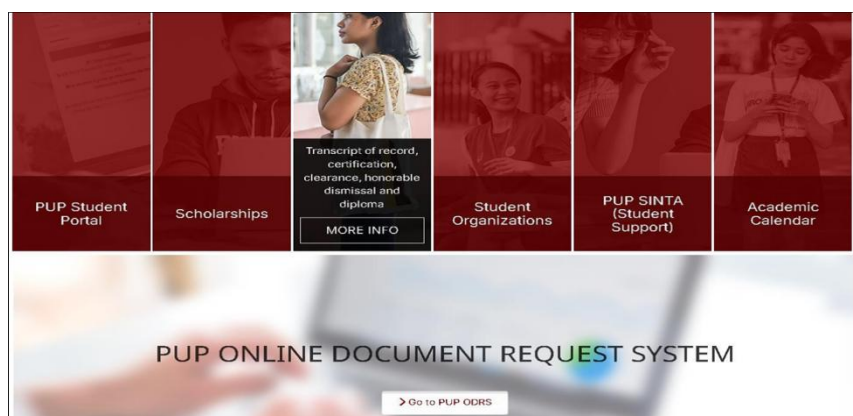
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VitalChek is an online service that enables individuals to request and receive official vital records, including birth, death, marriage, and divorce certificates, by partnering with over 450 government agencies to ensure secure and efficient processing and delivery. In comparison, the proposed Document Request System will allow residents to request civil documents while offering

additional features, such as real-time status updates, account registration, feedback submission, and pickup notifications, to enhance transparency and user satisfaction. The system will also include a Walk-In Request Form to accommodate residents who prefer to process their documents onsite, ensuring that both online and in-person requests are accurately recorded.

**Figure 5**

*PUP Online Document Request System*



<https://www.pup.edu.ph/students/>

The Polytechnic University of the Philippines (PUP) offers an Online Document Request System that allows students to request academic documents, such as transcripts, certificates, and diplomas, without needing to visit the campus, thereby making the process more efficient and accessible. In comparison, the proposed Document Request System will enable residents to request civil documents while placing greater emphasis on real-time status updates, account registration, feedback submission, and pickup notifications to ensure transparency and convenience. The system will also provide a Walk-In Request Form to

accommodate residents who prefer to process their requests in person, ensuring that both online and onsite transactions are accurately recorded. Additionally, the system will incorporate K-means clustering, data analytics, and a color-coded duration indicator to highlight document requests that have not changed status within three days.

**Figure 6**

*Business Name Registration System*



<https://bnrs.dti.gov.ph/request-certification>

The Department of Trade and Industry (DTI) offers an online platform for business owners to request certifications related to their registered business names, providing an efficient and convenient method for handling certification requests without visiting DTI offices. In comparison, the proposed Document Request System will enable residents to request civil documents while emphasizing real-time status updates, account registration, feedback submission, and pickup notifications to enhance transparency. While DTI's platform ensures

secure encryption and protects user data, the Document Request System will integrate technologies such as Optical Character Recognition (OCR), K-means clustering, and data analytics to improve service accuracy and administrative efficiency.

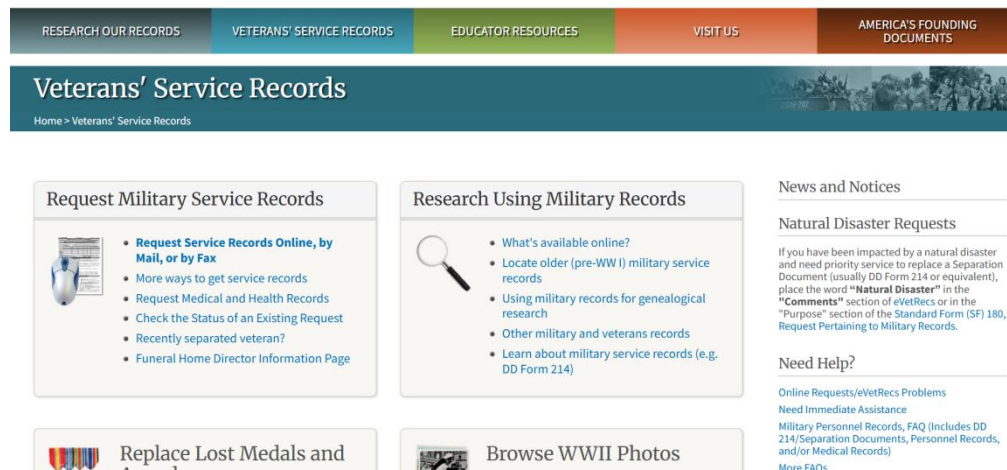
### Figure 7

#### *National Library of the Philippines*



<https://web.nlp.gov.ph/category/request-for-quotation/>

The National Library of the Philippines (NLP) offers a “Request for Quotation” page on its website, providing a transparent platform for suppliers to submit quotes for goods and services required by the library, ensuring compliance with government procurement policies. In comparison, the proposed Document Request System will enable residents to request civil documents, emphasizing real-time status updates, account registration, feedback submission, and pickup notifications to enhance transparency. While NLP’s platform ensures accountability in procurement, the Document Request System will incorporate K-means clustering and data analytics to improve accuracy and efficiency.

**Figure 8***National Archives*

<https://www.archives.gov/veterans>

The National Archives (NARA) offers an online platform that allows veterans and their families to request military records, such as DD214 forms and discharge papers, making the retrieval process faster and more accessible. Its system emphasizes secure document delivery and reliable request tracking. In comparison, the proposed Document Request System will enable residents to request civil documents while offering additional features, such as real-time status updates, account registration, feedback submission, and notifications when documents are ready for pickup, to ensure greater transparency and user convenience. The system will also provide a Walk-In Request Form to accommodate individuals who prefer to process their document requests onsite, ensuring that both online and in-person transactions are accurately recorded. Additionally, the system will incorporate K-means clustering, data analytics, and a

color-coded duration indicator to highlight document requests whose status has not changed within three days.

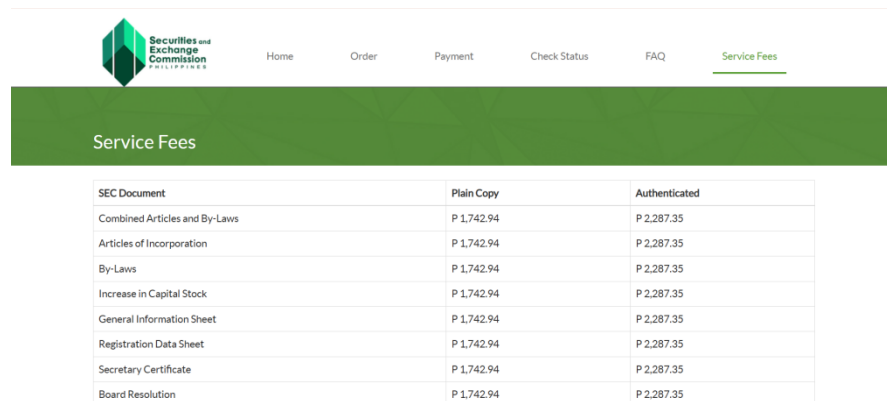
**Figure 9**

*Docs Vault*

The screenshot displays the 'Create & Send Document Request Link' interface. It includes a 'Request Link' section with a location path and a generated URL. Below this, 'Request Options' allow users to configure request parameters like sender information, expiry date, file size, and number of files. A 'Notification' section lists users to be notified. The 'Email Request Link' section provides fields for email address, subject, and a detailed message body. The message body contains a sample request for 'Bids and Proposals' from ACME Company, detailing the need for a website and software products, and setting a deadline for Friday, October 30.

<https://www.docsvault.com/features/document-request-links/>

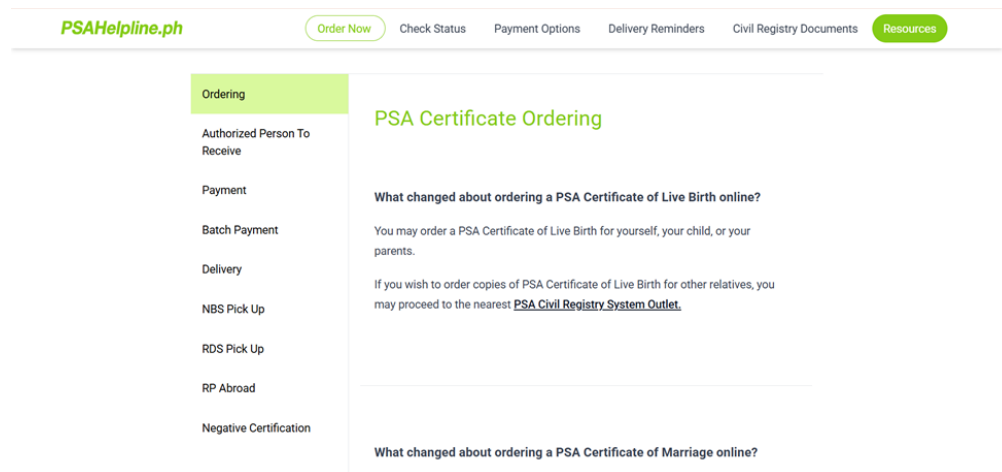
DocsVault offers a “Document Request Links” feature, allowing users to create personalized links for requesting documents, simplifying the retrieval of important files through secure and efficient requests. In contrast, the proposed Document Request System will focus on enabling residents to request civil documents while providing features such as real-time status updates, account registration, feedback submission, and notifications when documents are ready for pickup, thereby enhancing transparency and user experience. While DocsVault emphasizes secure and customizable document requests, the Document Request System will incorporate advanced technologies, such as K-means clustering and data analytics, to improve the accuracy and efficiency of document management.

**Figure 10***SEC Express*


| SEC Document                  | Plain Copy | Authenticated |
|-------------------------------|------------|---------------|
| Combined Articles and By-Laws | P 1,742.94 | P 2,287.35    |
| Articles of Incorporation     | P 1,742.94 | P 2,287.35    |
| By-Laws                       | P 1,742.94 | P 2,287.35    |
| Increase in Capital Stock     | P 1,742.94 | P 2,287.35    |
| General Information Sheet     | P 1,742.94 | P 2,287.35    |
| Registration Data Sheet       | P 1,742.94 | P 2,287.35    |
| Secretary Certificate         | P 1,742.94 | P 2,287.35    |
| Board Resolution              | P 1,742.94 | P 2,287.35    |

<https://secexpress.ph/service-fees>

The SEC Express System provides a clear list of service fees on its website, displaying the costs for various SEC-related requests, such as document submissions and business registrations. In comparison, the proposed Document Request System will also promote transparency by enabling residents to track real-time status updates, submit requests, provide feedback, and receive notifications when their civil documents are ready for pickup. Additionally, the system will offer a Walk-In Request Form, ensuring that both online and in-person transactions are accurately logged and managed.

**Figure 11***PSA Helpline*

<https://orders.psahelpline.ph/order/choose-type>

The PSA Helpline website offers a straightforward document request system, allowing users to easily request vital records, such as birth and marriage certificates, thereby simplifying the process of obtaining official documents. In comparison, the proposed Document Request System will also enable residents to request civil documents while offering additional features, including real-time status updates, account registration, feedback submission, and pickup notifications to improve transparency. While the PSA Helpline focuses on secure and efficient document retrieval, the Document Request System will integrate K-means clustering and data analytics to enhance accuracy, organization, and service quality.

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|  | <div>Project Synthesis</div> <div> <p>The “Development of Document Request System with K-Means Clustering for the Municipal Civil Registrar of the Municipality of Calauan” focuses on creating a document request system for the Municipal Registrar of Calauan to improve civil registration services through better document management and digital transformation. A study by Celeste and Tinagan (2020) found that although people are aware of the need to register, they do not always follow through, partly due to various barriers. Another study by Lascuña and Junsay (2023) highlighted that service quality directly affects customer satisfaction, emphasizing the need for improved service delivery. These findings support the development of a system that addresses these issues.</p> <p>The project uses K-Means clustering, a technique that helps organize and improve document retrieval. Bartosova et al. (2022) demonstrated that AI could enhance document searches by making them more efficient, inspiring the use of K-Means clustering in this system. The aim is to make document retrieval faster, personalize service delivery, and improve overall system performance, setting it apart from existing studies. While other studies have examined document request systems—for example, Abang et al. (2022) developed an online paper request system for the City College of Tagaytay, and Casicas et al. (2025) used Agile methods to create a system for PHINMA College of Cagayan—these studies demonstrate the benefits of digital systems but do not incorporate machine learning to predict trends in document requests, which this project does.</p> </div> |    |
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|  | <div> <p>Data security and accessibility are key concerns in digital systems. Okoth (2023) discussed the security threats in digital civil registration systems, while Latupeirissa et al. (2024) emphasized the need to protect user privacy and ensure accessibility for all users. This system will prioritize security and accessibility, including for marginalized communities, in line with the goals of modernizing government services (Trofimets, 2021).</p> <p>This project is not a copy or repetition of existing work but represents a novel approach that combines established methods to address the specific needs of the Municipal Registrar of Calauan. By using K-Means clustering and addressing concerns about security and accessibility, the project aims to create a system that meets the unique requirements of a municipal civil registry and supports the digital transformation of local government services.</p> </div> |    |
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## **PAMANTASAN NG LUNGSOD NG SAN PABLO**

### **CHAPTER III**

#### **DESIGN AND METHODOLOGY**

This chapter includes requirement analysis, documentation, system and software design, development, testing, and implementation of the proposed system.

##### **Requirement Analysis**

Requesting documents from a municipal civil registrar, such as birth, marriage, or death certificates, is primarily done in person. A recent survey of thirty respondents showed that all of them (100%) relied on walk-in transactions because the municipality does not provide alternative options, such as online requests. This lack of alternatives reduces convenience for residents and creates additional challenges. Many also experienced delays: thirteen respondents (43.3%) waited four to seven days, twelve (40%) waited one to three days, and six (20%) had to wait over a week. Only seven respondents (23.3%) received their documents on the same day. The main issues reported were long processing times (73.3%) and inconvenient office hours (53.3%).

These findings indicate significant inefficiencies in the current system. It depends heavily on manual processes, which slows service delivery. The absence of online options requires residents to visit the office during limited operating hours, which is particularly challenging for those living in distant areas. To address these issues, the municipality could introduce online services, extend office hours, and

streamline processes to make document requests faster and more convenient for all residents.

### Current Technical Situation in Municipal Civil Registrar of Calauan

This section presents the existing document request procedure currently followed by the Municipal Civil Registrar of Calauan. This traditional approach involves multiple steps that require manual verification and record handling, which often result in delays, errors, and inefficiencies.

**Figure 12**

#### *Document Request Procedure*

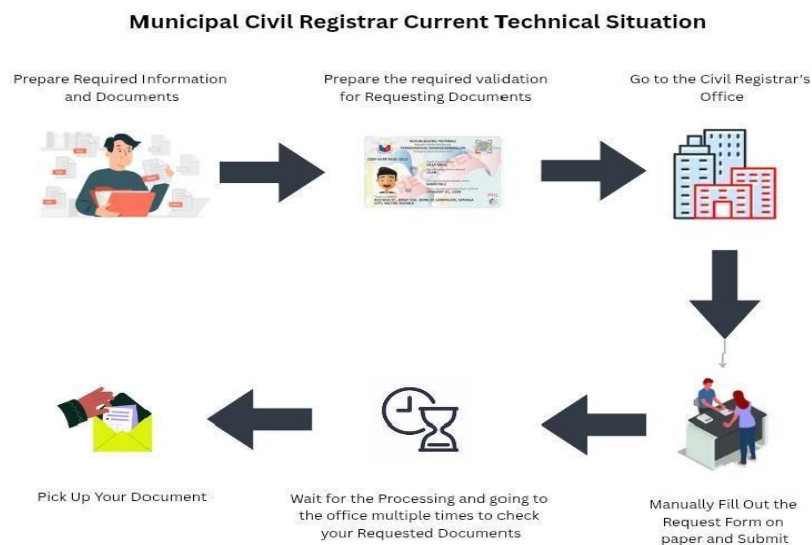


Figure 12 shows that the traditional process of requesting documents from the Civil Registrar's Office is lengthy and requires significant time and effort. It begins with the individual preparing the necessary items, such as an ID and

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|  | <p>supporting documents, to ensure the request is valid and complies with regulations. Next, the person must visit the Civil Registrar’s Office in person, which can be challenging for those who live far away or have limited time.</p> <p>At the office, the individual completes a paper request form, providing the details of the document needed. This step requires accuracy, as mistakes can lead to delays or even rejection of the request. After submitting the form, the office staff manually process the request, checking records and verifying details. Often, the requester must return to the office multiple times to follow up or provide additional information, which adds to the inconvenience, particularly for those with busy schedules or distant residences. Finally, the individual must return to pick up the completed document.</p> <p>While this system functions, it is slow and cumbersome. Relying on paper and in-person visits frequently results in delays and errors. As the number of document requests increases, this process becomes more challenging for the office to manage. Staff must handle numerous requests, which can further slow service and reduce efficiency. Consequently, both office personnel and residents experience frustration. To address these issues, the system could be improved through digital solutions, such as online form submissions, real-time tracking of requests, and automated updates, reducing the need for in-person visits. This approach would make the process faster, more convenient, and more accurate, benefiting both the Civil Registrar’s Office and the residents it serves.</p> | 45          |
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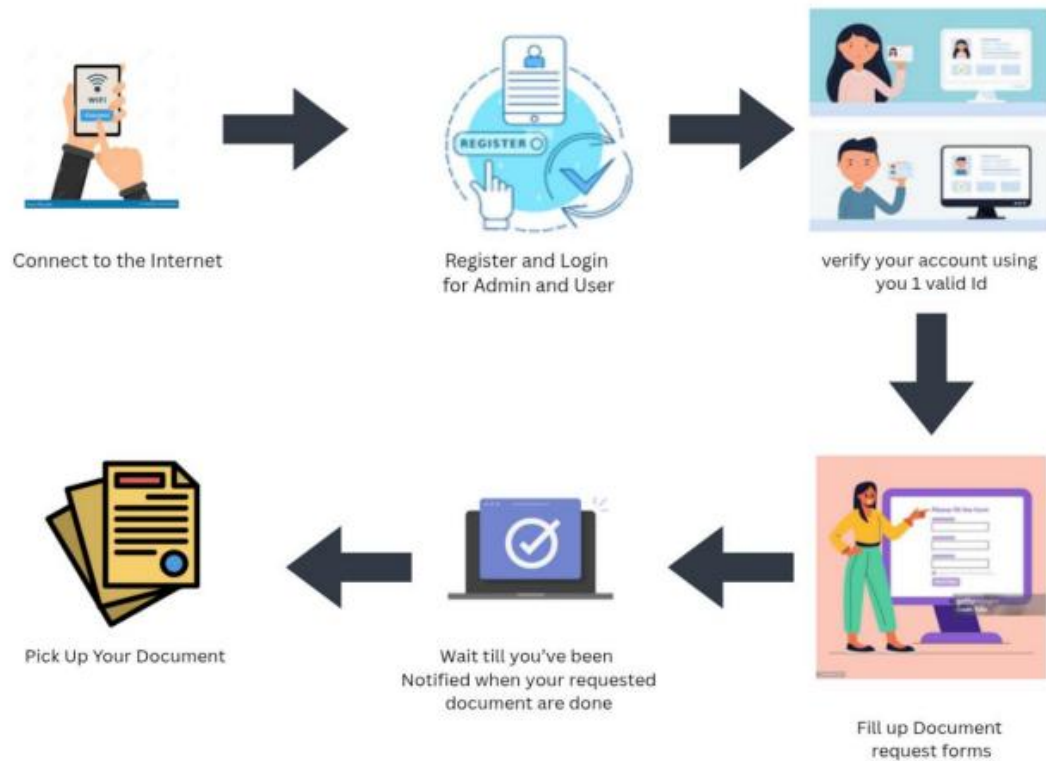
**Figure 13***New Technical Situation of User*

Figure 13 illustrates the new, streamlined process for requesting documents from the Civil Registrar's Office, which is faster and more convenient compared to the traditional method. The system allows residents to submit requests online from their homes, reducing the need for multiple visits to the office—particularly beneficial for those living far away or with busy schedules.

The process begins with the user accessing the internet via a computer or mobile device and visiting the Civil Registrar's platform. New users must register by providing personal information, including their name, contact details, and a valid

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|  | <div> <div>ID to verify residency, and then set up a username and password. Existing users can simply log in with their credentials.</div> <div> <p>Once logged in, the system requires account verification by uploading a valid ID, ensuring that only authorized users can access sensitive documents. After verification, users fill out an online document request form specifying the type of document needed—such as a birth, marriage, or death certificate—and include all relevant details like names and dates. The system allows users to review their entries before submission, reducing errors commonly associated with handwritten forms.</p> <p>After submission, the system processes the request automatically, eliminating the need for residents to repeatedly visit the office. Users receive notifications via email or the platform when their documents are ready for release. This ensures a smooth and convenient retrieval process.</p> <p>Compared to the traditional method, this new online system saves time, reduces errors, and improves overall efficiency. For the Civil Registrar’s Office, it decreases paperwork, minimizes processing errors, and enhances service delivery. Overall, this modernized approach benefits both residents and staff by making the document request process faster, simpler, and more reliable.</p> <div> <div>Flowchart</div> <div>Presents the basic flow of the Document Request Management System, from login to document processing and reporting.</div> </div> </div> </div> |    |
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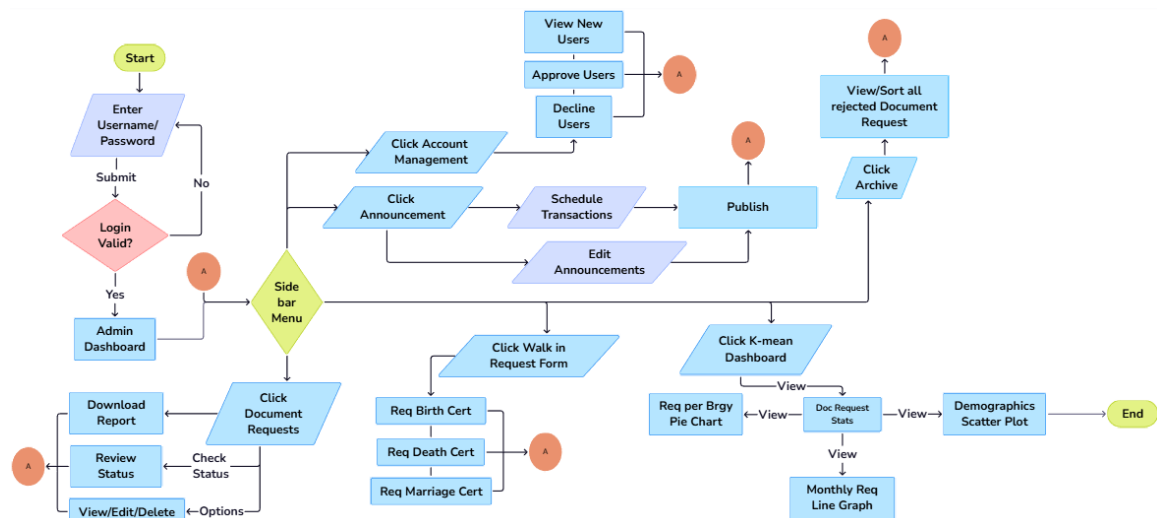
**Figure 14***Admin Flowchart*

Figure 14 illustrates the flowchart of the admin's workflow in the system from start to finish. The process begins when the admin logs in using a username and password. After account verification, the admin is directed to the Admin Dashboard, which provides access to various sections via the sidebar: Announcements, Account Management, Document Requests, Walk-In Request Form, K-Means Data Analytics Dashboard, and Archive.

In the Announcements section, the admin can create, edit, and schedule messages visible to system users. The Account Management section allows the admin to approve or decline new user registrations and view user details.

The Document Requests section enables the admin to review all user-submitted requests, update their status to pending, processing, ready, for release, or rejected, and provide reasons for any rejections to guide users.

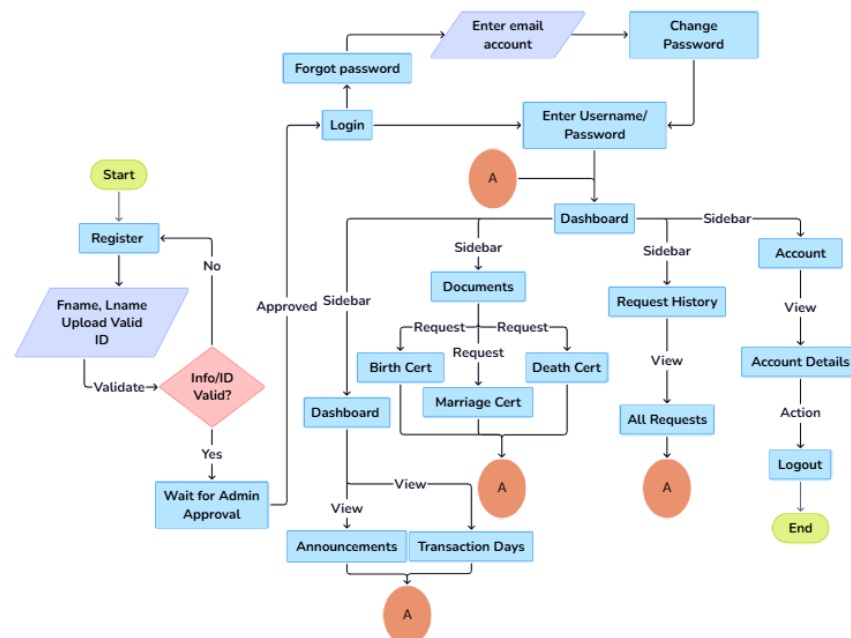
The Walk-In Request Form allows the admin to manually encode requests submitted by in-person clients, ensuring that these transactions are properly recorded, tracked, and integrated with online requests.

The K-Means Data Analytics Dashboard presents graphs and charts that provide insights into system activity, including the number of document requests, monthly trends, the most requested document types, and average processing times.

Finally, the Archive section stores all rejected requests, allowing the admin to maintain a record of past submissions and review them as needed. Overall, the flowchart demonstrates how the system's modules work together to help the admin efficiently manage users, documents, and operational data.

**Figure 15**

*Resident Flowchart*



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
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|  | <div> <p>Figure 15 presents the complete process flow of the system from the user or resident’s perspective. The process begins with the user registration, where the resident provides their first and last name, uploads a valid ID, and sets a username and password. The system validates the submitted information and ID. If the details are correct, the account enters a waiting stage for administrator approval. Once approved, the user can log in to the system.</p> <p>If the user forgets their password, the system provides a “Forgot Password” feature. By selecting this option, the user enters their email to receive a password reset link. After creating a new password, the user can log in and continue using the system.</p> <p>Upon successful login, the user is directed to the Dashboard, which includes a sidebar providing access to key sections such as Documents, Request History, and Account. The Dashboard also displays announcements and transaction schedules for quick reference.</p> <p>In the Documents section, users can request civil registry documents, including Birth Certificates, Marriage Certificates, and Death Certificates, by filling out the corresponding request forms. The Request History section allows users to view past and ongoing requests and track their status in real time. In the Account section, users can view and update their personal information as needed. Once activities are completed, users can securely log out from the system.</p> </div> |    |
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|  | <div>Requirement Documentation</div> <div> <p>After analyzing the current situation at the Municipal Civil Registrar of Calauan, the developers identified the need to leverage technology to streamline the document request process. To address this, a Document Request System was developed, providing an online platform integrated with K-Means clustering to improve service management, efficiency, and decision-making. The system is designed with the following key features and requirements:</p> <p>a.) Administrator Account; b.) Resident (User) Account; c.) Request Document; d.) Email Notifications; e.) Recent Document Requests; f.) Dashboard; g.) Managing, Approving and Disapproving of Requests.</p> <div>User Registration Module</div> <p>This module serves as the gateway to the system, ensuring that only authorized residents can access its features. Residents are required to create an account or log in to proceed. During registration, users must submit essential information, including full name, address, valid identification, and contact details. These details are verified by the Municipal Registrar’s Office to ensure authenticity. Only verified accounts are permitted to make official document requests, maintaining the system’s security, accountability, and integrity.</p> <div>Home Module</div> <p>Upon successful login, users are redirected to the Home Module, which serves as the starting point for navigating the system. This module provides a</p> </div> |    |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |

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|  | <p>clear, step-by-step guide on how to submit a document request. By presenting concise instructions and visual cues, it helps users understand the process, minimizes errors during submission, and improves overall efficiency in requesting civil registry documents.</p> <p><b><i>Service Selection Module</i></b></p> <p>This module allows users to select and request specific civil registry documents, including Birth Certificates, Marriage Certificates, and Death Certificates. Each request form includes the required fields that must be completed by the user to ensure accurate processing. Once a request is submitted, the Document Tracking Module enables users to monitor the progress of their application in real time. Users can view the current status of their requests—such as pending, processing, ready for release, or completed—ensuring transparency and keeping them informed throughout the entire process.</p> <p><b><i>Document Tracking Module</i></b></p> <p>This feature provides real-time status updates such as “Pending,” “Processing,” “Ready”, “For release”, “Released”. By offering visibility into the request’s status, the module promotes transparency and reduces the need for follow-up visits or calls to the municipal office.</p> <p><b><i>Email Notification Module</i></b></p> <p>The Email Notification Module ensures timely communication by automatically sending emails to users. Notifications include account approval</p> | 52          |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |

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|  | <div> <p>confirmations and other important updates, keeping users informed without requiring them to repeatedly log in. This feature enhances user convenience and contributes to a more efficient and user-friendly system.</p> <p><b>Admin Log-In Module</b></p> <p>The Admin Log-In Module provides a secure access point for municipal staff to manage and oversee the Document Request System. Each administrator is assigned a unique username and password, ensuring that only authorized personnel can log in. Upon successful authentication, the administrator is directed to the Admin Dashboard, which grants access to all management features and tools. This module safeguards sensitive civil registration data while enabling administrators to efficiently perform their tasks within the system.</p> <p><b>Admin Dashboard</b></p> <p>The Admin Dashboard serves as the central control panel where administrators can monitor the entire document request process. It provides a real-time summary of all requests, including those that are pending, in process, ready, for release, released, or rejected. By displaying up-to-date information, the dashboard enables administrators to make quick and informed decisions, improving efficiency and ensuring smooth management of document requests.</p> <p><b>Request Management Module</b></p> <p>In this module, administrators can view the complete list of requests submitted by residents. They can review the details of each request, including the</p> </div> |    |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |

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|  | <div> <p>type of document requested and any attached requirements. Administrators have the ability to approve or reject requests, with an option to provide feedback for rejections. Once the requested documents have been processed, requests can be marked as “Ready” for pickup. The system also allows administrators to easily search, and filter requests based on criteria such as status, resident name, date, or document type, streamlining request management and improving efficiency.</p> <p><b><i>Lookup and Management Module</i></b></p> <p>This module is composed of two sub-modules: <b>Resident Management</b> and <b>Forms Management</b>. In the <b>Resident Management</b> sub-module, administrators can oversee the list of authorized users, ensuring that only eligible residents can access the system. They have the ability to add, verify, and update user accounts to maintain system security and proper access control. In the <b>Forms Management</b> sub-module, administrators can manage the different types of documents available in the system. This includes updating document requirements or modifying existing forms to align with the municipality’s policies and procedures.</p> <p><b><i>K-Means Clustering Analysis Module</i></b></p> <p>This module enhances the system’s data analytics by utilizing the K-Means clustering algorithm to categorize user requests. Clustering is performed based on three key attributes: age, type of requested document, and barangay. By grouping requesters with similar characteristics, the system can uncover meaningful patterns, such as which age groups frequently request certain documents or which barangays generate the highest volume of requests. These insights enable</p> </div> |    |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |

administrators to make data-driven decisions, improve service allocation, and optimize resources according to user demographics and location.

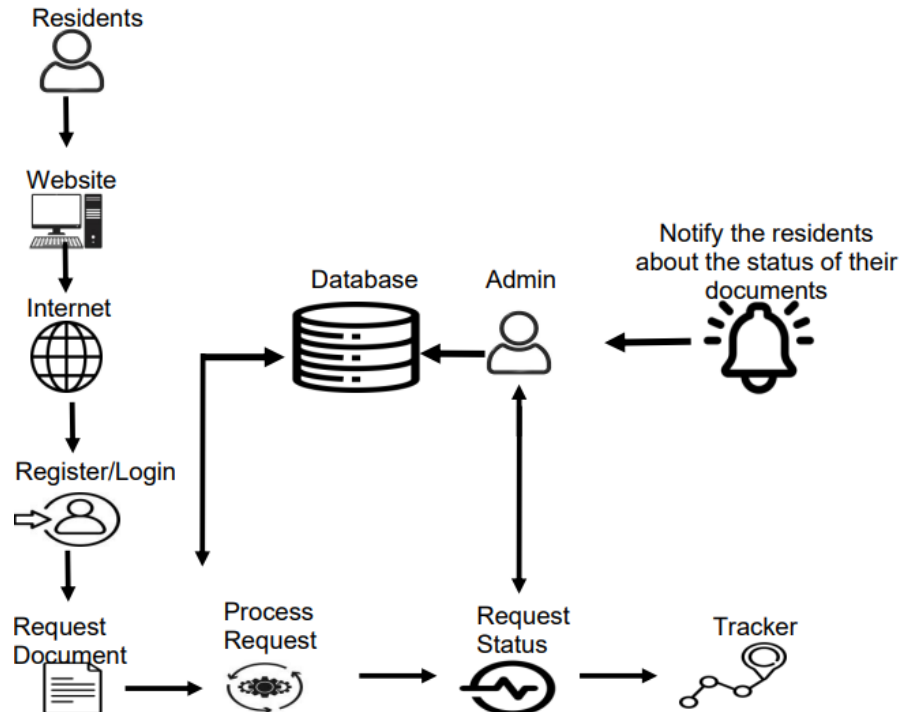
### Design of Software, Systems, Product and/ or Processes

This section explains how the developed system is designed according to accepted standards. It covers the design of the software, system architecture, and processes, illustrating how the system functions and how its components interact to achieve the intended objectives.

### System Architecture

**Figure 16**

*System Architecture of Document Request System*



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
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|  | <div> <p>Figure 16 shows the system architecture of the Document Request System, which accommodates two types of users: Admin and Resident. Residents interact with the system by registering or logging in through the website and submitting requests for specific civil documents. These requests are transmitted over the internet and stored in a centralized database. The Admin accesses the same system to retrieve, review, and process the requests. Once a request is completed, the status is updated in the database, and a notification is sent to inform the Resident of the progress or completion of their request. Additionally, a tracking feature is integrated, allowing both users and the system to monitor the status of each document request in real time.</p> <p><b>Use Case Diagram</b></p> <p>This section presents the use case diagram of the Document Request System, illustrating how the two primary users—the Resident and the Admin—interact with the system to perform various tasks. The diagram highlights the main functions of the system from the users’ perspective, providing a clear overview of their roles and interactions.</p> </div> |    |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |

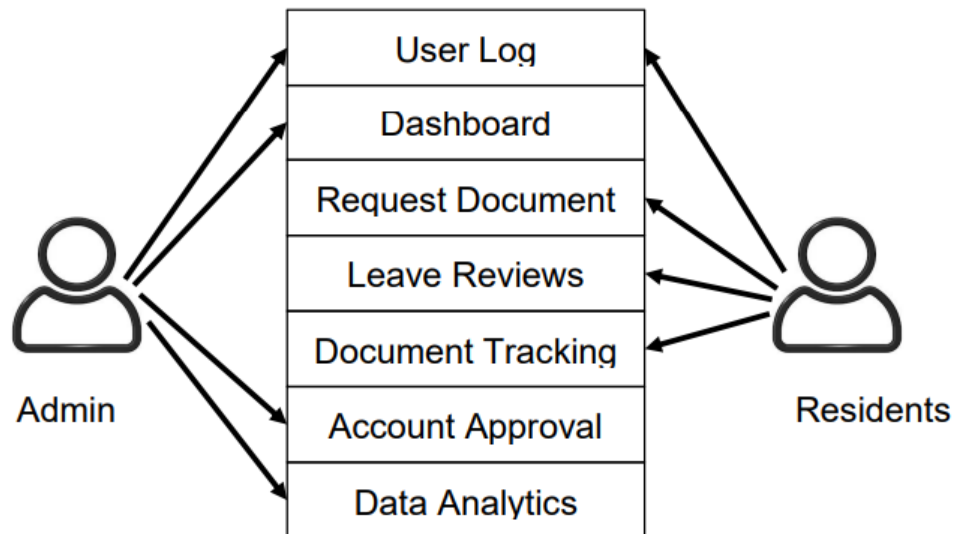
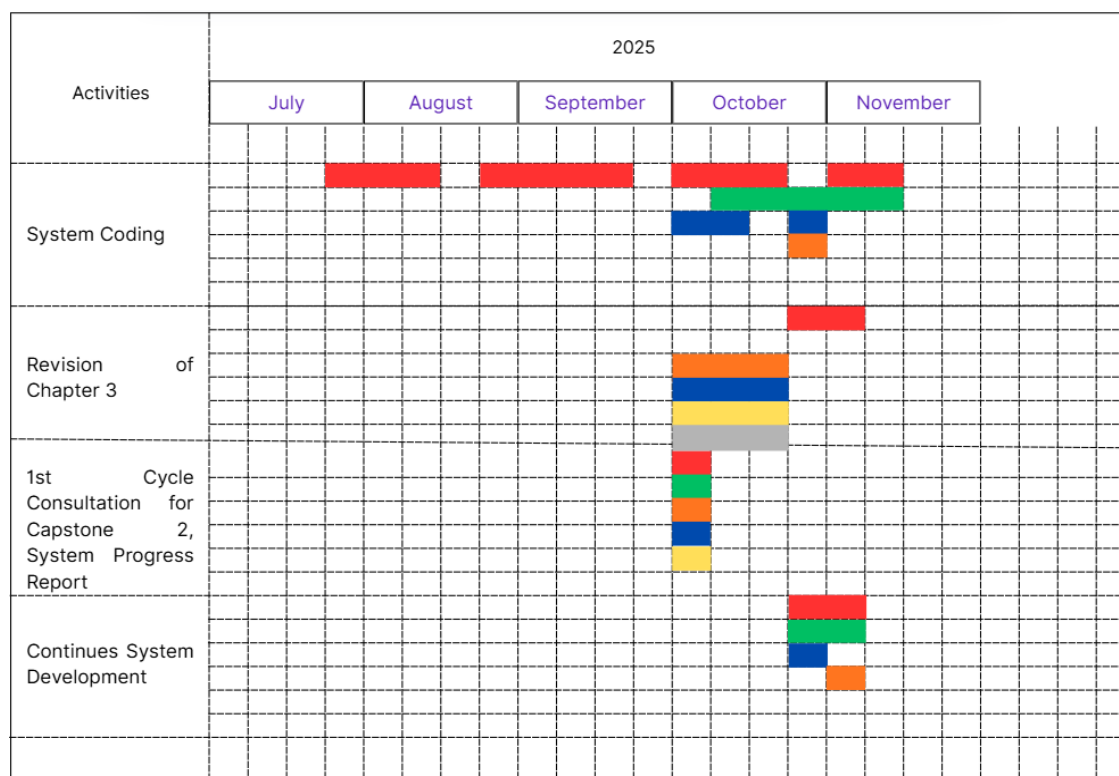
**Figure 17***Use Case Diagram of Document Request System*

Figure 17 illustrates a use case diagram for the Document Request System, featuring two primary users: Admin and Resident. The diagram outlines the actions each user can perform within the system. Residents can request documents, track the status of their requests, and leave reviews, while Admin users have additional privileges, such as approving accounts, managing user logs, viewing data analytics, and accessing the dashboard. This diagram provides a clear and concise overview of how each type of user interacts with the system.

### Gantt Chart



| Activities                                                                          | 2025  |  |  |                                                   |  |  |            |  |  |                                                        |  |      |                                                   |     |  |
|-------------------------------------------------------------------------------------|-------|--|--|---------------------------------------------------|--|--|------------|--|--|--------------------------------------------------------|--|------|---------------------------------------------------|-----|--|
|                                                                                     | March |  |  | April                                             |  |  | May        |  |  | June                                                   |  | July |                                                   | Aug |  |
| Development of UI Design and Start of Coding                                        |       |  |  | [Red Bar]                                         |  |  |            |  |  |                                                        |  |      |                                                   |     |  |
|                                                                                     |       |  |  |                                                   |  |  | [Blue Bar] |  |  |                                                        |  |      |                                                   |     |  |
| Revision of Chapter 1 and 2                                                         |       |  |  | [Stacked Bars: Green, Orange, Blue, Yellow, Grey] |  |  |            |  |  |                                                        |  |      |                                                   |     |  |
| 3rd Cycle of Consultation with revised Chapter 1 and 2, and submission of Chapter 3 |       |  |  |                                                   |  |  |            |  |  | [Stacked Bars: Red, Green, Orange, Blue, Yellow, Grey] |  |      |                                                   |     |  |
| Revision of Chapter 1 to 3 and                                                      |       |  |  |                                                   |  |  |            |  |  | [Stacked Bars: Green, Orange, Blue, Yellow, Grey]      |  |      | [Stacked Bars: Green, Orange, Blue, Yellow, Grey] |     |  |



Continuation of Table 1

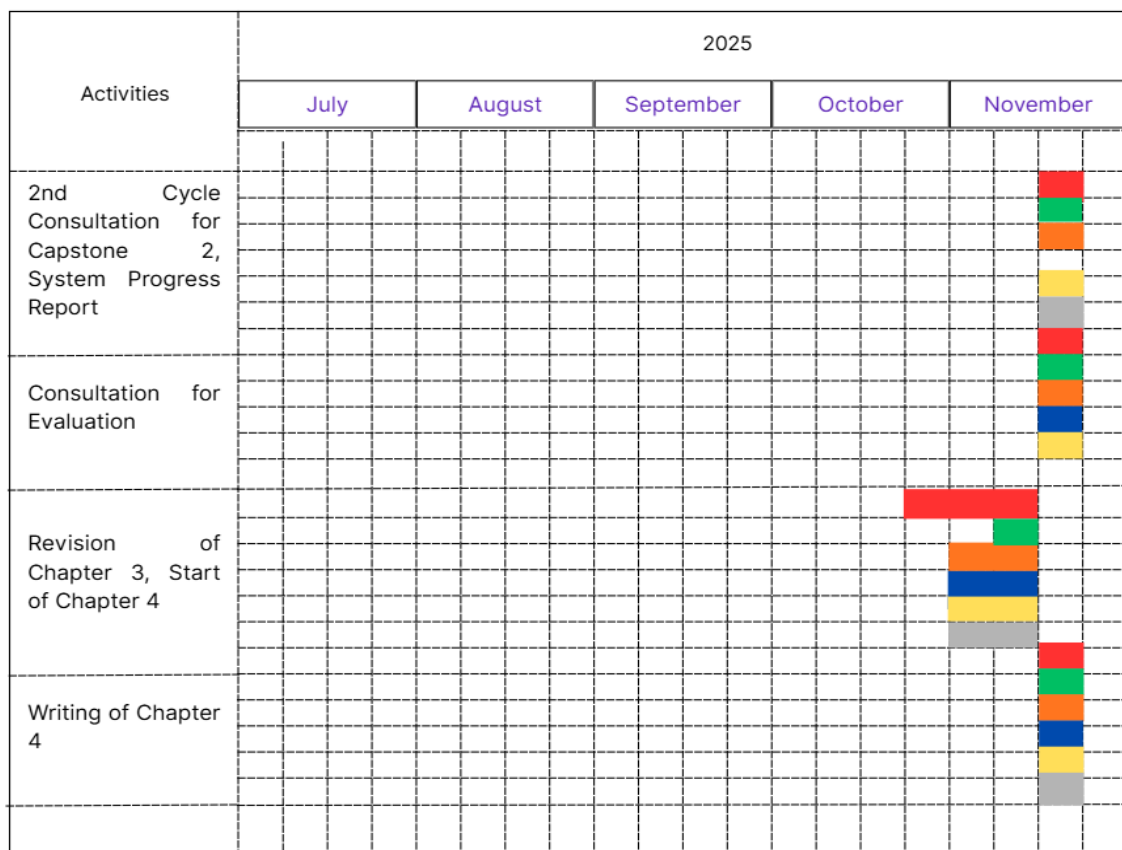


Table 1 presents the Gantt chart illustrating the project timeline from March to November, highlighting the major activities and each team member's involvement. The planning phase, capstone title selection, and data gathering were completed in March, while formal development began after the project title defense. The team conducted consultations, prepared chapters, designed the system, performed coding, and carried out revisions, with each member contributing according to their assigned roles.

In April, the group focused on consultations and the preparation of Chapter 1, followed by the second consultation cycle and submission of Chapter 2. By May and June, efforts shifted to user interface design and initial system coding, led by

the project manager with support from the technical members. This period also included revisions of Chapters 1 and 2, submission of Chapter 3, and the third consultation.

From July to September, the team concentrated on major system coding, continuing collaboration in October alongside revisions to Chapter 3 and the first consultation cycle for Capstone 2. In November, the team finalized system development, conducted the second consultation, prepared for evaluation, revised Chapter 3, began Chapter 4, and completed the writing of Chapter 4.

Overall, the Gantt chart reflects a systematic and collaborative workflow, demonstrating a balanced distribution of tasks and responsibilities throughout the project's duration.

**Table 2**

*ISO25010*

| CHARACTERISTIC       | SUBCHARACTERISTIC | Definition                                                                                                                       | 1 | 2 | 3 | 4 | 5 |
|----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY | FUNCTIONALITY     | The set of function covers all the specified tasks and user objectives                                                           |   |   |   |   |   |
|                      | CORRECTNESS       | The function provides the correct results with the needed degree of precision                                                    |   |   |   |   |   |
|                      | APPROPRIATENESS   | The function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   |   |
| RELIABILITY          | MATURITY          | A system, product or component meets for reliability under normal operation                                                      |   |   |   |   |   |
|                      | AVAILABILITY      | A product or system is operational and accessible when required for use                                                          |   |   |   |   |   |
|                      | FAULT TOLERANCE   | A system, product or component operates as intended despite the presence of hardware or software results                         |   |   |   |   |   |
|                      | RECOVERABILITY    | In the event of an interruption or a failure, a product or system can recover the data establish the desired state of the system |   |   |   |   |   |
|                      | ADAPTABILITY      | A prroduct or sytem can effectively and efficiently be adapted for different or                                                  |   |   |   |   |   |

|                           |                                    |                                                                                                                                 |  |  |  |  |  |
|---------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|                           |                                    | evolving hardware, software or other operational or usage environments                                                          |  |  |  |  |  |
| PORTABILITY               | INSTALLABILITY                     | A product or system can be successfully installed and/or uninstalled in a specified environment                                 |  |  |  |  |  |
|                           | REPLACEABILITY                     | A product can replace another specified software product for the same purpose in the same environment                           |  |  |  |  |  |
| USABILITY                 | APPROPRIATENESS<br>RECOGNIZABILITY | Users can recognize whether a product or a system is appropriate for their needs                                                |  |  |  |  |  |
|                           | LEARNABILITY                       | A product or system enables the user to learn how to use it with effectiveness, efficiency in emergency situations              |  |  |  |  |  |
|                           | OPERABILITY                        | A product or system is easy to operate, control and appropriate to use                                                          |  |  |  |  |  |
|                           | USER ERROR<br>PROTECTION           | A product or system protects user against making errors                                                                         |  |  |  |  |  |
|                           | USER INTERFACE<br>AESTHETICS       | A user interface enables pleasing and satisfying interactions for the user                                                      |  |  |  |  |  |
|                           | ACCESSIBILITY                      | A product or system can be used by people with the widest range of characteristics and capabilities to achieved specified goals |  |  |  |  |  |
| PERFORMANCE<br>EFFICIENCY | TIME-BEHAVIOR                      | The response and processing times and throughout rates of product or system, when performing its, functions meet requirements   |  |  |  |  |  |
|                           | RESOURCE<br>UTILIZATION            | The amounts and types of resources used by a product or system, when performing its functions, meet requirements                |  |  |  |  |  |
|                           | CAPACITY                           | The maximum limits of the product or system, parameter meet requirements                                                        |  |  |  |  |  |
| SECURITY                  | CONFIDENTIALITY                    | The prototype ensures that data is accessible only to those authorized to have access                                           |  |  |  |  |  |
|                           | INTEGRITY                          | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                   |  |  |  |  |  |
|                           | NON-REPUDATION                     | Actions or events can be proven to have taken place, so that the events or actions cannot be repudiated later                   |  |  |  |  |  |
|                           | ACCOUNTABILITY                     | The actions of an entity can be traced uniquely to the entity                                                                   |  |  |  |  |  |
|                           | AUTHENTICITY                       | The identity of a subject or resources can be proved to be the one claimed                                                      |  |  |  |  |  |
| COMPATIBILITY             | CO-EXISTENCE                       | A product can perform its required functions efficiently while sharing a common environment and resources with other products   |  |  |  |  |  |
|                           | INTEROPERABILITY                   | Two or more systems, products or components can exchange information                                                            |  |  |  |  |  |

|                 |               |                                                                                                                                            |  |  |  |  |  |
|-----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|                 |               | and use the information that has been exchanged                                                                                            |  |  |  |  |  |
| MAINTAINABILITY | MODULARITY    | A system or computer program is composed of discrete components such that a change to one component has minimal impact on other components |  |  |  |  |  |
|                 | REUSABILITY   | An asset can be used in more than one system                                                                                               |  |  |  |  |  |
|                 | ANALYZABILITY | It is possible to assess the impact on a product or system of an intended change to one or more of its parts                               |  |  |  |  |  |
|                 | MODIFIABILITY | A product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality          |  |  |  |  |  |
|                 | TESTABILITY   | Test criteria can be established for a system, product or components                                                                       |  |  |  |  |  |

**Table 3***Rating Scale Criteria*

| Range       | Verbal Interpretation |
|-------------|-----------------------|
| 4.51 - 5.00 | Highly Acceptable     |
| 3.51 - 4.50 | Very Acceptable       |
| 2.51 - 3.50 | Acceptable            |
| 1.51 - 2.50 | Fairly Acceptable     |
| 1.50 below  | Not Acceptable        |

Table 3 presents the rating scale used by the developers to evaluate the system. The mean score was computed to determine the average response of the evaluators for each item, based on the five-point rating scale derived from the ISO 25010 standards: 5 (Highly Acceptable), 4 (Very Acceptable), 3 (Acceptable), 2 (Fairly Acceptable), and 1 (Not Acceptable).

**Block Diagram**

Shows the interaction between the user, admin, database, and K-means clustering in processing and analyzing document requests.

**Figure 18**

*Block Diagram of Document Request System*

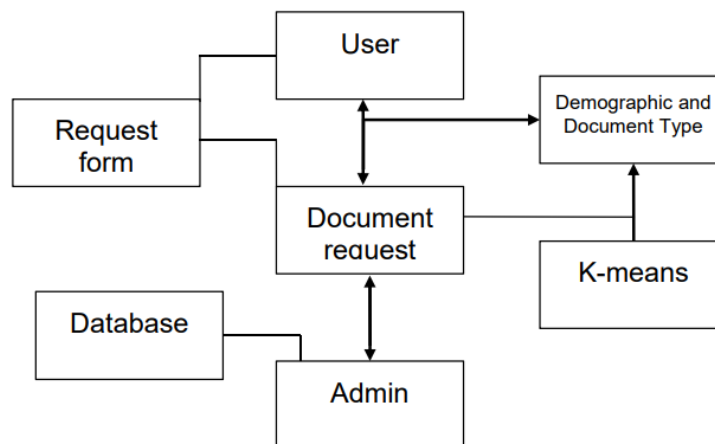


Figure 18 shows the block diagram illustrating the flow of the Document Request System integrated with K-Means clustering for the Municipal Civil Registrar of Calauan, Laguna. Users submit their requests through a form, which is then processed by the system and stored in the database. The K-Means clustering algorithm groups similar document requests into clusters, enabling faster processing and more insightful analysis. Administrators can manage incoming requests and monitor clustering data to optimize service efficiency and improve decision-making.

## Development and Testing

In developing the Document Request System with K-Means Clustering for the Municipal Civil Registrar of Calauan, the Agile methodology was employed to guide the project from planning through deployment. This iterative and collaborative approach ensured flexibility, facilitated rapid feedback loops, and enabled continuous refinement to align the system with the client's needs.

## Project Development

The developers followed the Agile methodology to guide the project from planning through deployment. This approach allowed the system to adapt efficiently to changes, support continuous improvement, and remain aligned with the client's specific requirements.

**Figure 19**

*Agile Model*



*[https://www.researchgate.net/figure/Agile-Methodology-in-System-Development-source-Okeke2021-retrieved-from\\_fig1\\_354310848](https://www.researchgate.net/figure/Agile-Methodology-in-System-Development-source-Okeke2021-retrieved-from_fig1_354310848)*

Figure 19 illustrates how the Agile methodology was applied throughout the development of the system. The process began with the **planning phase**, during which project objectives, scope, and system requirements were identified. The development team collaborated closely with the Municipal Civil Registrar to analyze the existing challenges in the manual document request process and determine essential system features, including online document requests, user account management, status tracking, and data clustering using the K-Means algorithm.

The **design phase** focused on creating the structure and layout of the system. Database models, system architecture, and user interface designs were prepared to ensure the system would be user-friendly, secure, and efficient. Special attention was given to designing intuitive forms, dashboards, and navigation menus so that both residents and municipal staff could interact with the system with minimal difficulty.

During the **development phase**, the system was built in small, manageable iterations. Individual modules—such as user registration, document request forms, admin approval tools, notification features, and the K-Means clustering module—were implemented and tested incrementally. This allowed the team to identify and resolve issues early, while ensuring that new features integrated seamlessly with existing ones.

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
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|  | <div> <p>The <b>testing phase</b> was conducted after each development cycle. Functional testing, usability testing, performance testing, and security testing were carried out to ensure that the system met all required standards. This phase helped identify bugs, errors, and performance issues, enabling the developers to continuously refine and improve the system.</p> <p>The <b>review and feedback phase</b> involved close coordination with the Municipal Civil Registrar’s Office. Users and staff were allowed to interact with the system and provide feedback regarding usability, accuracy, and overall performance. Their suggestions were incorporated to enhance the system’s features, interface, and workflow in subsequent development cycles.</p> <p>Each iteration concluded with testing and evaluation to identify errors, performance concerns, and areas for refinement. Feedback from the Municipal Civil Registrar’s Office played a critical role in improving the system throughout the development process.</p> <p>Overall, applying the Agile methodology enabled the team to develop a functional, flexible, and high-quality system that effectively supports the daily operations of the Municipal Civil Registrar.</p> <p><b>Data Analysis Plan</b></p> <p>To ensure accurate and reliable data presentation, analysis, and interpretation, the proponents used the arithmetic mean as a statistical tool.</p> <p>The following is the formal mathematical formula for the arithmetic mean.</p> </div> |    |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |

$$\bar{x}_A = \frac{1}{n} \sum_{i=1}^n x_i$$

Where:

A = average (or arithmetic mean)

n = the number of terms (e.g., the number of items or numbers being averaged)

Xi = the value of each individual item in the list of numbers being averaged

### **Different Types of Testing**

The following are the forms of testing that should be used:

#### **Integration Testing**

The researchers conducted integration testing to ensure that all system modules functioned together properly. This process involved verifying the smooth flow of data between the front-end interface and the back-end database, as well as the connections among the request forms, tracking features, and the K-Means clustering module.

#### **Usability Testing**

Usability testing was performed by allowing selected users, including municipal staff and residents, to interact with the system. The researchers gathered feedback on the layout, navigation, and overall user experience. Suggestions from users helped the team improve the system's design, making it more accessible, user-friendly, and efficient for processing document requests.

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|  | <b>PAMANTASAN NG LUNGSOD NG SAN PABLO</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Page</b> |
|  | <div> <div>69</div> <div> <div> <b>Performance Testing</b> </div> <div> <p>The researchers tested the system’s performance under different conditions, such as multiple users submitting requests simultaneously. They measured the system’s speed, response time, and stability to ensure that it could handle numerous transactions and records without delays or crashes, especially during peak usage periods.</p> </div> <div> <b>Security Testing</b> </div> <div> <p>Security testing was carried out to protect sensitive information from unauthorized access. The researchers tested features such as user authentication, role-based access control, and data encryption. These measures were implemented to ensure that all personal and document-related data stored in the system remained secure and confidential.</p> </div> <div> <b>Algorithm Testing</b> </div> <div> <p>The K-Means clustering algorithm was tested to evaluate its accuracy and reliability in grouping document requests. The algorithm categorizes requests based on the mode of submission, whether made online or through walk-in transactions. The results contributed to optimizing the system’s data and improving the efficiency of service delivery within the registrar’s office.</p> </div> </div> </div> |             |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |

**System Testing**

A full system test was conducted to verify that all modules and functions operated seamlessly as a complete system. This testing ensured that document request submission, approval, tracking, and clustering all worked together without errors, meeting both functional and non-functional requirements.

**Acceptance Testing**

Finally, acceptance testing was conducted with actual municipal employees and selected residents. The purpose was to confirm that the system met user expectations and functioned according to the registrar's operational needs. Feedback from this phase validated the system's readiness for deployment and guided the final adjustments for smooth implementation.

**Implementation Plan**

This plan outlines the step-by-step process for implementing the system. It helps ensure that the system is properly deployed, tested, and adopted by the office staff.

**Table 4***Organized Implementation*

| TASK                     | DURATION | COMMENTS          |
|--------------------------|----------|-------------------|
| Identify Project Manager | 1 day    | Assign leadership |

Continuation of Table 4

|                                                           |        |                                    |
|-----------------------------------------------------------|--------|------------------------------------|
| Familiarize team with system objectives                   | 3 days | Discuss the benefits of the system |
| Recognize essential stakeholders involved in the project. | 2 days | Coordinate roles                   |
| Set project meeting schedule                              | 1 day  | Weekly check-ins                   |
| Prepare implementation budget                             | 3 days | Determine the cost                 |
| Plan internal communication.                              | 1 day  | Ensure clarity of updates          |

Table 5

*System implementation*

| TASK                                                     | DURATION | COMMENTS                                            |
|----------------------------------------------------------|----------|-----------------------------------------------------|
| Identify existing manual processes for document requests | 1 week   | Analyze how document requests are currently handled |
| Map current workflows and forms used                     | 2 weeks  | For system conversion                               |
| List all data points needed for clustering               | 2 weeks  | Basis for k-means                                   |

Continuation of Table 5

|                                                |        |                                    |
|------------------------------------------------|--------|------------------------------------|
| Define new workflows                           | 1 week | Simply request flow                |
| Assess need for consultant of external trainer | 3 days | Optional, if clustering is complex |

Table 6

*System Deployment and Transition*

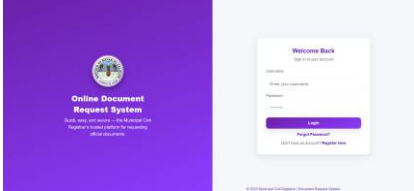
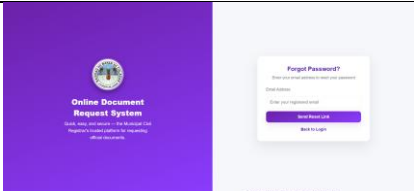
| TASK                                                | DURATION | COMMENTS                |
|-----------------------------------------------------|----------|-------------------------|
| Install and test system on-site                     | 1 week   | Resolve bugs            |
| Conduct staff training sessions                     | 1 week   | Use test environment    |
| Perform pilot test                                  | 2 weeks  | Collect feedback        |
| Adjust system based on feedback                     | 1 week   | Apply necessary changes |
| Full deployment in Municipal Civil Registrar Office | 3 days   | Launch phase            |

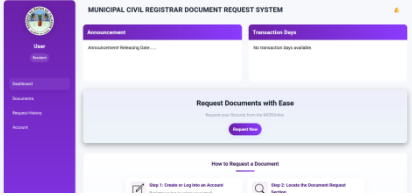
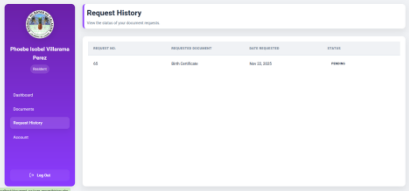
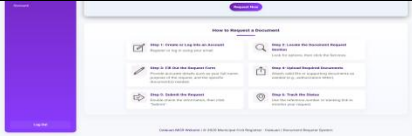
## Test Plan

This section presents the test plan for the user side of the Document Request System. It outlines the different actions that residents perform in the system and the expected results for each action. The purpose of this test plan is to ensure that all user functions, such as logging in, submitting document requests, and viewing request statuses, work correctly and provide a smooth and reliable experience for the users.

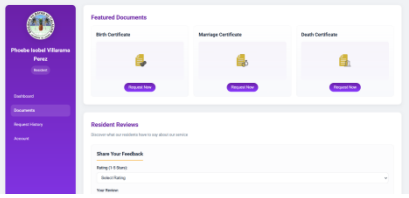
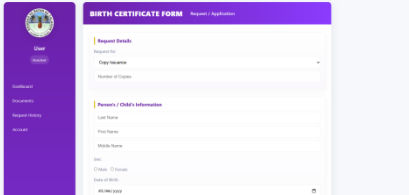
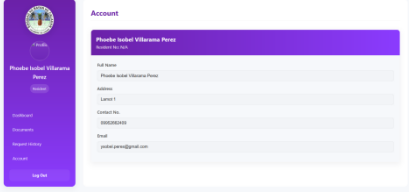
**Table 7**

### Test Plan - USER SIDE

| MODULE                                                                              | ACTION                                                                                          | EXPECTED OUTPUT                                                                                                                         | ACTUAL OUTPUT                                                                                 | REMARKS       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------|
|  | User enters valid username and password, then clicks the Login button.                          | The system successfully verifies the credentials and redirects the user to the dashboard or homepage                                    | The system successfully logs in and redirects the user dashboard                              | <b>PASSED</b> |
|  | User clicks Forgot Password, enters their registered email address, and clicks Send Reset Link. | The system validates the email and successfully sends a password reset link to the user's email account. A confirmation message appears | The system successfully sends the reset link and displays a confirmation message to the user. | <b>PASSED</b> |

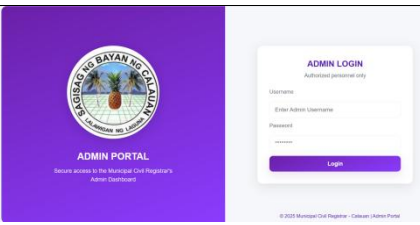
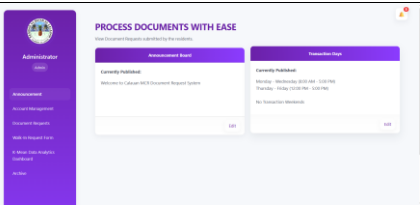
|                                                                                     |                                                   |                                                                                                                                    |                                                               |               |
|-------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------|
|                                                                                     |                                                   | (e.g., "Reset link sent").                                                                                                         |                                                               |               |
|    | After successful login, user views the dashboard. | The system displays user information, navigation side bar, and summary of recent document requests                                 | Dashboard loads properly with all components visible          | <b>PASSED</b> |
|    | User can check there Document Request History     | The system displays the list of document requests with their respective status (e.g., Pending, Approved, Released)                 | The system successfully displays the document request history | <b>PASSED</b> |
|  | User clicks Request Document on the dashboard     | The system redirects to the document selection page showing available request options (e.g., Birth, Marriage, Death Certificate s) | The system successfully displays the document request options | <b>PASSED</b> |

## Continuation of Table 7

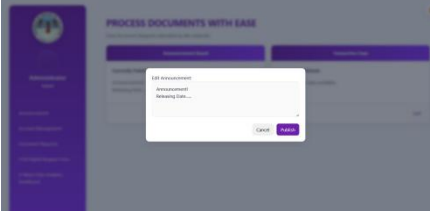
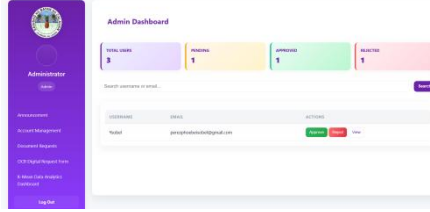
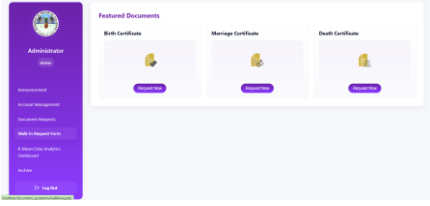
|                                                                                    |                                                                    |                                                                                         |                                                                      |               |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------|
|   | User selects e.g., Birth, Marriage, Death Certificates)            | The user will be redirected to there chosen Certificate                                 | The form is submitted successfully and confirmation message appears. | <b>PASSED</b> |
|   | and fills out the form with required information, then submits.    | The system validates inputs and confirms successful submission of the document request. | The form is submitted successfully and confirmation message appears. | <b>PASSED</b> |
|  | User opens the Account/Profile page to view or update information. | The system displays user profile details                                                | Account information is displayed                                     | <b>PASSED</b> |

## Table 8

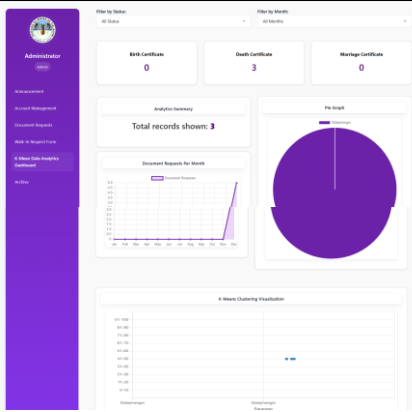
## Test Plan - ADMIN SIDE

| MODULE                                                                              | ACTION                                                    | EXPECTED OUTPUT                                                                     | ACTUAL OUTPUT                                                        | REMARKS       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------|
|  | Admin Enters valid Username and Password and click Login. | The system successfully authenticates the user and redirects to the Admin Dashboard | The system successfully logs in and redirects to the admin dashboard | <b>PASSED</b> |
|  | After successful login and admin views the dashboard      | The system displays admin dashboard.                                                | The admin dashboard is successfully displayed after login            | <b>PASSED</b> |

Continuation of Table 8

|                                                                                    |                                                                                                                                                |                                                                                                                                                   |                                                                                                                 |               |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------|
|   | Click the Edit Announcement or Transaction button and input new announcement/transaction text                                                  | The system will display the edited Announcement/Transaction on user dashboard                                                                     | The edited Announcement/Transaction is successfully displayed on the user dashboard                             | <b>PASSED</b> |
|   | Can View the main Admin Dashboard Account request status.                                                                                      | The system displays the account request status section and can approved or reject                                                                 | The account request status section is successfully displayed, and the admin can approve or reject user requests | <b>PASSED</b> |
|  | The admin manually inputs the information of a walk-in client, selects the document type, fills out all required fields, and submits the form. | The system successfully records the walk-in request, saves all entered information, and adds it to the list of document requests for tracking and | The system successfully saves the walk-in request and displays it in the Document Requests list.                | <b>PASSED</b> |

## Continuation of Table 8

|                                                                                   |                                      |                                                                                                          |                                                                                                                         |               |
|-----------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|
|  | Click the K-means and Data Analytics | The system displays the K-Means and Data Analytics section showing clustered data and analytical results | The K-Means and Data Analytics section is successfully displayed, showing accurate clustered data and analytics results | <b>PASSED</b> |
|-----------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|

## CHAPTER IV RESULTS AND DISCUSSIONS

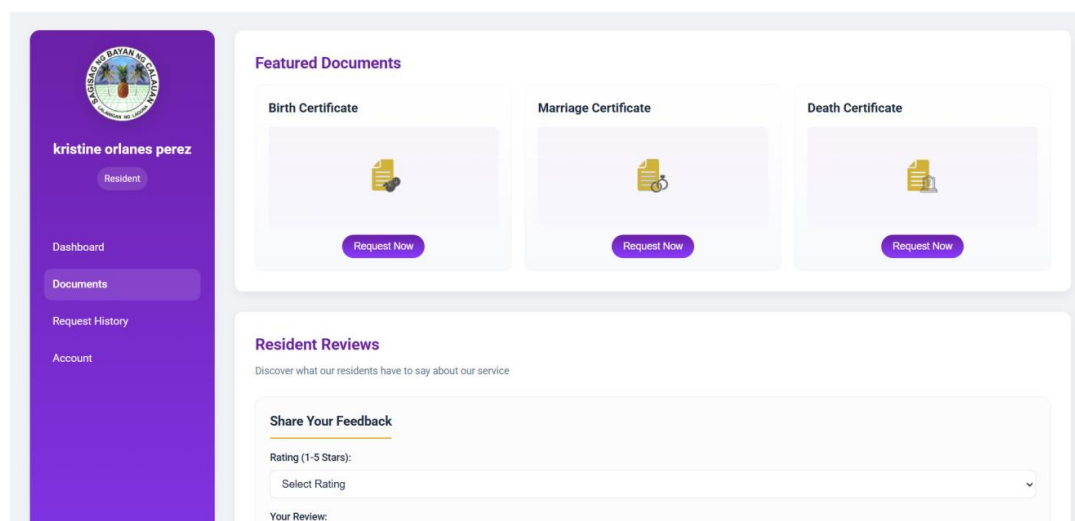
This chapter presents the results of the study, the project evaluation based on responses gathered from the participants, and the recommendations.

### Results

The primary aim of this project is to develop a Document Request System that enhances the services of the Municipal Civil Registrar by streamlining the submission and processing of requests for civil documents. Integrated with K-Means clustering and data analytics, the system provides administrators with a clearer understanding of the frequency and distribution of document requests, enabling more efficient management and planning.

### Figure 20

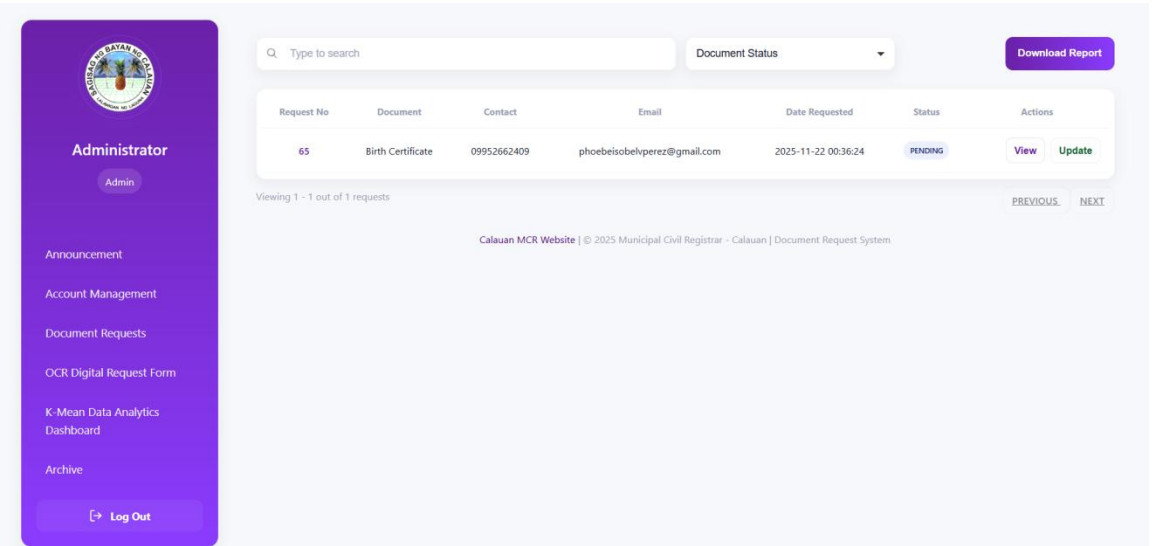
*User's Available Document Request*

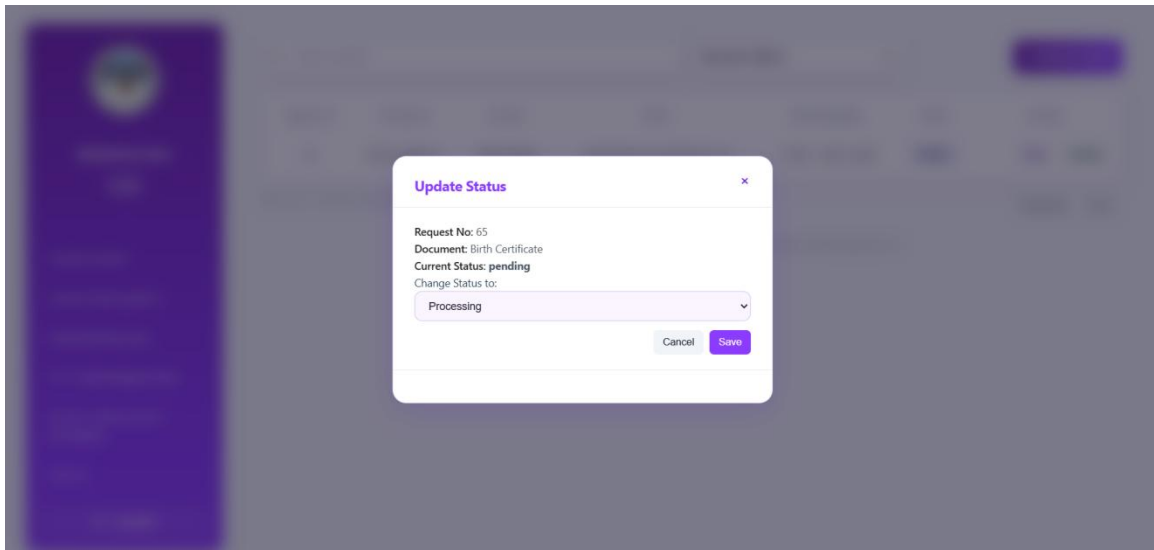


a) Provide Residents with a User-Friendly and Accessible Platform for Requesting Civil Registry Documents

The system allows residents to easily request documents online through a simple and organized interface. The “Request Now” buttons for Birth, Marriage, and Death Certificates enable quick online submission, while the clear navigation makes managing and tracking requests fast and convenient.

**Figure 21**  
*Admin Document Request Dashboard*

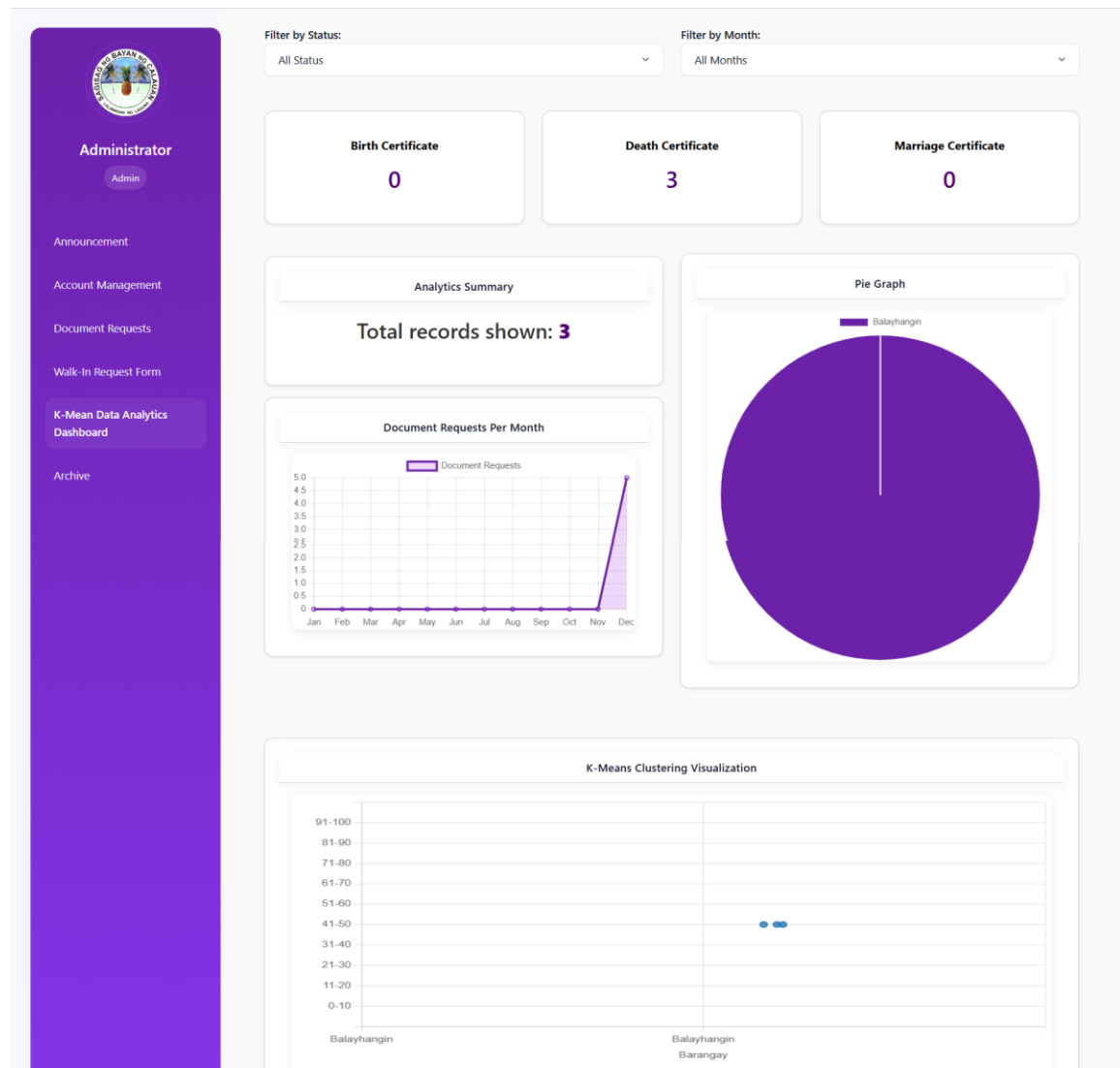




b) Provide a Digital Dashboard That Makes Staff Work Easier by Allowing Them to Quickly View, Approve, and Manage Document Requests

The digital dashboard displays real-time updates of pending, processing, and released requests. This enables staff to manage workloads efficiently, reduce processing time, and maintain organized transaction records.

Figure 22

*Admin K-means and Data Analytics*

c) Integrate Data Analytics and K-Means Clustering to Group Requests by Submission Method and Identify Request Patterns

This module enhances the system's data analytics by applying the K-Means clustering algorithm to categorize user requests based on three key attributes: age, type of requested document, and barangay. By grouping requesters with similar

characteristics, the system can identify meaningful patterns, such as which age groups frequently request specific documents and which barangays generate the highest volume of requests. These insights enable administrators to make informed decisions, improve service allocation, and optimize resources according to user demographics and location-based demand.

### **Project Evaluation**

The Development of Document Request System with K-Means Clustering for the Municipal Civil Registrar of the Municipality of Calauan was evaluated using the ISO 25010 evaluation instrument. The evaluation involved three (3) types of respondents: ten (10) IT practitioners, one (1) client, and five (5) residents/user. The assessment was conducted using a questionnaire structured on a numerical 5-point Likert scale. Mean values were computed from the responses of these groups and interpreted according to the defined numerical ranges.

#### **IT Practitioners**

The following are the results gathered from the ten (10) IT Practitioners conducted by the developers

**Table 9**

*IT Practitioners' Evaluation Based on Functional Suitability*

| <b>Functional Stability</b> | <b>Average Mean</b> | <b>Interpretation</b> |
|-----------------------------|---------------------|-----------------------|
| <b>Functionality</b>        | 4.30                | Very Acceptable       |

Continuation of Table 9

|                        |             |                        |
|------------------------|-------------|------------------------|
| <b>Correctness</b>     | 4.50        | Highly Acceptable      |
| <b>Appropriateness</b> | 4.50        | Highly Acceptable      |
| <b>Mean</b>            | <b>4.40</b> | <b>Very Acceptable</b> |

Table 9 shows the evaluation of the ten (10) IT Practitioners based on Functional Stability. The overall mean is 4.4, which is **“Very Acceptable”**. Functionality received a mean of 4.3, while Correctness and Appropriateness both received 4.5, also interpreted as **“Very Acceptable”**.

**Table 10**

*IT Practitioners' Evaluation Based on Reliability*

| <b>Reliability</b>     | <b>Average Mean</b> | <b>Interpretation</b> |
|------------------------|---------------------|-----------------------|
| <b>Maturity</b>        | 4.20                | Very Acceptable       |
| <b>Availability</b>    | 4.20                | Very Acceptable       |
| <b>Fault Tolerance</b> | 4.00                | Very Acceptable       |
| <b>Recoverability</b>  | 4.00                | Very Acceptable       |
| <b>Adaptability</b>    | 4.00                | Very Acceptable       |

Continuation of Table 10

|      |      |                 |
|------|------|-----------------|
| Mean | 4.10 | Very Acceptable |
|------|------|-----------------|

Table 10 shows the evaluation of the ten (10) IT Practitioners based on Reliability. The overall mean is 4.1, which is interpreted as **“Very Acceptable.”** Maturity and Availability both received a mean of 4.2, while Fault Tolerance, Recoverability, and Adaptability each received 4.0, all of which are also rated as **“Very Acceptable.”**

Table 11

*IT Practitioners' Evaluation Based on Portability*

| Portability    | Average Mean | Interpretation  |
|----------------|--------------|-----------------|
| Installability | 4.20         | Very Acceptable |
| Replaceability | 4.00         | Very Acceptable |
| Mean           | 4.10         | Very Acceptable |

Table 11 shows the evaluation of the ten (10) IT Practitioners based on Portability. The overall mean is 4.1, which is interpreted as **“Very Acceptable.”** Installability received a mean of 4.2, while Replaceability received 4.0, both also rated as **“Very Acceptable.”**

Table 12

*IT Practitioners' Evaluation Based on Usability*

| Usability                       | Average Mean | Interpretation    |
|---------------------------------|--------------|-------------------|
| Appropriateness Recognizability | 4.50         | Highly Acceptable |
| Learnability                    | 4.50         | Highly Acceptable |
| Operability                     | 4.30         | Very Acceptable   |
| User Error Protection           | 4.10         | Very Acceptable   |
| User Interface Aesthetics       | 4.20         | Very Acceptable   |
| Accessibility                   | 4.20         | Very Acceptable   |
| Mean                            | 4.30         | Very Acceptable   |

Table 12 shows the evaluation of the ten (10) IT practitioners based on Usability. The overall mean score is 4.3, interpreted as “**Very Acceptable.**” Appropriateness Recognizability and Learnability both received the highest mean score of 4.5, interpreted as “**Highly Acceptable.**” Operability scored 4.3, while User Error Protection and User Interface Aesthetics received mean scores of 4.1, and Accessibility scored 4.2, all interpreted as “**Very Acceptable.**”

**Table 13***IT Practitioners' Evaluation Based on Performance Efficiency*

| Performance Efficiency | Average Mean | Interpretation         |
|------------------------|--------------|------------------------|
| Time-Behavior          | 4.10         | Very Acceptable        |
| Resource Utilization   | 4.30         | Very Acceptable        |
| Capacity               | 4.30         | Very Acceptable        |
| Mean                   | 4.20         | <b>Very Acceptable</b> |

Table 13 shows the evaluation of the ten (10) IT Practitioners based on Performance Efficiency. The overall mean is 4.2, which is interpreted as “**Very Acceptable.**” Time-Behavior received a mean of 4.1, while Resource Utilization and Capacity both scored 4.3, all of which are also rated as “**Very Acceptable.**”

**Table 14***IT Practitioners' Evaluation Based on Security*

| Security        | Average Mean | Interpretation  |
|-----------------|--------------|-----------------|
| Confidentiality | 4.30         | Very Acceptable |

Continuation of Table 14

|                        |             |                        |
|------------------------|-------------|------------------------|
| <b>Integrity</b>       | 4.20        | Very Acceptable        |
| <b>Non-Repudiation</b> | 4.20        | Very Acceptable        |
| <b>Accountability</b>  | 4.00        | Very Acceptable        |
| <b>Authenticity</b>    | 4.10        | Very Acceptable        |
| <b>Mean</b>            | <b>4.20</b> | <b>Very Acceptable</b> |

Table 14 shows the evaluation of the ten (10) IT Practitioners based on Security. The overall mean is 4.2, which is interpreted as **“Very Acceptable.”** Confidentiality received a mean of 4.3, while Integrity and Non-Repudiation both scored 4.2. Accountability obtained a mean of 4.0, and Authenticity received 4.1, all of which are also rated as **“Very Acceptable.”**

**Table 15**

*IT Practitioners' Evaluation Based on Compatibility*

| <b>Compatibility</b> | <b>Average Mean</b> | <b>Interpretation</b> |
|----------------------|---------------------|-----------------------|
| <b>Co-Existence</b>  | 4.20                | Very Acceptable       |

Continuation of Table 15

|                  |      |                 |
|------------------|------|-----------------|
| Interoperability | 4.10 | Very Acceptable |
| Mean             | 4.20 | Very Acceptable |

Table 15 shows the evaluation of the ten (10) IT Practitioners based on Compatibility. The overall mean is 4.2, which is interpreted as “**Very Acceptable.**” Co-Existence received a mean of 4.2, while Interoperability scored 4.1, both of which are also rated as “**Very Acceptable.**”

Table 16

*IT Practitioners' Evaluation Based on Maintainability*

| Maintainability | Average Mean | Interpretation  |
|-----------------|--------------|-----------------|
| Modularity      | 4.10         | Very Acceptable |
| Reusability     | 4.30         | Very Acceptable |
| Analyzability   | 4.00         | Very Acceptable |
| Modifiability   | 4.00         | Very Acceptable |

Continuation of Table 16

|             |      |                 |
|-------------|------|-----------------|
| Testability | 4.10 | Very Acceptable |
| Mean        | 4.10 | Very Acceptable |

Table 16 shows the evaluation of the ten (10) IT Practitioners based on Maintainability. The overall mean is 4.1, which is interpreted as “**Very Acceptable.**” Modularity and Testability both received a mean of 4.1, while Reusability obtained the highest mean of 4.3. Analyzability and Modifiability each scored 4.0, and all indicators are rated as “**Very Acceptable.**”

Table 17

*Summary of IT Practitioners' Evaluation*

| Software Quality Criteria | Average Mean | Interpretation  |
|---------------------------|--------------|-----------------|
| Functional Stability      | 4.40         | Very Acceptable |
| Reliability               | 4.10         | Very Acceptable |
| Portability               | 4.10         | Very Acceptable |
| Usability                 | 4.30         | Very Acceptable |

Continuation of Table 17

|                        |      |                 |
|------------------------|------|-----------------|
| Performance Efficiency | 4.20 | Very Acceptable |
| Security               | 4.20 | Very Acceptable |
| Compatability          | 4.20 | Very Acceptable |
| Maintability           | 4.10 | Very Acceptable |
| Mean                   | 4.20 | Very Acceptable |

Table 17 shows the overall evaluation of the system based on the Software Quality Criteria. The overall mean is 4.2, which is interpreted as “**Very Acceptable.**” Functional Stability received the highest mean of 4.4, while Reliability, Portability, and Maintainability each scored 4.1. Usability scored 4.3, and both Performance Efficiency and Security received 4.2. Compatibility also obtained a mean of 4.2. All criteria were rated as “**Very Acceptable**”.

**Municipal Civil Registrar**

The following are the results gathered from the one (1) Client conducted by the developers.

**Table 18**

*Client's Evaluation Based on Functional Suitability*

| <b>Functional Stability</b> | <b>Average Mean</b> | <b>Interpretation</b>    |
|-----------------------------|---------------------|--------------------------|
| <b>Functionality</b>        | 5.00                | Highly Acceptable        |
| <b>Correctness</b>          | 5.00                | Highly Acceptable        |
| <b>Appropriateness</b>      | 5.00                | Highly Acceptable        |
| <b>Mean</b>                 | <b>5.00</b>         | <b>Highly Acceptable</b> |

Table 18 shows the clients' evaluation based on Functional Suitability. The overall mean is 5.0, which is interpreted as "**Highly Acceptable.**" All three criteria Functionality, Correctness, and Appropriateness received perfect mean scores of 5.0, also interpreted as "**Highly Acceptable.**"

Table 19

*Client's Evaluation Based on Reliability*

| Reliability     | Average Mean | Interpretation    |
|-----------------|--------------|-------------------|
| Maturity        | 5.00         | Highly Acceptable |
| Availability    | 5.00         | Highly Acceptable |
| Fault Tolerance | 4.00         | Very Acceptable   |
| Recoverability  | 4.00         | Very Acceptable   |
| Adaptability    | 4.00         | Very Acceptable   |
| Mean            | 4.40         | Very Acceptable   |

Table 19 shows the clients' evaluation based on Reliability. The overall mean is 4.4, which is interpreted as **“Very Acceptable.”** Maturity and Availability both received perfect mean scores of 5.0, interpreted as **“Highly Acceptable.”** Meanwhile, Fault Tolerance, Recoverability, and Adaptability each received a mean of 4.0, all rated as **“Very Acceptable.”**

**Table 20***Client's Evaluation Based on Portability*

| Portability    | Average Mean | Interpretation  |
|----------------|--------------|-----------------|
| Installability | 4.00         | Very Acceptable |
| Replaceability | 4.00         | Very Acceptable |
| Mean           | 4.00         | Very Acceptable |

Table 20 shows the clients' evaluation based on Portability. The overall mean is 4.0, which is interpreted as **“Very Acceptable.”** Both Installability and Replaceability received a mean score of 4.0, also rated as **“Very Acceptable.”**

**Table 21***Client's Evaluation Based on Usability*

| Usability                          | Average Mean | Interpretation       |
|------------------------------------|--------------|----------------------|
| Appropriateness<br>Recognizability | 5.00         | Highly<br>Acceptable |
| Learnability                       | 5.00         | Highly<br>Acceptable |
| Operability                        | 5.00         | Highly<br>Acceptable |
| User Error Protection              | 4.00         | Very<br>Acceptable   |

Continuation of Table 21

|                           |      |                   |
|---------------------------|------|-------------------|
| User Interface Aesthetics | 5.00 | Highly Acceptable |
| Accessibility             | 5.00 | Highly Acceptable |
| Mean                      | 4.80 | Highly Acceptable |

Table 21 shows the clients' evaluation based on Usability. The overall mean is 4.8, which is interpreted as **“Highly Acceptable.”** Appropriateness, Recognizability, Learnability, Operability, User Interface Aesthetics, and Accessibility all received perfect mean scores of 5.0, interpreted as **“Highly Acceptable.”** User Error Protection received a mean of 4.0, which is rated as **“Very Acceptable.”**

Table 22

*Client's Evaluation Based on Performance Efficiency*

| Performance Efficiency | Average Mean | Interpretation    |
|------------------------|--------------|-------------------|
| Time-Behavior          | 5.00         | Highly Acceptable |
| Resource Utilization   | 5.00         | Highly Acceptable |
| Capacity               | 4.00         | Very Acceptable   |
| Mean                   | 4.70         | Highly Acceptable |

Table 22 shows the clients' evaluation based on Performance Efficiency. The overall mean is 4.7, which is interpreted as **"Highly Acceptable."** Time-Behavior and Resource Utilization both received perfect mean scores of 5.0, indicating **"Highly Acceptable."** Capacity received a mean of 4.0, which is interpreted as **"Very Acceptable."**

**Table 23**

*Client's Evaluation Based on Security*

| Security        | Average Mean | Interpretation    |
|-----------------|--------------|-------------------|
| Confidentiality | 5.00         | Highly Acceptable |
| Integrity       | 5.00         | Highly Acceptable |
| Non-Repudiation | 4.00         | Very Acceptable   |
| Accountability  | 4.00         | Very Acceptable   |
| Authenticity    | 5.00         | Highly Acceptable |
| Mean            | 4.60         | Highly Acceptable |

Table 23 shows the clients' evaluation based on Security. The overall mean is 4.6, which is interpreted as **"Highly Acceptable."** Confidentiality, Integrity, and Authenticity all received perfect mean scores of 5.0, indicating that clients find the system highly secure in protecting data and ensuring authenticity. Non-

Repudiation and Accountability both received a mean of 4.0, interpreted as “**Very Acceptable.**”

**Table 24**

*Client’s Evaluation Based on Compatibility*

| Compatibility    | Average Mean | Interpretation    |
|------------------|--------------|-------------------|
| Co-Existence     | 5.00         | Highly Acceptable |
| Interoperability | 5.00         | Highly Acceptable |
| Mean             | 5.00         | Highly Acceptable |

Table 24 shows the clients’ evaluation based on Compatibility. The overall mean is 5.0, which is interpreted as “**Highly Acceptable.**” Both Co-Existence and Interoperability received perfect mean scores of 5.0, also rated as “**Highly Acceptable.**”

**Table 25**

*Client’s Evaluation Based on Maintainability*

| Maintainability | Average Mean | Interpretation    |
|-----------------|--------------|-------------------|
| Modularity      | 5.00         | Highly Acceptable |
| Reusability     | 5.00         | Highly Acceptable |

Continuation of Table 25

|                      |             |                          |
|----------------------|-------------|--------------------------|
| <b>Analyzability</b> | 5.00        | Highly Acceptable        |
| <b>Modifiability</b> | 4.00        | Very Acceptable          |
| <b>Testability</b>   | 5.00        | Highly Acceptable        |
| <b>Mean</b>          | <b>4.80</b> | <b>Highly Acceptable</b> |

Table 25 shows the clients' evaluation based on Maintainability. The overall mean is 4.8, which is interpreted as **“Highly Acceptable.”** Modularity, Reusability, Analyzability, and Testability all received perfect mean scores of 5.0, indicating **“Highly Acceptable.”** Modifiability received a mean of 4.0, which is interpreted as **“Very Acceptable.”**

Table 26

*Summary of Client's Evaluation*

| <b>Software Quality Criteria</b> | <b>Average Mean</b> | <b>Interpretation</b> |
|----------------------------------|---------------------|-----------------------|
| <b>Functional Stability</b>      | 5.00                | Very Acceptable       |
| <b>Reliability</b>               | 4.40                | Very Acceptable       |

Continuation of Table 26

|                               |             |                          |
|-------------------------------|-------------|--------------------------|
| <b>Portability</b>            | 4.00        | Very Acceptable          |
| <b>Usability</b>              | 4.80        | Very Acceptable          |
| <b>Performance Efficiency</b> | 4.60        | Very Acceptable          |
| <b>Security</b>               | 4.60        | Very Acceptable          |
| <b>Compatibility</b>          | 5.00        | Very Acceptable          |
| <b>Maintainability</b>        | 4.80        | Very Acceptable          |
| <b>Mean</b>                   | <b>4.70</b> | <b>Highly Acceptable</b> |

Table 26 presents the clients' overall evaluation based on the Software Quality Criteria. The overall mean is 4.7, which is interpreted as **"Highly Acceptable."** Functional Stability received a perfect score of 5.0, while Reliability scored 4.4 and Portability scored 4.0, all interpreted as **"Very Acceptable."** Usability and Maintainability both obtained 4.8, and Performance Efficiency and Security each

scored 4.6, all rated as “**Very Acceptable.**” Compatibility also received a mean of 5.0, interpreted as “**Very Acceptable.**”

### Residents

The following are the results gathered from the five (5) Residents conducted by the developers.

**Table 27**

*Residents’ Evaluation Based on Functional Suitability*

| Functional Stability | Average Mean | Interpretation    |
|----------------------|--------------|-------------------|
| Functionality        | 5.00         | Highly Acceptable |
| Correctness          | 5.00         | Highly Acceptable |
| Appropriateness      | 5.00         | Highly Acceptable |
| Mean                 | 5.00         | Highly Acceptable |

Table 27 shows the residents’ evaluation based on Functional Suitability. The overall mean is 5.0, which is interpreted as “**Highly Acceptable.**” All three criteria Functionality, Correctness, and Appropriateness received perfect mean scores of 5.0, also interpreted as “**Highly Acceptable.**”

Table 28

*Residents' Evaluation Based on Reliability*

| Reliability     | Average Mean | Interpretation    |
|-----------------|--------------|-------------------|
| Maturity        | 5.00         | Highly Acceptable |
| Availability    | 5.00         | Highly Acceptable |
| Fault Tolerance | 4.40         | Very Acceptable   |
| Recoverability  | 4.20         | Very Acceptable   |
| Adaptability    | 4.60         | Very Acceptable   |
| Mean            | 4.64         | Very Acceptable   |

Table 28 shows the residents' evaluation based on Reliability. The overall mean is 4.64, which is interpreted as **“Very Acceptable.”** Maturity and Availability both received perfect scores of 5.0, interpreted as **“Highly Acceptable.”** Fault Tolerance received a mean of 4.4, Recoverability scored 4.2, and Adaptability obtained 4.6, all of which are interpreted as **“Very Acceptable.”**

Table 29

*Residents' Evaluation Based on Portability*

| Portability    | Average Mean | Interpretation  |
|----------------|--------------|-----------------|
| Installability | 4.20         | Very Acceptable |
| Replaceability | 4.00         | Very Acceptable |
| Mean           | 4.10         | Very Acceptable |

Table 29 shows the residents' evaluation based on Portability. The overall mean is 4.1, which is interpreted as “**Very Acceptable.**” Installability received a mean score of 4.2, while Replaceability scored 4.0, both interpreted as “**Very Acceptable.**”

Table 30

*Residents' Evaluation Based on Usability*

| Usability                          | Average Mean | Interpretation       |
|------------------------------------|--------------|----------------------|
| Appropriateness<br>Recognizability | 5.00         | Highly<br>Acceptable |
| Learnability                       | 5.00         | Highly<br>Acceptable |
| Operability                        | 5.00         | Highly<br>Acceptable |
| User Error Protection              | 5.00         | Highly<br>Acceptable |

Continuation of Table 30

|                           |      |                   |
|---------------------------|------|-------------------|
| User Interface Aesthetics | 5.00 | Highly Acceptable |
| Accessibility             | 5.00 | Highly Acceptable |
| Mean                      | 5.00 | Highly Acceptable |

Table 30 shows the residents' evaluation based on Usability. The overall mean is 5.0, which is interpreted as **“Highly Acceptable.”** All usability indicators Appropriateness Recognizability, Learnability, Operability, and User Error Protection received perfect mean scores of 5.0, also interpreted as **“Highly Acceptable.”**

Table 31

*Residents' Evaluation Based on Performance Efficiency*

| Performance Efficiency | Average Mean | Interpretation    |
|------------------------|--------------|-------------------|
| Time-Behavior          | 4.60         | Highly Acceptable |
| Resource Utilization   | 5.00         | Highly Acceptable |
| Capacity               | 5.00         | Highly Acceptable |
| Mean                   | 4.87         | Highly Acceptable |

Table 31 shows the residents' evaluation based on Performance Efficiency. The overall mean is 4.87, which is interpreted as **“Highly Acceptable.”** Time-Behaviour received a mean of 4.6, while Resource Utilization and Capacity both received perfect scores of 5.0, also interpreted as **“Highly Acceptable.”**

**Table 32**

*Residents' Evaluation Based on Security*

| Security        | Average Mean | Interpretation    |
|-----------------|--------------|-------------------|
| Confidentiality | 5.00         | Highly Acceptable |
| Integrity       | 4.60         | Highly Acceptable |
| Non-Repudiation | 4.80         | Highly Acceptable |
| Accountability  | 4.60         | Highly Acceptable |
| Authenticity    | 4.60         | Highly Acceptable |
| Mean            | 4.72         | Highly Acceptable |

Table 32 shows the residents' evaluation based on Security. The overall mean is 4.72, which is interpreted as **“Highly Acceptable.”** Confidentiality received a perfect score of 5.0, while Integrity, Accountability, and Authenticity each scored 4.6. Non-Repudiation obtained 4.8, and all criteria are interpreted as **“Highly Acceptable.”**

**Table 33***Residents' Evaluation Based on Compatibility*

| Compatibility    | Average Mean | Interpretation    |
|------------------|--------------|-------------------|
| Co-Existence     | 5.00         | Highly Acceptable |
| Interoperability | 5.00         | Highly Acceptable |
| Mean             | 5.00         | Highly Acceptable |

Table 33 shows the residents' evaluation based on Compatibility. The overall mean is **5.0**, which is interpreted as Highly **Acceptable.**" Co-Existence received a perfect score of **5.0**, also rated Highly **Acceptable.**"

**Table 34***Residents' Evaluation Based on Maintainability*

| Maintainability | Average Mean | Interpretation    |
|-----------------|--------------|-------------------|
| Modularity      | 4.60         | Highly Acceptable |
| Reusability     | 4.60         | Highly Acceptable |

Continuation of Table 34

|                      |             |                          |
|----------------------|-------------|--------------------------|
| <b>Analyzability</b> | 4.80        | Highly Acceptable        |
| <b>Modifiability</b> | 5.00        | Very Acceptable          |
| <b>Testability</b>   | 5.00        | Highly Acceptable        |
| <b>Mean</b>          | <b>5.00</b> | <b>Highly Acceptable</b> |

Table 34 shows the residents' evaluation based on Maintainability. The overall mean score is 5.0, interpreted as **"Highly Acceptable."** Modularity and Reusability both received a mean score of 4.6, while Analyzability scored 4.8, all interpreted as **"Highly Acceptable."** Modifiability and Testability both received perfect scores of 5.0, also rated as **"Highly Acceptable."**

**Table 35**

*Summary of Residents' Evaluation*

| <b>Software Quality Criteria</b> | <b>Average Mean</b> | <b>Interpretation</b> |
|----------------------------------|---------------------|-----------------------|
| <b>Functional Stability</b>      | 5.00                | Highly Acceptable     |
| <b>Reliability</b>               | 4.64                | Highly Acceptable     |

Continuation of Table 35

|                               |             |                          |
|-------------------------------|-------------|--------------------------|
| <b>Portability</b>            | 4.10        | Very Acceptable          |
| <b>Usability</b>              | 5.00        | Highly Acceptable        |
| <b>Performance Efficiency</b> | 4.87        | Highly Acceptable        |
| <b>Security</b>               | 4.72        | Highly Acceptable        |
| <b>Compatibility</b>          | 5.00        | Highly Acceptable        |
| <b>Maintainability</b>        | 4.80        | Highly Acceptable        |
| <b>Mean</b>                   | <b>4.80</b> | <b>Highly Acceptable</b> |

Table 35 presents a summary of residents' evaluations across all software quality criteria. The overall mean score is 4.8, interpreted as “**Highly Acceptable.**” Functional Stability, Usability, Compatibility, and Maintainability all received high

ratings, while Reliability, Performance Efficiency, and Security were also rated as “**Highly Acceptable.**” Portability received a mean score of 4.1, interpreted as “**Very Acceptable.**” Overall, residents perceive the system as highly functional, secure, efficient, and easy to use and maintain.

**Table 36**

*Overall Evaluation Results*

| Software Quality Criteria | IT Practitioners | Client | Resident | Average Mean | Interpretation    |
|---------------------------|------------------|--------|----------|--------------|-------------------|
| Functional Suitability    | 4.40             | 5.00   | 5.00     | 4.8          | Highly Acceptable |
| Reliability               | 4.10             | 4.40   | 4.64     | 4.38         | Very Acceptable   |
| Portability               | 4.10             | 4.00   | 4.10     | 4.0          | Very Acceptable   |
| Usability                 | 4.30             | 4.80   | 5.00     | 4.70         | Highly Acceptable |
| Performance Efficiency    | 4.20             | 4.60   | 4.87     | 4.55         | Highly Acceptable |
| Security                  | 4.20             | 4.60   | 4.72     | 4.50         | Very Acceptable   |
| Compatibility             | 4.20             | 5.00   | 5.00     | 4.73         | Highly Acceptable |

Continuation of Table 36

|                 |      |      |      |      |                   |
|-----------------|------|------|------|------|-------------------|
| Maintainability | 4.10 | 4.80 | 4.80 | 4.56 | Highly Acceptable |
| Mean            | 4.20 | 4.70 | 4.80 | 4.53 | Highly Acceptable |

Table 36 presents the combined evaluation of ten (10) IT practitioners, one (1) client, and five (5) residents across all software quality criteria. The overall mean score is 4.53, which is interpreted as **“Highly Acceptable.”** Functional Suitability, Usability, Performance Efficiency, Compatibility, and Maintainability all received high ratings, indicating strong system performance in these areas. Meanwhile, Reliability, Portability, and Security were rated as **“Very Acceptable.”** Overall, the results demonstrate that the system is well-received by all user groups and performs effectively across the major software quality standards.

**Recommendations**

The following recommendations are proposed for further improvement. These suggestions aim to enhance user convenience, streamline processes, and ensure faster and more reliable service delivery for both residents and system administrators.

**a) Allow Requests from Non-Residents or External Addresses**

Enhance the system to allow former residents or individuals living outside Calauan to request documents remotely, eliminating the need for in-person visits.

**b) Enable SMS Notifications**

Configure the system to automatically notify users via SMS about important updates, such as pickup schedules and document releases.

**c) Implement OCR for Walk-In Clients (Auto-Fill Feature)**

Incorporate an OCR feature that enables the system to scan documents submitted by walk-in clients. The system will automatically detect and populate key information, such as name, address, and date of birth, thereby reducing manual data entry, minimizing errors, and improving processing efficiency.

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|  | <b>PAMANTASAN NG LUNGSOD NG SAN PABLO</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Page</b> |
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|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |

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|  | <b>PAMANTASAN NG LUNGSOD NG SAN PABLO</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Page</b> |
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|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |

**APPENDICES**

## APPENDIX:

## A. Letter for an Interview and Data Gathering to the Client

March 17, 2025

HON. ROSSELER G. CARATIHAN  
Municipal Mayor  
Thru: MR. DARIO C. ESCAREZ  
Municipal Registrar

OFFICE OF THE MUNICIPAL MAYOR  
Municipality of Calauan  
Calauan, Laguna

Dear Sir,

Greetings!

We, third year BSIT students from Pamantasan ng Lungsod ng San Pablo is conducting a proposed research entitled: **"Development of Document Request System for Municipal Civil Registrar of Municipality of Calauan"** In line with this, we kindly request your good office to allow us to conduct an interview related to our proposed research.

Your responses are highly valuable and will significantly contribute to our study. Please be assured that all answers will remain strictly confidential and used solely for academic purposes. We greatly appreciate your participation. Thank you for your time and support.

Sincerely yours,

*P. Isobel*  
PHOEBE ISOBEL V. PEREZ  
VINCE KYLE C. CALAYAG  
KRISTINE O. PEREZ  
LESTER YURIE J. SALCEDO  
ADONIS A. BALAQUE  
RONNEL GERVACIO  
Researchers

Noted:

*[Signature]*  
SHERWIN D. QUIZON, MSIT  
Capstone Project Adviser

*B. Letter of the Developers to the IT Practitioners*

**PAMANTASAN NG LUNGSOD NG SAN PABLO**  
**College of Computing Sciences and Engineering**  
Brgy. San Jose, San Pablo City  
Tel No.: (049) 536-7830  
Email Address: [plspofficial@plsp.edu.ph](mailto:plspofficial@plsp.edu.ph)



November 18, 2025

**IT Practitioners**

Good day!

We, Fourth-year BSIT students from Pamantasan ng Lungsod ng San Pablo, currently in the evaluation phase of our capstone project entitled “**Development of Document Request System with K-Means Clustering for the Municipal Civil Registrar of the Municipality of Calauan.**” This is part of our **Chapter 4: Project Evaluation** requirement.

In this, we would like to request your assistance in evaluating our system. Your expertise in Information Technology will be valuable in assessing its performance, features, and usability.

With your permission, we would like to provide a short demonstration of our website, along with an evaluation form based on the **ISO 25010 software quality model**, at your most convenient time.

Thank you very much for your time, support, and guidance.

Sincerely,

*P. Isobel*  
**PHOEBE ISOBEL V. PEREZ**  
Project Manager

Noted by:

*Sherwin Quizon*  
**MR. SHERWIN QUIZON**  
Capstone Project Adviser

*C. Letter of the Developers to the Client*

**PAMANTASAN NG LUNGSOD NG SAN PABLO**  
*independencia sa pag-unlad sa kalinangan*  
College of Computing Sciences and Engineering  
Brgy. San Jose, San Pablo City  
Tel No.: (049) 536-7830  
Email Address: [plspofficial@plsp.edu.ph](mailto:plspofficial@plsp.edu.ph)



November 18, 2025

**HON. ROSSELER G. CARATIHAN**  
Municipal Mayor  
Municipality of Calauan, Laguna

**Thru: MR. DARIO C. ESCAREZ**  
Municipal Registrar

Dear Sir,

Good day!

We, Fourth-year BSIT students from **Pamantasan ng Lungsod ng San Pablo**, currently in the evaluation phase of our capstone project entitled **"Development of Document Request System with K-Means Clustering for the Municipal Civil Registrar of the Municipality of Calauan."** This activity is part of our **Chapter 4: Project Evaluation** requirement.

In this, we would like to request your assistance in evaluating our system. Your expertise in Information Technology will be valuable in assessing its performance, features, and usability.

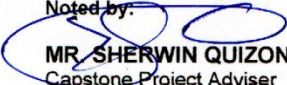
With your permission, we would like to provide a short demonstration of our website, along with an evaluation form based on the **ISO 25010 software quality model**, at your most convenient time.

Thank you very much for your time, support, and guidance.

Sincerely,

  
**PHOEBE ISOBEL V. PEREZ**  
Project Manager

Noted by:

  
**MR. SHERWIN QUIZON**  
Capstone Project Adviser

*D. Pictures Taken During Evaluation**Actual Evaluation and Testing of the Developed Document Request System*



*Actual Evaluation and Testing of the Developed Document Request System at the Municipal Civil Registrar of Calauan.*



## E. Sample Software Quality Evaluation Form



## PAMANTASAN NG LUNGSOD NG SAN PABLO

College of Computing Sciences and Engineering

Brgy. San Jose, San Pablo City

Tel No.: (049) 536-7838

Email Address: [plgoffice@plgs.edu.ph](mailto:plgoffice@plgs.edu.ph)



## IT Practitioner

Name: \_\_\_\_\_

Date: \_\_\_\_/\_\_\_\_/\_\_\_\_

Work: \_\_\_\_\_

Company: \_\_\_\_\_

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                         | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives.                                                            |   |   |   |   |   |
|                        | CORRECTNESS           | The function provides the correct results with the needed degree of precision.                                                     |   |   |   |   |   |
|                        | APPROPRIATENESS       | The function facilitates the accomplishment of specified tasks and objectives.                                                     |   |   |   |   |   |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation.                                                       |   |   |   |   |   |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use.                                                           |   |   |   |   |   |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of _____.                                                 |   |   |   |   |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data.                                            |   |   |   |   |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware.                                 |   |   |   |   |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment.                                   |   |   |   |   |   |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose.                                                     |   |   |   |   |   |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs.                                                  |   |   |   |   |   |
|                        | LEARNABILITY          | A product or system enables the user to learn how to use it with effectiveness.                                                    |   |   |   |   |   |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use.                                                            |   |   |   |   |   |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors.                                                                           |   |   |   |   |   |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user.                                                        |   |   |   |   |   |
|                        | AESTHETICS            | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals.    |   |   |   |   |   |
| PERFORMANCE EFFICIENCY | TIME-BEHAVIOR         | The response and processing times and throughput rates of product or system.                                                       |   |   |   |   |   |
|                        | RESOURCE UTILIZATION  | The amounts and types of resources used by a product or system, when performing its functions, meet requirements.                  |   |   |   |   |   |
|                        | CAPACITY              | The maximum limits of the product or system, parameter meet requirements.                                                          |   |   |   |   |   |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have.                                                    |   |   |   |   |   |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data.                     |   |   |   |   |   |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events or actions cannot be repudiated later.                     |   |   |   |   |   |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity.                                                                     |   |   |   |   |   |
|                        | AUTHENTICITY          | The identity of a subject or resources can be proved to be the one claimed.                                                        |   |   |   |   |   |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products.     |   |   |   |   |   |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information and use the information that has been exchanged.              |   |   |   |   |   |
| MAINTAINABILITY        | MODULARITY            | A system or computer program is composed of discrete components such that _____.                                                   |   |   |   |   |   |
|                        | REUSABILITY           | An asset can be used in more than one system.                                                                                      |   |   |   |   |   |
|                        | ANALYZABILITY         | It is possible to assess the impact on a product or system of an intended change.                                                  |   |   |   |   |   |
|                        | MODIFIABILITY         | A product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality. |   |   |   |   |   |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components.                                                              |   |   |   |   |   |

## F. Filled-up Software Quality Evaluation Instrument



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Professionalism

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Tel No.: (049) 536-7830

Email Address: [plsofficial@plsp.edu.ph](mailto:plsofficial@plsp.edu.ph)



## Client

Name: Dario C. Escarce  
Position: Municipal Civil Registrar I HEAD

Date: 11 / 19 / 25

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC         | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY             | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   |   |
|                        | CORRECTNESS               | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   |   |
|                        | APPROPRIATENESS           | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   |   |
| RELIABILITY            | MATURITY                  | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   |   |
|                        | AVAILABILITY              | A product or system is operational and accessible when required for use                                                        |   |   |   |   |   |
|                        | FAULT TOLERANCE           | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   |   |
|                        | RECOVERABILITY            | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   |   |
|                        | ADAPTABILITY              | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   |   |
| PORTABILITY            | INSTALLABILITY            | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   |   |
|                        | REPLACEABILITY            | A product can replace another specified software product for the same purpose                                                  |   |   |   |   |   |
| USABILITY              | APPROPRIATENESS           | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   |   |
|                        | RECOGNIZABILITY           | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   |   |
|                        | LEARNABILITY              | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   |   |
|                        | OPERABILITY               | A product or system protects user against making errors                                                                        |   |   |   |   |   |
|                        | USER ERROR PROTECTION     | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   |   |
|                        | USER INTERFACE AESTHETICS | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   |   |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY             | or system, when performing its functions meet requirements                                                                     |   |   |   |   |   |
|                        | TIME-BEHAVIOR             | when performing its functions, meet requirements                                                                               |   |   |   |   |   |
|                        | RESOURCE UTILIZATION      | performing its functions, meet requirements                                                                                    |   |   |   |   |   |
|                        | CAPACITY                  | A maximum limits of the product or system, parameter meet requirements                                                         |   |   |   |   |   |
| SECURITY               | CONFIDENTIALITY           | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   |   |
|                        | INTEGRITY                 | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   |   |
|                        | NON-REPUDIATION           | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   |   |
|                        | ACCOUNTABILITY            | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   |   |
|                        | AUTHENTICITY              | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   |   |
| COMPATIBILITY          | CO-EXISTENCE              | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   |   |
|                        | INTEROPERABILITY          | Two or more systems, products or components can exchange information                                                           |   |   |   |   |   |
| MAINTAINABILITY        | MODULARITY                | such that a change to one component has minimal impact on other                                                                |   |   |   |   |   |
|                        | REUSABILITY               | An asset can be used in more than one system                                                                                   |   |   |   |   |   |
|                        | ANALYZABILITY             | intended change to one or more of its parts                                                                                    |   |   |   |   |   |
|                        | MODIFIABILITY             | without introducing defects or degrading existing product quality                                                              |   |   |   |   |   |
|                        | TESTABILITY               | Test criteria can be established for a system, product or components                                                           |   |   |   |   |   |



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College of Computing Sciences and Engineering

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### IT Practitioner

Name: Dorothy Mei Cabuyao

Date: 11 / 19 / 25

Work: Part-time College Instructor

Company: LSPU RPCC

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | / |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | / |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   | / |   |   |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   | / |   |   |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   | / |   |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   | / |   |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   | / |   |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   | / |   |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   | / |   |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   | / |   |   |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   | / |   |
|                        | LEARNABILITY          | A product or system is easy to learn, understand, and use effectively, efficiency in emergency situations                      |   |   | / |   |   |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   | / |   |   |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   | / |   |   |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   | / |   |   |
|                        | AESTHETICS            | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   | / |   |   |
| PERFORMANCE EFFICIENCY | TIME-BEHAVIOR         | A product or system, when performing its functions meet requirements                                                           |   |   | / |   |   |
|                        | RESOURCE UTILIZATION  | A product or system, when performing its functions, meet requirements                                                          |   |   | / |   |   |
|                        | CAPACITY              | A product or system, maximum limits of the product or system, parameter meet requirements                                      |   |   | / |   |   |
|                        | SECURITY              | A product or system, ensures that data is accessible only to those authorized to have                                          |   |   | / |   |   |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   | / |   |   |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   | / |   |   |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events                                                        |   |   | / |   |   |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   | / |   |   |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   | / |   |   |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   | / |   |   |
| MAINTAINABILITY        | MODULARITY            | A product or system, such that a change to one component has minimal impact on other                                           |   |   | / |   |   |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   | / |   |   |
|                        | ANALYZABILITY         | A product or system, intended change to one or more of its parts                                                               |   |   | / |   |   |
|                        | MODIFIABILITY         | A product or system, without introducing defects or degrading existing product quality                                         |   |   | / |   |   |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   | / |   |   |



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### IT Practitioner

Name: Marlon Dungs

Date: 11 / 19 / 25

Work: Part-time Faculty

Company: LSPU - SPCC

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | / |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | / |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | / |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | / |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   |   | / |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | / |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | / |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | / |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   | / |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | / |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | / |
|                        | RECOGNIZABILITY       | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   | / |
|                        | LEARNABILITY          | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | / |
|                        | OPERABILITY           | A product or system protects user against making errors                                                                        |   |   |   |   | / |
|                        | USER ERROR PROTECTION | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | / |
|                        | USER INTERFACE        | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | / |
| PERFORMANCE EFFICIENCY | AESTHETICS            | or system, when performing its functions meet requirements                                                                     |   |   |   |   | / |
|                        | ACCESSIBILITY         | when performing its functions, meet requirements                                                                               |   |   |   |   | / |
|                        | TIME-BEHAVIOR         | maximum limits of the product or system, parameter meet requirements                                                           |   |   |   |   | / |
|                        | RESOURCE UTILIZATION  |                                                                                                                                |   |   |   |   | / |
| SECURITY               | CAPACITY              | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | / |
|                        | CONFIDENTIALITY       | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | / |
|                        | INTEGRITY             | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | / |
|                        | NON-REPUDIATION       | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | / |
|                        | ACCOUNTABILITY        | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | / |
| COMPATIBILITY          | AUTHENTICITY          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | / |
|                        | CO-EXISTENCE          | Two or more systems, products or components can exchange information                                                           |   |   |   |   | / |
| MAINTAINABILITY        | INTEROPERABILITY      | such that a change to one component has minimal impact on other                                                                |   |   |   |   | / |
|                        | MODULARITY            | An asset can be used in more than one system                                                                                   |   |   |   |   | / |
|                        | REUSABILITY           | intended change to one or more of its parts                                                                                    |   |   |   |   | / |
|                        | ANALYZABILITY         | without introducing defects or degrading existing product quality                                                              |   |   |   |   | / |
|                        | MODIFIABILITY         | Test criteria can be established for a system, product or components                                                           |   |   |   |   | / |
|                        | TESTABILITY           |                                                                                                                                |   |   |   |   | / |



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### IT Practitioner

Name: Biem Marc Arista

Date: 11/19/2025

Work: Developer

Company: FDS Asya Philippines Inc.

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | / |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | / |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | / |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | / |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   | / |   |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   | / |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   | / |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   | / |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                |   |   |   |   | / |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | / |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | / |
|                        | RECOGNIZABILITY       | A product or system is easy to learn and use                                                                                   |   |   |   |   | / |
|                        | LEARNABILITY          | A product or system is easy to learn and use                                                                                   |   |   |   |   | / |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | / |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   |   | / |   |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | / |
| PERFORMANCE EFFICIENCY | AESTHETICS            | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   | / |   |
|                        | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   | / |   |
|                        | TIME BEHAVIOR         | A system, product or component, when performing its functions meet requirements                                                |   |   |   |   | / |
|                        | RESOURCE UTILIZATION  | A system, product or component, when performing its functions, meet requirements                                               |   |   |   |   | / |
| SECURITY               | CAPACITY              | A system, product or component, when performing its functions, meet requirements                                               |   |   |   |   | / |
|                        | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have access                                          |   |   |   |   | / |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   | / |   |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events cannot be denied                                       |   |   |   | / |   |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   | / |   |
| COMPATIBILITY          | AUTHENTICITY          | The identity of a subject or resources can be proved to be the one claimed                                                     |   |   |   | / |   |
|                        | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   | / |   |
| MAINTAINABILITY        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   | / |   |
|                        | MODULARITY            | A product or system is designed such that a change to one component has minimal impact on other components                     |   |   |   | / |   |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   |   | / |   |
|                        | ANALYZABILITY         | A product or system is designed to allow for intended change to one or more of its parts                                       |   |   |   | / |   |
|                        | MODIFIABILITY         | A product or system is designed without introducing defects or degrading existing product quality                              |   |   |   | / |   |
| TESTABILITY            | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   | / |   |



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### IT Practitioner

Name: Alvin C. Tundialan

Date: 11 / 19 / 2024

Work: ITC ASYA PHILIPPINES - FDSAP / Product Specialist

Company: ITC ASYA PHILIPPINES

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC         | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY             | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | / |
|                        | CORRECTNESS               | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | / |
|                        | APPROPRIATENESS           | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | / |
| RELIABILITY            | MATURITY                  | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | / |
|                        | AVAILABILITY              | A product or system is operational and accessible when required for use                                                        |   |   |   |   | / |
|                        | FAULT TOLERANCE           | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | / |
|                        | RECOVERABILITY            | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | / |
|                        | ADAPTABILITY              | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | / |
| PORTABILITY            | INSTALLABILITY            | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   | / |
|                        | REPLACEABILITY            | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | / |
| USABILITY              | APPROPRIATENESS           | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | / |
|                        | RECOGNIZABILITY           | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   | / |
|                        | LEARNABILITY              | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | / |
|                        | OPERABILITY               | A product or system protects user against making errors                                                                        |   |   |   |   | / |
|                        | USER ERROR PROTECTION     | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | / |
|                        | USER INTERFACE AESTHETICS | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | / |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY             | TIME-BEHAVIOR                                                                                                                  |   |   |   |   | / |
|                        |                           | or system, when performing its functions meet requirements                                                                     |   |   |   |   | / |
|                        |                           | RESOURCE UTILIZATION                                                                                                           |   |   |   |   | / |
|                        |                           | when performing its functions, meet requirements                                                                               |   |   |   |   | / |
| SECURITY               | CONFIDENTIALITY           | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | / |
|                        | INTEGRITY                 | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | / |
|                        | NON-REPUDIATION           | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | / |
|                        | ACCOUNTABILITY            | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | / |
|                        | AUTHENTICITY              | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | / |
| COMPATIBILITY          | CO-EXISTENCE              | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | / |
|                        | INTEROPERABILITY          | Two or more systems, products or components can exchange information                                                           |   |   |   |   | / |
| MAINTAINABILITY        | MODULARITY                | such that a change to one component has minimal impact on other                                                                |   |   |   |   | / |
|                        | REUSABILITY               | An asset can be used in more than one system                                                                                   |   |   |   |   | / |
|                        | ANALYZABILITY             | intended change to one or more of its parts                                                                                    |   |   |   |   | / |
|                        | MODIFIABILITY             | without introducing defects or degrading existing product quality                                                              |   |   |   |   | / |
|                        | TESTABILITY               | Test criteria can be established for a system, product or components                                                           |   |   |   |   | / |



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## IT Practitioner

Name: Raphael Agichano

Date: 11/19/2023

Work: Developer

Company: Bakawan Data Analytics Inc.

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC         | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY             | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | ✓ |
|                        | CORRECTNESS               | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | ✓ |
|                        | APPROPRIATENESS           | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | ✓ |
| RELIABILITY            | MATURITY                  | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | ✓ |
|                        | AVAILABILITY              | A product or system is operational and accessible when required for use                                                        |   |   |   |   | ✓ |
|                        | FAULT TOLERANCE           | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | ✓ |
|                        | RECOVERABILITY            | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | ✓ |
|                        | ADAPTABILITY              | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | ✓ |
| PORTABILITY            | INSTALLABILITY            | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   | ✓ |
|                        | REPLACEABILITY            | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | ✓ |
| USABILITY              | APPROPRIATENESS           | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | RECOGNIZABILITY           | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   | ✓ |
|                        | LEARNABILITY              | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | ✓ |
|                        | OPERABILITY               | A product or system protects user against making errors                                                                        |   |   |   |   | ✓ |
|                        | USER ERROR PROTECTION     | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | ✓ |
|                        | USER INTERFACE AESTHETICS | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | ✓ |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY             | or system, when performing its functions meet requirements                                                                     |   |   |   |   | ✓ |
|                        | TIME BEHAVIOR             | when performing its functions, meet requirements                                                                               |   |   |   |   | ✓ |
|                        | RESOURCE UTILIZATION      | performing its functions, meet requirements                                                                                    |   |   |   |   | ✓ |
|                        | CAPACITY                  | A maximum limits of the product or system, parameter meet requirements                                                         |   |   |   |   | ✓ |
| SECURITY               | CONFIDENTIALITY           | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | ✓ |
|                        | INTEGRITY                 | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | ✓ |
|                        | NON-REPUDIATION           | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | ✓ |
|                        | ACCOUNTABILITY            | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | ✓ |
|                        | AUTHENTICITY              | The identity of a subject or resources can be proved to be the one claimed                                                     |   |   |   |   | ✓ |
| COMPATIBILITY          | CO-EXISTENCE              | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | ✓ |
|                        | INTEROPERABILITY          | Two or more systems, products or components can exchange information                                                           |   |   |   |   | ✓ |
| MAINTAINABILITY        | MODULARITY                | such that a change to one component has minimal impact on other                                                                |   |   |   |   | ✓ |
|                        | REUSABILITY               | An asset can be used in more than one system                                                                                   |   |   |   |   | ✓ |
|                        | ANALYZABILITY             | intended change to one or more of its parts                                                                                    |   |   |   |   | ✓ |
|                        | MODIFIABILITY             | without introducing defects or degrading existing product quality                                                              |   |   |   |   | ✓ |
|                        | TESTABILITY               | Test criteria can be established for a system, product or components                                                           |   |   |   |   | ✓ |



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### IT Practitioner

Name: Jacell Aunio

Date: 19 / 11 / 25

Work: Technical IT staff

Company: LSPM SC

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | / |
|                        | CORRECTNESS           | function provides the correct results with the needed degree of precision                                                      |   |   |   |   | / |
|                        | APPROPRIATENESS       | function facilitates the accomplishment of specified tasks and objectives                                                      |   |   |   |   | / |
| RELIABILITY            | MATURITY              | system, product or component meets for reliability under normal operation                                                      |   |   |   |   | / |
|                        | AVAILABILITY          | product or system is operational and accessible when required for use                                                          |   |   |   |   | / |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | / |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   | / |   |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | / |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   | / |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | / |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | / |
|                        | RECOGNIZABILITY       | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   | / |
|                        | LEARNABILITY          | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | / |
|                        | OPERABILITY           | A product or system protects user against making errors                                                                        |   |   |   |   | / |
|                        | USER ERROR PROTECTION | user interface enables pleasing and satisfying interactions for the user                                                       |   |   |   |   | / |
|                        | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | / |
| PERFORMANCE EFFICIENCY | TIME-BEHAVIOR         | or system, when performing its functions meet requirements                                                                     |   |   |   |   | / |
|                        | RESOURCE UTILIZATION  | when performing its functions, meet requirements                                                                               |   |   |   |   | / |
|                        | CAPACITY              | maximum limits of the product or system, parameter meet requirements                                                           |   |   |   |   | / |
|                        |                       |                                                                                                                                |   |   |   |   | / |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | / |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | / |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | / |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | / |
|                        | AUTHENTICITY          | the identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | / |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | / |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   |   | / |
| MAINTAINABILITY        | MODULARITY            | such that a change to one component has minimal impact on other                                                                |   |   |   |   | / |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   |   |   | / |
|                        | ANALYZABILITY         | intended change to one or more of its parts                                                                                    |   |   |   |   | / |
|                        | MODIFIABILITY         | without introducing defects or degrading existing product quality                                                              |   |   |   |   | / |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   |   | / |



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### Client

Name: Christlyn Lami

Date: 11 / 19 / 25

Position: Product Specialist BAKAWAN Data Analytics Inc.

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | ✓ |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | ✓ |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | ✓ |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | ✓ |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   |   | ✓ |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | ✓ |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | ✓ |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | ✓ |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                |   |   |   |   | ✓ |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | ✓ |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | LEARNABILITY          | Users can learn to use a product or system effectively, efficiently in emergency situations                                    |   |   |   |   | ✓ |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | ✓ |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   |   |   | ✓ |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | ✓ |
|                        | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | ✓ |
| PERFORMANCE EFFICIENCY | TIME-BEHAVIOR         | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
|                        | RESOURCE UTILIZATION  | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
|                        | CAPACITY              | A system, product or component meets requirements at maximum limits of the product or system, parameter meet requirements      |   |   |   |   | ✓ |
|                        |                       |                                                                                                                                |   |   |   |   | ✓ |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have access                                          |   |   |   |   | ✓ |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | ✓ |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events are not denied                                         |   |   |   |   | ✓ |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | ✓ |
|                        | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | ✓ |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | ✓ |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   |   | ✓ |
| MAINTAINABILITY        | MODULARITY            | A system, product or component is such that a change to one component has minimal impact on other components                   |   |   |   |   | ✓ |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   |   |   | ✓ |
|                        | ANALYZABILITY         | A system, product or component is such that a change to one or more of its parts can be made without affecting the system      |   |   |   |   | ✓ |
|                        | MODIFIABILITY         | A system, product or component is such that a change to one or more of its parts can be made without affecting the system      |   |   |   |   | ✓ |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   |   | ✓ |



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### IT Practitioner

Name: Jessica T. Biano

Date: 11 / 19 / 25

Work: product specialist 1

Company: FDs ASIA PHILIPPINES INC.

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                      | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives.                                                         |   |   |   |   | ✓ |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision.                                                    |   |   |   |   | ✓ |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives.                                                    |   |   |   |   | ✓ |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation.                                                    |   |   |   |   | ✓ |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use.                                                        |   |   |   |   | ✓ |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults.                                             |   |   |   | ✓ |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data.                                         |   |   |   | ✓ |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware.                              |   |   |   | ✓ |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment.                                |   |   |   | ✓ |   |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose.                                                  |   |   |   | ✓ |   |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs.                                               |   |   |   |   | ✓ |
|                        | LEARNABILITY          | A product or system is easy to learn, understand, and use effectively, efficiency in emergency situations.                      |   |   |   |   | ✓ |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use.                                                         |   |   |   |   | ✓ |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors.                                                                        |   |   |   |   | ✓ |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user.                                                     |   |   |   | ✓ |   |
|                        | AESTHETICS            | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals. |   |   |   | ✓ |   |
| PERFORMANCE EFFICIENCY | TIME BEHAVIOR         | A system, product or component meets requirements when performing its functions.                                                |   |   |   |   | ✓ |
|                        | RESOURCE UTILIZATION  | A system, product or component meets requirements when performing its functions.                                                |   |   |   |   | ✓ |
|                        | CAPACITY              | A system, product or component meets requirements when performing its functions.                                                |   |   |   |   | ✓ |
|                        | SECURITY              | A system, product or component meets requirements when performing its functions.                                                |   |   |   |   | ✓ |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have access.                                          |   |   |   |   | ✓ |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data.                  |   |   |   |   | ✓ |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events are not denied.                                         |   |   |   | ✓ |   |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity.                                                                  |   |   |   | ✓ |   |
|                        | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed.                                                      |   |   |   | ✓ |   |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products.  |   |   |   | ✓ |   |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information.                                                           |   |   |   | ✓ |   |
| MAINTAINABILITY        | MODULARITY            | A system, product or component is designed such that a change to one component has minimal impact on other components.          |   |   |   | ✓ |   |
|                        | REUSABILITY           | An asset can be used in more than one system.                                                                                   |   |   |   | ✓ |   |
|                        | ANALYZABILITY         | A system, product or component is designed to one or more of its parts.                                                         |   |   |   | ✓ |   |
|                        | MODIFIABILITY         | A system, product or component is designed without introducing defects or degrading existing product quality.                   |   |   |   | ✓ |   |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components.                                                           |   |   |   | ✓ |   |



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## IT Practitioner

Name: REYMAR V. MANALOTO

Date: 11/19/25

Work: \_\_\_\_\_

Company: LSPL

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | ✓ |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | ✓ |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | ✓ |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   | ✓ |   |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   | ✓ |   |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   | ✓ |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   | ✓ |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   | ✓ |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                |   |   |   | ✓ |   |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   | ✓ |   |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | RECOGNIZABILITY       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | LEARNABILITY          | Users can learn to use a product or system effectively, efficiently in emergency situations                                    |   |   |   |   | ✓ |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | ✓ |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   |   | ✓ |   |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   | ✓ |   |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | ✓ |
|                        | TIME BEHAVIOR         | A system or system, when performing its functions meet requirements                                                            |   |   |   |   | ✓ |
|                        | RESOURCE UTILIZATION  | A system or system, when performing its functions, meet requirements                                                           |   |   |   |   | ✓ |
|                        | CAPACITY              | A system or system, the maximum limits of the product or system, parameter meet requirements                                   |   |   |   |   | ✓ |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have access                                          |   |   |   |   | ✓ |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | ✓ |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events cannot be denied                                       |   |   |   |   | ✓ |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | ✓ |
|                        | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | ✓ |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | ✓ |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   |   | ✓ |
| MAINTAINABILITY        | MODULARITY            | A system or system, such that a change to one component has minimal impact on other components                                 |   |   |   |   | ✓ |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   |   |   | ✓ |
|                        | ANALYZABILITY         | A system or system, intended change to one or more of its parts                                                                |   |   |   |   | ✓ |
|                        | MODIFIABILITY         | A system or system, without introducing defects or degrading existing product quality                                          |   |   |   |   | ✓ |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   |   | ✓ |



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## IT Practitioner

Name: MARK CHRISTOPHER M. MENDIADO

Date: 11/1/2011

Work: ADMINISTRATIVE AIDE III

Company: LSPU SPC

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                                   | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                                       |   |   |   |   |   |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                                  |   |   |   |   |   |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                                  |   |   |   |   |   |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                                  |   |   |   |   |   |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                                      |   |   |   |   |   |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                                           |   |   |   |   |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                                       |   |   |   |   |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                                            |   |   |   |   |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                              |   |   |   |   |   |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                                |   |   |   |   |   |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                                             |   |   |   |   |   |
|                        | RECOGNIZABILITY       | Users can recognize whether a product or a system is appropriate for their needs                                                             |   |   |   |   |   |
|                        | LEARNABILITY          | Users can learn to use a product or system effectively, efficiently in emergency situations                                                  |   |   |   |   |   |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                                       |   |   |   |   |   |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                                      |   |   |   |   |   |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                                   |   |   |   |   |   |
| PERFORMANCE EFFICIENCY | AESTHETICS            | A user interface enables pleasing and satisfying interactions for the user                                                                   |   |   |   |   |   |
|                        | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals               |   |   |   |   |   |
|                        | TIME-BEHAVIOR         | A system, product or component meets requirements when performing its functions                                                              |   |   |   |   |   |
|                        | RESOURCE UTILIZATION  | A system, product or component meets requirements when performing its functions                                                              |   |   |   |   |   |
| SECURITY               | CAPACITY              | A system, product or component meets requirements when performing its functions                                                              |   |   |   |   |   |
|                        | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have                                                               |   |   |   |   |   |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                                |   |   |   |   |   |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events                                                                      |   |   |   |   |   |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                                |   |   |   |   |   |
| COMPATIBILITY          | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed                                                                    |   |   |   |   |   |
|                        | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products                |   |   |   |   |   |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                                         |   |   |   |   |   |
| MAINTAINABILITY        | MODULARITY            | A product or system is designed such that a change to one component has minimal impact on other components                                   |   |   |   |   |   |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                                 |   |   |   |   |   |
|                        | ANALYZABILITY         | A system, product or component can be analyzed to determine intended change                                                                  |   |   |   |   |   |
|                        | MODIFIABILITY         | A system, product or component can be modified to one or more of its parts without introducing defects or degrading existing product quality |   |   |   |   |   |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                                         |   |   |   |   |   |



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### Resident

Name: Jendryx Bautista

Date: 11 / 21 / 25

Occupation: Municipal IT staff

Address: Imok

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC         | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY             | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | ✓ |
|                        | CORRECTNESS               | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | ✓ |
|                        | APPROPRIATENESS           | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | ✓ |
| RELIABILITY            | MATURITY                  | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | ✓ |
|                        | AVAILABILITY              | A product or system is operational and accessible when required for use                                                        |   |   |   |   | ✓ |
|                        | FAULT TOLERANCE           | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | ✓ |
|                        | RECOVERABILITY            | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | ✓ |
|                        | ADAPTABILITY              | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | ✓ |
| PORTABILITY            | INSTALLABILITY            | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   | ✓ |
|                        | REPLACEABILITY            | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | ✓ |
| USABILITY              | APPROPRIATENESS           | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | RECOGNIZABILITY           | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   | ✓ |
|                        | LEARNABILITY              | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | ✓ |
|                        | OPERABILITY               | A product or system protects user against making errors                                                                        |   |   |   |   | ✓ |
|                        | USER ERROR PROTECTION     | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | ✓ |
|                        | USER INTERFACE AESTHETICS | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | ✓ |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY             | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | ✓ |
|                        | TIME-BEHAVIOR             | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
|                        | RESOURCE UTILIZATION      | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
|                        | CAPACITY                  | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
| SECURITY               | CONFIDENTIALITY           | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | ✓ |
|                        | INTEGRITY                 | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | ✓ |
|                        | NON-REPUDIATION           | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | ✓ |
|                        | ACCOUNTABILITY            | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | ✓ |
|                        | AUTHENTICITY              | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | ✓ |
| COMPATIBILITY          | CO-EXISTENCE              | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | ✓ |
|                        | INTEROPERABILITY          | Two or more systems, products or components can exchange information                                                           |   |   |   |   | ✓ |
| MAINTAINABILITY        | MODULARITY                | A product or system is designed such that a change to one component has minimal impact on other components                     |   |   |   |   | ✓ |
|                        | REUSABILITY               | An asset can be used in more than one system                                                                                   |   |   |   |   | ✓ |
|                        | ANALYZABILITY             | A system, product or component can be analyzed to determine intended change to one or more of its parts                        |   |   |   |   | ✓ |
|                        | MODIFIABILITY             | A system, product or component can be modified without introducing defects or degrading existing product quality               |   |   |   |   | ✓ |
|                        | TESTABILITY               | Test criteria can be established for a system, product or components                                                           |   |   |   |   | ✓ |



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## Resident

Name: Mark Alegado

Date: 11 / 21 / 25

Occupation: Calavan DSWD staff

Address: Lamot 2

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | ✓ |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | ✓ |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | ✓ |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | ✓ |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   |   | ✓ |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | ✓ |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | ✓ |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | ✓ |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                |   |   |   |   | ✓ |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | ✓ |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | RECOGNIZABILITY       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | ✓ |
|                        | LEARNABILITY          | Users can learn to use a product or system effectively, efficiently in emergency situations                                    |   |   |   |   | ✓ |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | ✓ |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   |   |   | ✓ |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | ✓ |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | ✓ |
|                        | TIME-BEHAVIOR         | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
|                        | RESOURCE UTILIZATION  | A system, product or component meets requirements when performing its functions                                                |   |   |   |   | ✓ |
|                        | CAPACITY              | A maximum limits of the product or system, parameter meet requirements                                                         |   |   |   |   | ✓ |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | ✓ |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | ✓ |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | ✓ |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | ✓ |
|                        | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | ✓ |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | ✓ |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   |   | ✓ |
| MAINTAINABILITY        | MODULARITY            | A system, product or component is designed such that a change to one component has minimal impact on other components          |   |   |   |   | ✓ |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   |   |   | ✓ |
|                        | ANALYZABILITY         | A system, product or component is designed to allow intended change to one or more of its parts                                |   |   |   |   | ✓ |
|                        | MODIFIABILITY         | A system, product or component is designed without introducing defects or degrading existing product quality                   |   |   |   |   | ✓ |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   |   | ✓ |



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## PAMANTASAN NG LUNGSOD NG SAN PABLO

College of Computing Sciences and Engineering

Brgy. San Jose, San Pablo City

Tel No.: (049) 536-7830

Email Address: [plspofficial@plsp.edu.ph](mailto:plspofficial@plsp.edu.ph)



### Resident

Name: Journalyn Mendoza

Date: 11 / 21 / 25

Occupation: Sari Sari store owner

Address: Balayhangin

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                                               | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                                                   |   |   |   |   | / |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                                              |   |   |   |   | / |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                                              |   |   |   |   | / |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                                              |   |   |   |   | / |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                                                  |   |   |   |   | / |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                                                       |   |   |   |   | / |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                                                   |   |   |   |   | / |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                                                        |   |   |   |   | / |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                                          |   |   |   |   | / |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                                            |   |   |   |   | / |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                                                         |   |   |   |   | / |
|                        | RECOGNIZABILITY       | A product or system is easy to learn, use, and understand                                                                                                |   |   |   |   | / |
|                        | LEARNABILITY          | A product or system is easy to learn, use, and understand                                                                                                |   |   |   |   | / |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                                                   |   |   |   |   | / |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                                                  |   |   |   |   | / |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                                               |   |   |   |   | / |
|                        | AESTHETICS            | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals                           |   |   |   |   | / |
| PERFORMANCE EFFICIENCY | TIME-BEHAVIOR         | A system, product or component performs its functions when performing its functions, meet requirements                                                   |   |   |   |   | / |
|                        | RESOURCE UTILIZATION  | A system, product or component performs its functions, meet requirements when performing its functions, meet requirements                                |   |   |   |   | / |
|                        | CAPACITY              | A system, product or component performs its functions, meet requirements within the maximum limits of the product or system, parameter meet requirements |   |   |   |   | / |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have access                                                                    |   |   |   |   | / |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                                            |   |   |   |   | / |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events cannot be denied                                                                 |   |   |   |   | / |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                                            |   |   |   |   | / |
|                        | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed                                                                                |   |   |   |   | / |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products                            |   |   |   |   | / |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                                                     |   |   |   |   | / |
| MAINTAINABILITY        | MODULARITY            | A system, product or component is designed such that a change to one component has minimal impact on other components                                    |   |   |   |   | / |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                                             |   |   |   |   | / |
|                        | ANALYZABILITY         | A system, product or component is designed to facilitate intended change to one or more of its parts                                                     |   |   |   |   | / |
|                        | MODIFIABILITY         | A system, product or component is designed to facilitate change without introducing defects or degrading existing product quality                        |   |   |   |   | / |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                                                     |   |   |   |   | / |



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College of Computing Sciences and Engineering

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Tel No.: (049) 536-7830

Email Address: [plspofficial@plsp.edu.ph](mailto:plspofficial@plsp.edu.ph)



## Resident

Name: Emelyn Villanueva Garinga

Date: 11 / 21 / 25

Occupation: Sari Sari Store Owner

Address: Purok Ibaba Dayap

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   | / |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   | / |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   | / |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   | / |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   |   | / |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of faults                                             |   |   |   |   | / |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   | / |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware                              |   |   |   |   | / |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in a specified environment                                |   |   |   |   | / |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   | / |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | / |
|                        | RECOGNIZABILITY       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   | / |
|                        | LEARNABILITY          | Users can learn to use a product or system effectively, efficiently in emergency situations                                    |   |   |   |   | / |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   | / |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   |   |   | / |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   | / |
| PERFORMANCE EFFICIENCY | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   | / |
|                        | TIME-BEHAVIOR         | A system, product or component, when performing its functions meet requirements                                                |   |   |   |   | / |
|                        | RESOURCE UTILIZATION  | A system, product or component, when performing its functions, meet requirements                                               |   |   |   |   | / |
|                        | CAPACITY              | A system, product or component, maximum limits of the product or system, parameter meet requirements                           |   |   |   |   | / |
| SECURITY               | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   | / |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   | / |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   | / |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   | / |
|                        | AUTHENTICITY          | The identity of a subject or resource can be proved to be the one claimed                                                      |   |   |   |   | / |
| COMPATIBILITY          | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   | / |
|                        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   |   | / |
| MAINTAINABILITY        | MODULARITY            | A system, product or component such that a change to one component has minimal impact on other components                      |   |   |   |   | / |
|                        | REUSABILITY           | An asset can be used in more than one system                                                                                   |   |   |   |   | / |
|                        | ANALYZABILITY         | A system, product or component intended change to one or more of its parts                                                     |   |   |   |   | / |
|                        | MODIFIABILITY         | A system, product or component without introducing defects or degrading existing product quality                               |   |   |   |   | / |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   |   | / |



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College of Computing Sciences and Engineering

Brgy. San Jose, San Pablo City

Tel No.: (049) 536-7830

Email Address: [plspofficial@plsp.edu.ph](mailto:plspofficial@plsp.edu.ph)



## Resident

Name: Michelle Briz

Date: 11 / 21 / 25

Occupation: Meat Shop

Address: Brgy. Lamot 1

Choose on the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

| CHARACTERISTIC         | SUBCHARACTERISTIC     | Definition                                                                                                                     | 1 | 2 | 3 | 4 | 5 |
|------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| FUNCTIONAL STABILITY   | FUNCTIONALITY         | The set of function covers all the specified tasks and user objectives                                                         |   |   |   |   |   |
|                        | CORRECTNESS           | A function provides the correct results with the needed degree of precision                                                    |   |   |   |   |   |
|                        | APPROPRIATENESS       | A function facilitates the accomplishment of specified tasks and objectives                                                    |   |   |   |   |   |
| RELIABILITY            | MATURITY              | A system, product or component meets for reliability under normal operation                                                    |   |   |   |   |   |
|                        | AVAILABILITY          | A product or system is operational and accessible when required for use                                                        |   |   |   |   |   |
|                        | FAULT TOLERANCE       | A system, product or component operates as intended despite the presence of                                                    |   |   |   |   |   |
|                        | RECOVERABILITY        | In the event of an interruption or a failure, a product or system can recover the data                                         |   |   |   |   |   |
|                        | ADAPTABILITY          | A product or system can effectively and efficiently be adapted for different or evolving hardware.                             |   |   |   |   |   |
| PORTABILITY            | INSTALLABILITY        | A product or system can be successfully installed and/or uninstalled in                                                        |   |   |   |   |   |
|                        | REPLACEABILITY        | A product can replace another specified software product for the same purpose                                                  |   |   |   |   |   |
| USABILITY              | APPROPRIATENESS       | Users can recognize whether a product or a system is appropriate for their needs                                               |   |   |   |   |   |
|                        | RECOGNIZABILITY       |                                                                                                                                |   |   |   |   |   |
|                        | LEARNABILITY          | effectiveness, efficiency in emergency situations                                                                              |   |   |   |   |   |
|                        | OPERABILITY           | A product or system is easy to operate, control and appropriate to use                                                         |   |   |   |   |   |
|                        | USER ERROR PROTECTION | A product or system protects user against making errors                                                                        |   |   |   |   |   |
|                        | USER INTERFACE        | A user interface enables pleasing and satisfying interactions for the user                                                     |   |   |   |   |   |
| PERFORMANCE EFFICIENCY | AESTHETICS            |                                                                                                                                |   |   |   |   |   |
|                        | ACCESSIBILITY         | A product or system can be used by people with the widest range of characteristics and capabilities to achieve specified goals |   |   |   |   |   |
|                        | TIME-BEHAVIOR         | A product or system, when performing its functions meet requirements                                                           |   |   |   |   |   |
|                        | RESOURCE UTILIZATION  | A product or system, when performing its functions, meet requirements                                                          |   |   |   |   |   |
| SECURITY               | CAPACITY              | A product or system, when performing its functions, meet requirements                                                          |   |   |   |   |   |
|                        | CONFIDENTIALITY       | The prototype ensures that data is accessible only to those authorized to have                                                 |   |   |   |   |   |
|                        | INTEGRITY             | A system, product or component prevents unauthorized access to, or modification of, computer programs or data                  |   |   |   |   |   |
|                        | NON-REPUDIATION       | Actions or events can be proven to have taken place, so that the events                                                        |   |   |   |   |   |
|                        | ACCOUNTABILITY        | The actions of an entity can be traced uniquely to the entity                                                                  |   |   |   |   |   |
| COMPATIBILITY          | AUTHENTICITY          | The identity of a subject or resources can be proved to be the one claimed                                                     |   |   |   |   |   |
|                        | CO-EXISTENCE          | A product can perform its required functions efficiently while sharing a common environment and resources with other products  |   |   |   |   |   |
| MAINTAINABILITY        | INTEROPERABILITY      | Two or more systems, products or components can exchange information                                                           |   |   |   |   |   |
|                        | MODULARITY            | A product or system, when performing its functions, meet requirements                                                          |   |   |   |   |   |
|                        | REUSABILITY           | such that a change to one component has minimal impact on other                                                                |   |   |   |   |   |
|                        | ANALYZABILITY         | An asset can be used in more than one system                                                                                   |   |   |   |   |   |
|                        | MODIFIABILITY         | intended change to one or more of its parts without introducing defects or degrading existing product quality                  |   |   |   |   |   |
|                        | TESTABILITY           | Test criteria can be established for a system, product or components                                                           |   |   |   |   |   |

*G. User's Manual***Pamantasan ng Lungsod ng San Pablo**

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# User's Manual

DEVELOPMENT OF DOCUMENT REQUEST SYSTEM  
WITH K-MEANS CLUSTERING FOR MUNICIPAL CIVIL  
REGISTRAR OF MUNICIPALITY OF CALAUAN

Developed By:

Balague, Adonis A.

Calayag, Vince Kyle C.

Gervacio, Ronnel

Perez, Kristine O.

Perez, Phoebe Isobel V.

Salcedo, Lester Yurie J.



## Pamantasan ng Lungsod ng San Pablo

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### HARDWARE REQUIREMENTS

Processor : i3 and Up  
Hard Disk Drive : 512gb  
Memory Module : 16gb RAM  
Router : Any Available



## **Pamantasan ng Lungsod ng San Pablo**

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### **SOFTWARE REQUIREMENTS**

**Operating System : Windows**

**Application Software : Visual Studio Code**

**Database Software : SQL Management**

**Screen Resolution : 1366x768 px and Up for Desktop**

**Internet Provider : Any Available**



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### Installation Procedure

#### Step 1: Installation of Required Software

1. Download XAMPP from the official website.
2. Run the XAMPP installer and follow the on-screen installation instructions.
3. During the installation process, select Apache and MySQL as the required components.
4. After the installation is completed, open the XAMPP Control Panel.
5. Start the Apache and MySQL services by clicking the Start buttons.



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### Installation Procedure

#### Step 2: System File Setup

1. Copy the document\_sys folder.
2. Paste the project folder into the XAMPP htdocs directory: C:\xampp\htdocs\

#### Step 3: Database Configuration

1. Open a web browser and navigate to: <http://localhost/phpmyadmin>
2. Click New to create a new database.
3. Enter document as the database name.
4. Click Create.



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### Installation Procedure

5. Import the database file by following these steps:

- Select the newly created database.
- Click the Import tab.
- Choose the provided .sql file.
- Click Go to complete the database import.

#### Step 4: Running the System

1. Ensure that both Apache and MySQL services are running in the XAMPP Control Panel.
2. Open a web browser.



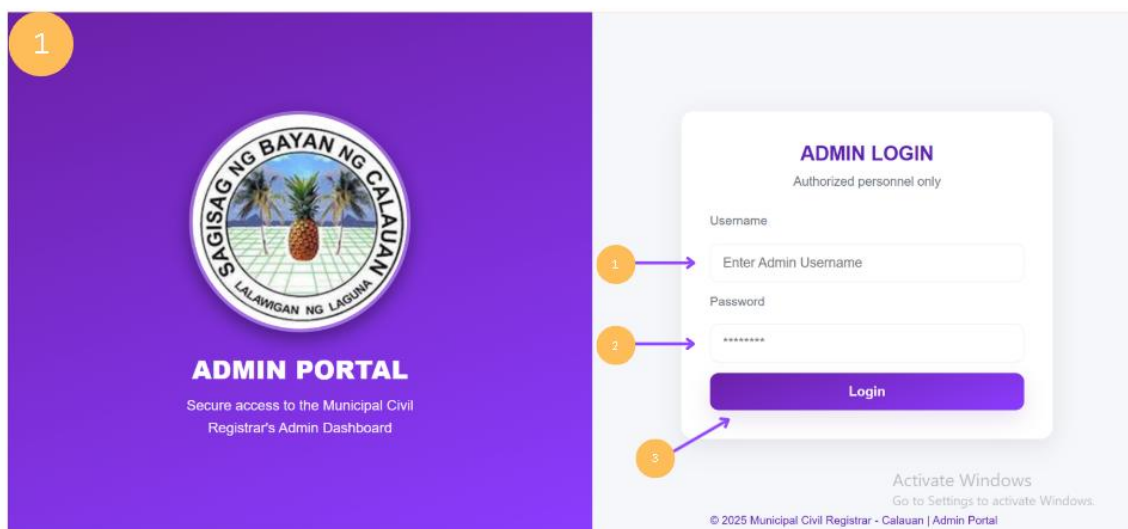
## Pamantasan ng Lungsod ng San Pablo

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### Installation Procedure

3. Enter the following URL in the address bar:  
`localhost/document_sys/landing.php`

4. The Document Request System landing page should appear, indicating that the installation was successful.

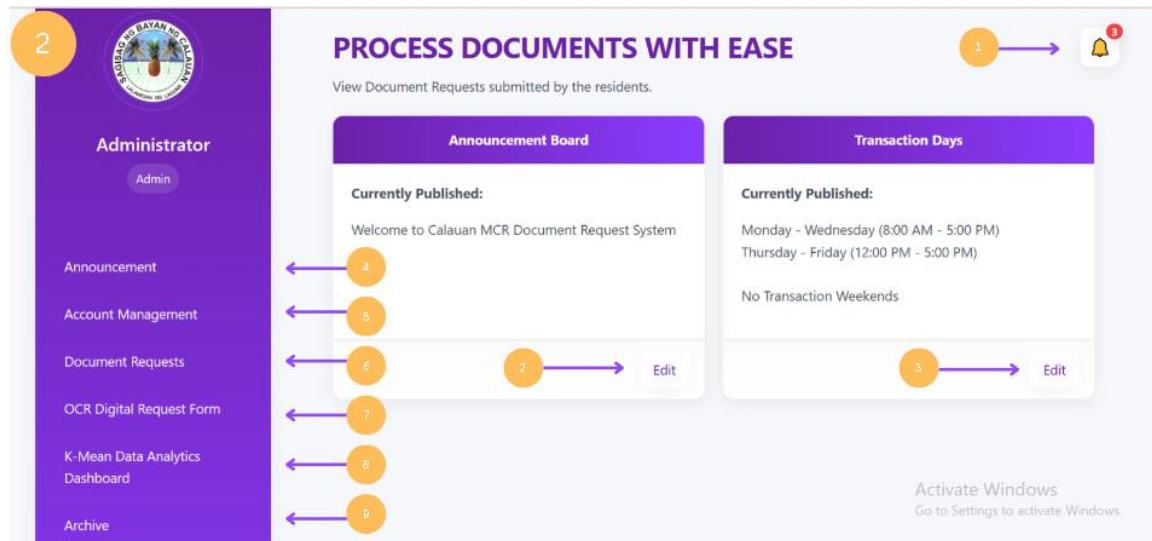


## 1. Admin Portal Log In

This is the secure access point for authorized personnel to manage the Municipal Civil Registrar's dashboard.

### Steps to Use (Admin Log In)

1. Enter the **Admin Username** in the first text field.
2. Enter the **Password** in the second text field.
3. Click the **Login** button to access the Admin Dashboard.



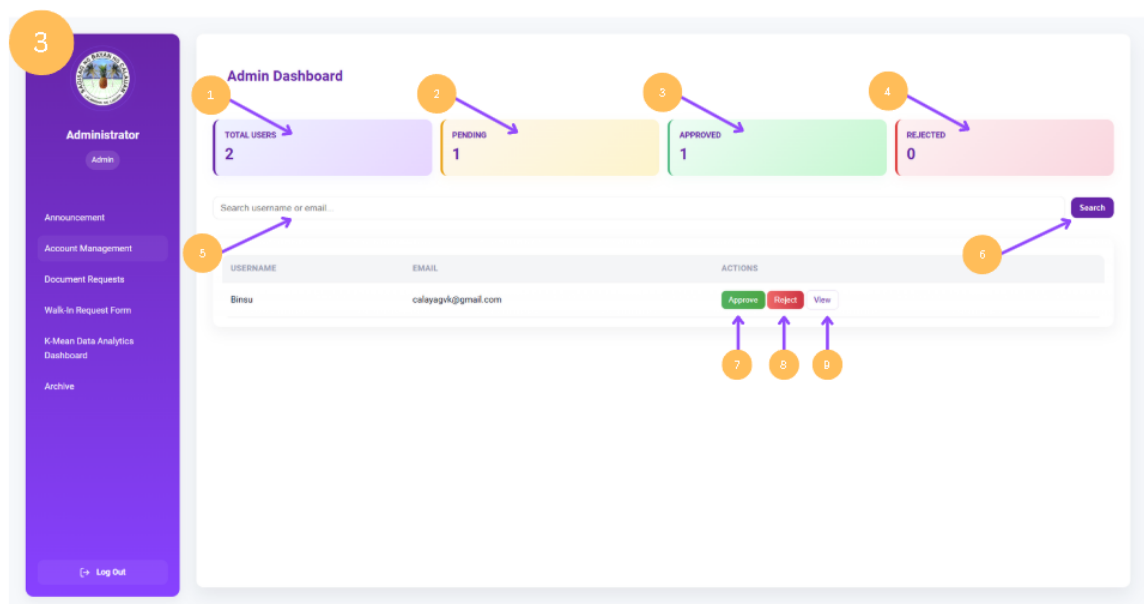
## 2. Admin Dashboard

This is the main admin page for processing documents and managing system settings.

### Steps to Use (Admin Dashboard)

1. Click the **Notification Bell** to view system alerts or new request updates.
2. Click the **Edit** button under the **Announcement Board** to update or change the welcome message.
3. Click the **Edit** button under **Transaction Days** to modify the official operating hours.

4. Click the **Announcement** link to manage and publish important system notices.
5. Click the **Account Management** link to manage user accounts and system permissions.
6. Click the **Document Requests** link to view and process all pending and submitted requests from residents.
7. Click the **OCR Digital Request Form** link for specific forms or processing tools.
8. Click the **K-Mean Data Analytics Dashboard** link to view statistical data and reports.
9. Click the **Archive** link to access or manage historical records and completed requests.



### 3. Admin Dashboard (Account Management)

This is the central view for the administrator to manage user accounts, showing key statistics and allowing user searching and action.

#### Steps to Use

1. Click the **Total Users** to see the total users accounts.
2. Click the **Pending** to see the total pending accounts.
3. Click the **Approved** to see the total approved accounts.
4. Click the **Rejected** to see the total rejected accounts.
5. Review the statistics boxes for the total number of **Total Users**, **Pending** accounts, **Approved** accounts, and **Rejected** accounts.
6. Use the **Search** bar to find a specific user by username or email, then click the **Search** button.
7. Click the **Approve** button to activate a pending user account.
8. Click the **Reject** button to deny a user's sign-up request.
9. Click the **View** button to see a user's complete profile details.

The screenshot displays the 'Document Requests' section of the PAMANTASAN NG LUNGSOD NG SAN PABLO system. The interface includes a sidebar with navigation options and a main table of document requests. A modal window for updating the status of a request is also shown.

**Document Requests Table:**

| Request No. | Document          | Contact    | Email                    | Date Requested      | Status  | Actions     |
|-------------|-------------------|------------|--------------------------|---------------------|---------|-------------|
| 42          | Death Certificate | 0999292099 | perckristine80@gmail.com | 2025-12-04 16:43:42 | PENDING | View Update |
| 41          | Death Certificate | 0999292099 | perckristine80@gmail.com | 2025-12-04 16:36:52 | PENDING | View Update |
| 40          | Death Certificate | 0999292099 | perckristine80@gmail.com | 2025-12-04 16:28:40 | PENDING | View Update |
| 39          | Death Certificate | 0999292099 | perckristine80@gmail.com | 2025-12-04 16:27:36 | PENDING | View Update |
| 38          | Death Certificate | 0999292099 | perckristine80@gmail.com | 2025-12-04 16:23:23 | PENDING | View Update |

**Update Status Modal:**

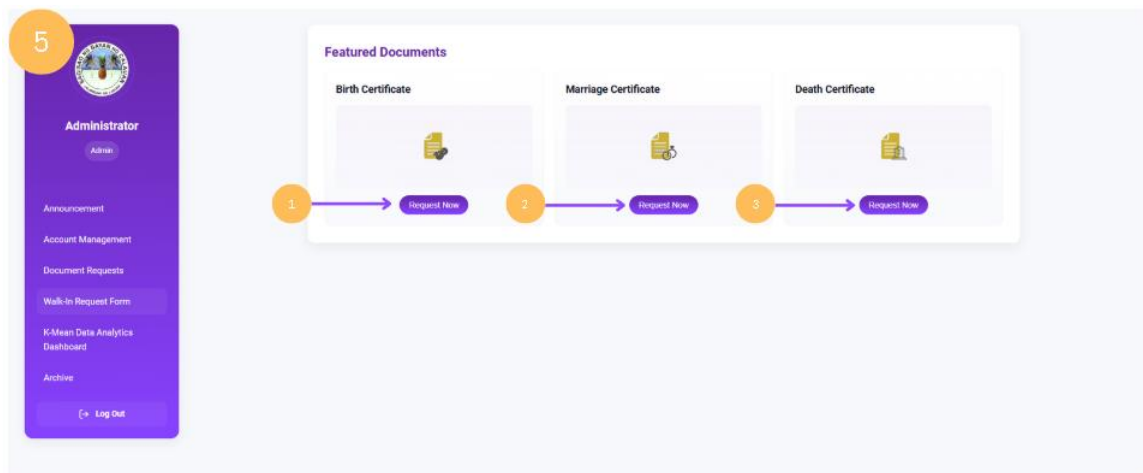
Request No: 42  
Document: Death Certificate  
Current Status: pending  
Change Status to:  
Processing  
Cancel Save

#### 4. Document Requests

This section displays a searchable and sortable table of all submitted resident document requests, allowing the administrator to track and process each application.

**Steps to Use**

1. Use the search bar or the **Document Status** dropdown to filter the list of requests.
2. Click the **Download Report** button to export the request data.
3. For each request:
4. Click **View** to see the full details of the request form and submitted requirements.
5. Click **Update** to change the status of the request (e.g., from PENDING to PROCESSING, READY, or RELEASED).
6. Click the **View** button (from the main request table) to open the **Request Details** pop-up, which displays a digital copy of the submitted form.
7. Click the **X** icon (or the blurred area outside the pop-up) to close the window.
8. Click the **Update** button (from the main request table) to open the **Update Status** pop-up.
9. Select the desired new status (e.g., Processing, Ready, For Release) from the **Change Status to** dropdown menu.
10. Click the **Cancel** button to close the pop-up without saving changes.
11. Click the **Save** button to apply the new status to the document request.



## 5. Walk-In Request Form

This section allows users to quickly access and initiate requests for commonly requested documents through clearly displayed options, making the request process faster and more convenient.

### Steps to Use

1. Review the **Featured Documents** panel to see the available document types (Birth Certificate, Marriage Certificate, Death Certificate).
2. Click the **Request Now** button under the desired document to start a new request.
3. Follow the on-screen form to provide the required information for the selected document.
4. Use the **side navigation menu** to move to other sections such as Document Requests, Walk-in Request Form, Dashboard, or Archive.

6



Administrator

Admin

Announcement

Account Management

Document Requests

Walk-In Request Form

K-Mean Data Analytics Dashboard

Archive

BIRTH CERTIFICATE FORM Request / Application

Request Details

Request for

Copy Issuance

Number of Copies

Person's / Child's Information

Last Name

First Name

Middle Name

Sex:

☐ Male ☐ Female

Date of Birth

dd/mm/yyyy

Place of Birth

City/Municipality and Province

Father's Name

Last Name

First Name

Middle Name

Mother's Name

Father's Name

Last Name

First Name

Middle Name

Mother's Name

Maiden Name

First Name

Middle Name

Upload Valid ID

Choose File No file chosen

Requester's Details

perez

kristine

o

Balayhangin

City/Municipality, Province

09992529399

Purpose of Request

The screenshot shows a web form for requesting a birth certificate. On the left is a purple sidebar with a 'Log Out' button. The main form area contains the following sections:

- Middle Name:** A text input field.
- Upload Valid ID:** A section with a 'Choose File' button and the text 'No file chosen'.
- Requester's Details:** A section with input fields for 'periz', 'kristine', 'o', 'Balayhargin', 'City/Municipality, Province', and '0995293399'.
- Purpose of Request:** A dropdown menu labeled 'Select Purpose'.
- Email:** An input field containing 'perickristine88@gmail.com'.
- Submit Request:** A purple button at the bottom.

Numbered callouts are present: '6' in the sidebar, '20' pointing to the 'Purpose of Request' dropdown, '21' pointing to the 'Email' input field, and '22' pointing to the 'Submit Request' button.

## 6. Birth Certificate Form

This is the application form users must accurately fill out to request a copy of a Birth Certificate, including details about the person, the requestor, the purpose, and submission.

### Steps to Use

1. Under **Request Details**, select the type of request (e.g., **Copy Issuance**) from the dropdown menu.
2. Input the **Number of Copies** you wish to request.
3. Under **Person's / Child's Information**, input the **Last Name**.
4. Input the **First Name**.
5. Input the **Middle Name**.
6. Select the **Sex** (Male or Female) using the radio buttons.

7. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
8. Input the **Place of Birth** (City/Municipality and Province).
9. Under **Father's Name**, input his **Last Name**.
10. Input his **First Name**.
11. Input his **Middle Name**.
12. Under **Mother's Name**, input her **Maiden Name**.
13. Input her **First Name**.
14. Input her **Middle Name**.
15. Under **Upload Valid ID**, click **Choose File** to upload the necessary identification document.
16. Under **Requester's Details**, input your **Last Name**.
17. Input your **First Name**.
18. Input your **Middle Initial**.
19. Input your **Address** (Barangay).
20. Input your **City/Municipality, Province** (and Country if abroad).
21. Input your **Mobile Number**.
22. (Continuation of Requester's Details) Input your **City/Municipality, Province** (and Country if abroad).
23. Input your **Mobile Number**.
24. Under **Purpose of Request**, specify the **purpose of your request** in the text area.
25. Verify your **Email** address (this field is usually pre-filled).

26. Click the **Submit Request** button to finalize and send your application.

The screenshot displays the 'MARRIAGE CERTIFICATE FORM' application page. On the left is a purple sidebar with a navigation menu. The main content area is white with a purple header. Numbered callouts (1-22) point to various form fields and buttons:

- 1**: Request for (dropdown menu)
- 2**: Number of Copies (text input)
- 3**: Husband's Name - Last Name (text input)
- 4**: Husband's Name - First Name (text input)
- 5**: Husband's Name - Middle Name (text input)
- 6**: Date of Birth (calendar icon)
- 7**: Wife's Maiden Name - Maiden Last Name (text input)
- 8**: Wife's Maiden Name - First Name (text input)
- 9**: Wife's Maiden Name - Middle Name (text input)
- 10**: Date of Birth (calendar icon)
- 11**: Date of Marriage (calendar icon)
- 12**: Place of Marriage - City/Municipality and Province (text input)
- 13**: Upload Valid ID - Choose File (button)
- 14**: Requester's Details - Last Name (text input)
- 15**: Requester's Details - First Name (text input)
- 16**: Requester's Details - Middle Initial (text input)
- 17**: Select Barangay (dropdown menu)
- 18**: City/Municipality, Province (text input)
- 19**: Mobile Number (text input)
- 20**: Purpose of Request - Select a purpose (dropdown menu)
- 21**: Email - Email (text input)
- 22**: Submit Request (button)

The sidebar menu includes: Administrator (Admin), Announcement, Account Management, Document Requests, Walk-In Request Form, K-Mean Data Analytics Dashboard, and Archive. The bottom of the sidebar has a 'Log Out' link.

## 7. Marriage Certificate Form

This is the application form users must accurately fill out to request a copy of a Marriage Certificate, covering details of the couple, the requestor, the purpose, and submission.

**Steps to Use**

1. Under **Request Details**, select the type of request (e.g., **Copy Issuance**) from the dropdown menu.
2. Input the **Number of Copies** you wish to request.
3. Under **Couple's Information** for **Husband's Name**, input his **Last Name**.
4. Input his **First Name**.
5. Input his **Middle Name**.
6. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
7. Under **Wife's Maiden Name**, input her **Maiden Last Name**.
8. Input her **First Name**.
9. Input her **Middle Name**.
10. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
11. Input the **Date of Marriage** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
12. Input the **Place of Marriage** (City/Municipality and Province).
13. Under **Upload Valid ID**, click **Choose File** to upload the necessary identification document.
14. Under **Requester's Details**, input your **Last Name**.
15. Input your **First Name**.
16. Input your **Middle Initial**.
17. Input your **Address** (House No., Street Name, Barangay).

18. Input your **City/Municipality, Province** (and Country if abroad).
19. Input your **Mobile Number**.
20. Under **Purpose of Request**, specify the **purpose of your request** in the text area.
21. Verify your **Email** address (this field is usually pre-filled).
22. Click the **Submit Request** button to finalize and send your application.

8

The screenshot displays the 'DEATH CERTIFICATE FORM' interface. On the left is a purple sidebar with the 'Administrator' menu. The main form area is titled 'Request / Application' and contains two sections: 'Request Details' and 'Deceased Person's Information'. Numbered arrows point to specific fields: 1 points to the 'Request for' dropdown (set to 'Copy Issuance'); 2 points to the 'Number of Copies' input field; 3 points to the 'Last Name' input field; 4 points to the 'First Name' input field; 5 points to the 'Middle Name' input field; 6 points to the 'Date of Birth' input field (format dd/mm/yyyy); 7 points to the 'Select Sex' dropdown; 8 points to the 'Date of Death' input field (format dd/mm/yyyy); and 9 points to the 'City/Municipality and Province' input field.

**Administrator**

- Admin
- Announcement
- Account Management
- Document Requests
- Walk-In Request Form
- K-Means Data Analytics Dashboard
- Archive

**DEATH CERTIFICATE FORM** Request / Application

**Request Details**

Request for: Copy Issuance

Number of Copies:

**Deceased Person's Information**

Last Name:

First Name:

Middle Name:

Date of Birth: dd/mm/yyyy

Sex: Select Sex

Date of Death: dd/mm/yyyy

Place of Death:

City/Municipality and Province:

The screenshot displays a web-based application form for a Death Certificate. The form is organized into several sections, each with a title and corresponding input fields. A purple sidebar on the left contains a 'Log Out' button. Numbered callouts (8-26) point to specific elements: 8 points to the sidebar; 11-16 point to the 'Parents' Information' section (Last Name, First Name, Middle Name for both Father and Mother); 17 points to the 'Upload Valid ID' section (Choose File button); 18-23 point to the 'Requester's Details' section (Last Name, First Name, Middle Initial, Barangay, Select Barangay dropdown, City/Municipality, Province, Mobile Number); 24 points to the 'Purpose of Request' section (Select Purpose dropdown); 25 points to the 'Email' section (Email input field); and 26 points to the 'Submit Request' button at the bottom.

**8**

**Parents' Information**

**Father's Name**

Last Name

First Name

Middle Name

**Mother's Maiden Name**

Maiden Last Name

First Name

Middle Name

**Upload Valid ID**

Choose File No file chosen

**Requester's Details**

Last Name

First Name

Middle Initial

Barangay

Select Barangay

City/Municipality, Province

Mobile Number

**Upload Valid ID**

Choose File No file chosen

**Requester's Details**

Last Name

First Name

Middle Initial

Barangay

Select Barangay

City/Municipality, Province

Mobile Number

**Purpose of Request**

Select Purpose

**Email**

Email

Log Out

Submit Request

## 8. Death Certificate Form

This is the application form users must accurately fill out to request a copy of a Death Certificate, covering details of the deceased, the requestor, the purpose, and submission.

**Steps to Use**

1. Under **Request Details**, select the type of request (e.g., **Copy Issuance**) from the dropdown menu.
2. Input the **Number of Copies** you wish to request.
3. Under **Deceased Person's Information**, input the **Last Name**.
4. Input the **First Name**.
5. Input the **Middle Name**.
6. Select the **Sex** from the dropdown menu.
7. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
8. Input the **Place of Birth** (City/Municipality and Province).
9. Input the **Date of Death** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
10. Input the **Place of Death** (City/Municipality and Province).
11. Under **Parents' Information** for **Father's Name**, input his **Last Name**.
12. Input his **First Name**.
13. Input his **Middle Name**.
14. Under **Mother's Maiden Name**, input her **Maiden Last Name**.
15. Input her **First Name**.
16. Input her **Middle Name**.
17. Under **Upload Valid ID**, click **Choose File** to upload the necessary identification document.
18. Under **Requester's Details**, input your **Last Name**.

19. Input your **First Name**.

20. Input your **Middle Initial**.

21. Input your **Address** (House No., Street Name, Barangay).

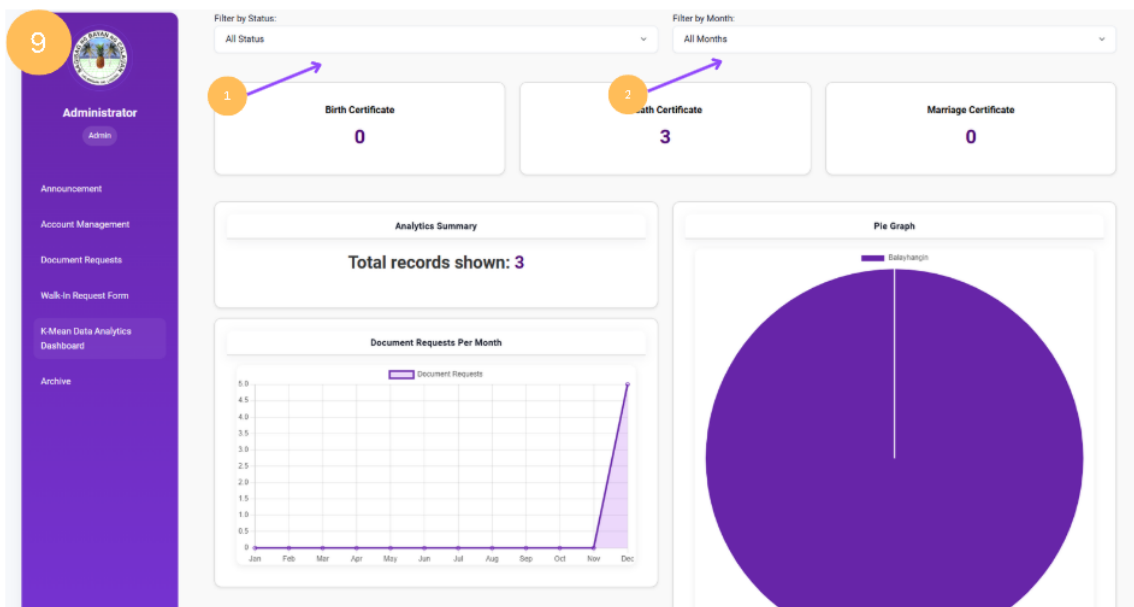
22. Input your **City/Municipality, Province** (and Country if abroad).

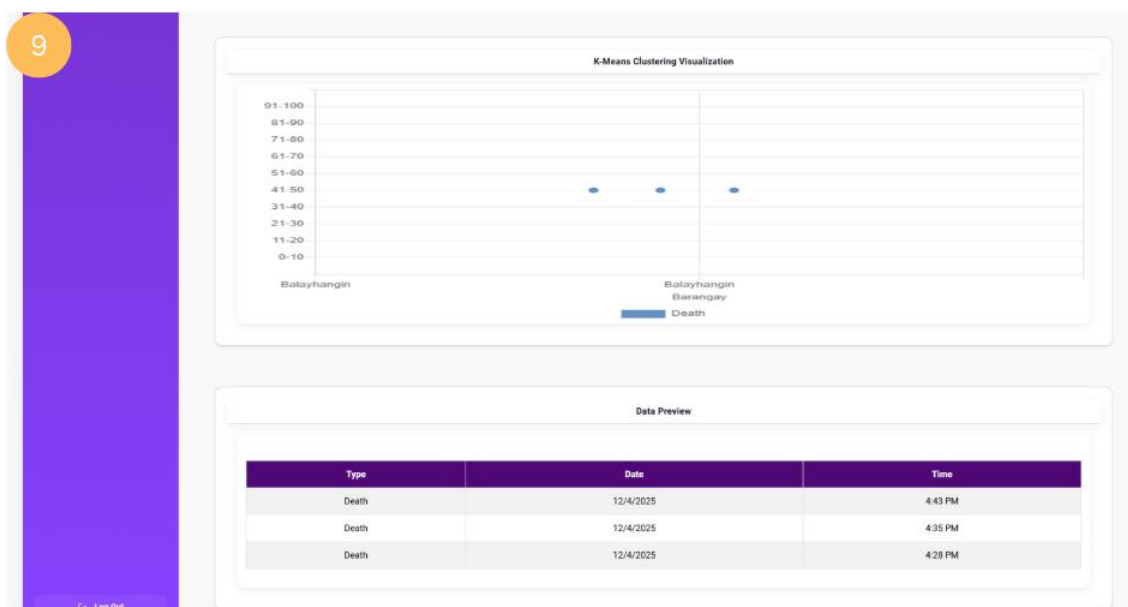
23. Input your **Mobile Number**.

24. Under **Purpose of Request**, specify the **purpose of your request** in the text area.

25. Verify your **Email** address (this field is usually pre-filled).

26. Click the **Submit Request** button to finalize and send your application.





## 9. K-Means Data Analytics Dashboard

This section provides an overview of document request statistics, allowing the administrator to monitor records by type, status, and month through summary cards, charts, and data previews.

### Steps to Use

1. Use the **Filter by Status** dropdown to display records based on their current status.
2. Use the **Filter by Month** dropdown to view document requests for a specific month.
3. Review the **summary cards** (Birth Certificate, Death Certificate, Marriage Certificate) to quickly see the total count per document type.

4. Check the **Analytics Summary** to see the total number of records currently shown based on the selected filters.
5. Analyze trends using the **Document Requests Per Month** line graph. View distribution data through the **Pie Graph** for a visual breakdown of requests.
6. Refer to the **Data Preview table** to see detailed entries including document type, date, and time.

10

| REQUEST NO | DOCUMENT          | STATUS   | CONTACT     | EMAIL                      | DATE REQUESTED | REJECTION REASON |
|------------|-------------------|----------|-------------|----------------------------|----------------|------------------|
| 40         | Birth Certificate | rejected | 09992929399 | perezkristine860@gmail.com | Nov 19, 2025   | tututru          |
| 38         | Birth Certificate | rejected | 09992929399 | perezkristine860@gmail.com | Nov 19, 2025   | tutuue           |
| 35         | Birth Certificate | Rejected | 09992929399 | perezkristine860@gmail.com | Nov 15, 2025   | sds              |
| 34         | Birth Certificate | Rejected | 09992929399 | perezkristine860@gmail.com | Nov 15, 2025   | ada              |
| 33         | Birth Certificate | Rejected | 09992929399 | perezkristine860@gmail.com | Nov 15, 2025   | cv               |
| 32         | Birth Certificate | Rejected | 09992929399 | perezkristine860@gmail.com | Nov 15, 2025   | a                |
| 31         | Birth Certificate | Rejected | 09992929399 | perezkristine860@gmail.com | Nov 15, 2025   | ass              |

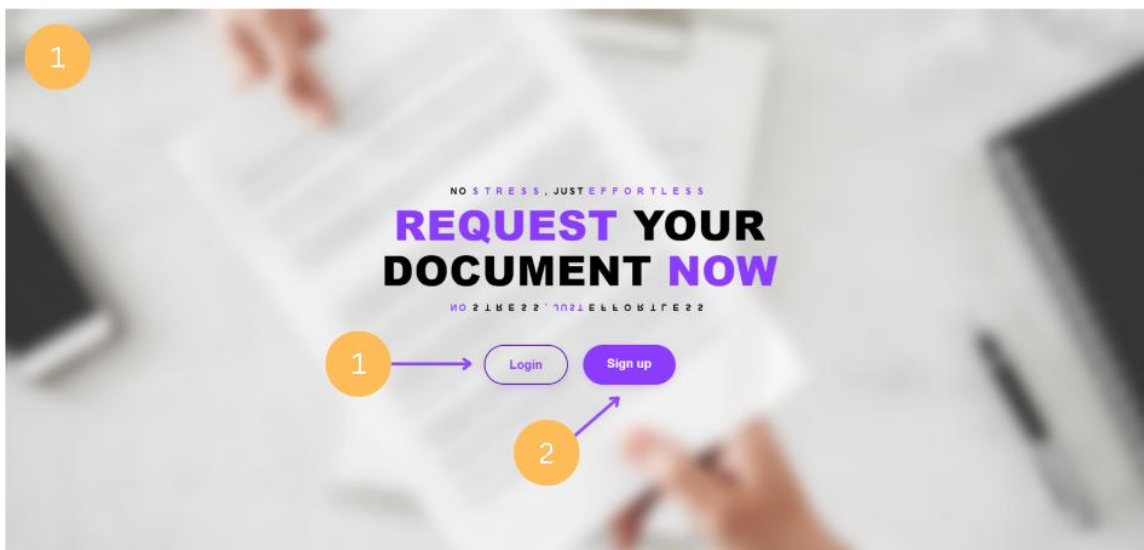
Activate Windows  
Go to Settings to activate Windows.

## 10. Archive

This section displays a searchable record of documents that have been rejected, along with the specific reason for the rejection, allowing the administrator to review past decisions.

**Steps to Use**

1. Use the **Search by Request No or Document...** bar to quickly find a specific record.
2. Use the **All-Documents** dropdown menu to filter the table by the type of document requested.
3. Review the **Rejection Reason** column to understand why a specific document request was rejected.



1

**MISSION**

To provide equally sound education, shelter, employment, and health care service opportunities; promote eco-tourism, agro-industry and commerce while enriching the natural environmental beauties, products, and culture; and establish a systematic approach to encourage participation of private individuals/sectors to communal, relevant, and transparent governance.

**VISION**

Calauan, the Pineapple Capital and tourist haven of Laguna, a tributary to nation building, is an economically stable, calamity-resilient, secured community, home of empowered, contributing, and God-fearing citizens with government trusted by the people, run by dynamic and competent public servant and noble leaders.

1

**How to Request a Document**

Follow these steps to successfully request a document.

**Step 1: Create or Log into An Account**

Register or log in using your email.

**Step 2: Locate the Document Request Section**

Look for options, click the Services.

**Step 3: Fill Out the Request Form**

Provide accurate details such as your full name, purpose of the request, and the specific document(s) needed.

**Step 4: Upload Required Documents**

Attach valid IDs or supporting documents as needed (e.g., authorization letter, etc.).

**Step 5: Submit the Request**

Double-check the information, then click "Submit".

**Step 6: Track the Status**

Use the reference number or tracking link to monitor your request.

1

**LOCATION**

## 1.User Landing Page

This section serves as the entry point of the system, allowing users to navigate the MCR Office location, log in to their accounts, and access a step-by-step guide for requesting documents—from account creation and form submission to uploading requirements and tracking request status—along with the map location of the MCR Office.

1. Click the Log In button to proceed on your Account.
2. Click the Sign Up button to register and create your Account.
3. This section provides the Mission and Visions of MCR.

The image shows two parts of the user interface. On the left, a 'Create Your Account' form is displayed with a light blue background. The form has a title 'Create Your Account' and a subtitle 'Fill out your details to get started'. It contains five input fields: 'Last Name', 'First Name', 'Middle Name', 'Email', and 'Contact No.'. To the left of the form, there are five orange circles numbered 1 through 5, each with a purple arrow pointing to one of the input fields. At the bottom right of the form is a 'Next' button, with a purple arrow pointing to it from an orange circle numbered 6. On the right, there is a purple rectangular area representing the landing page. It features the MCR Office logo at the top, followed by the text 'Online Document Request System' and a tagline 'Quick, easy, and secure — your trusted online document request system.'.

The image shows a user interface for an online document request system. On the left, a 'Create Your Account' form is displayed with the following fields and callouts:

- 2: Title 'Create Your Account' and subtitle 'Fill out your details to get started'
- 7: Address dropdown menu
- 8: Username text field
- 9: Upload Valid ID section with a 'Choose File' button and 'No file chosen' text
- 10: Password text field
- 11: Confirm Password text field
- 12: 'Back' button
- 13: 'Register' button

On the right, the 'Online Document Request System' landing page is shown with a purple background. It features the system logo, the title 'Online Document Request System', and the tagline 'Quick, easy, and secure — your trusted online document request system.'

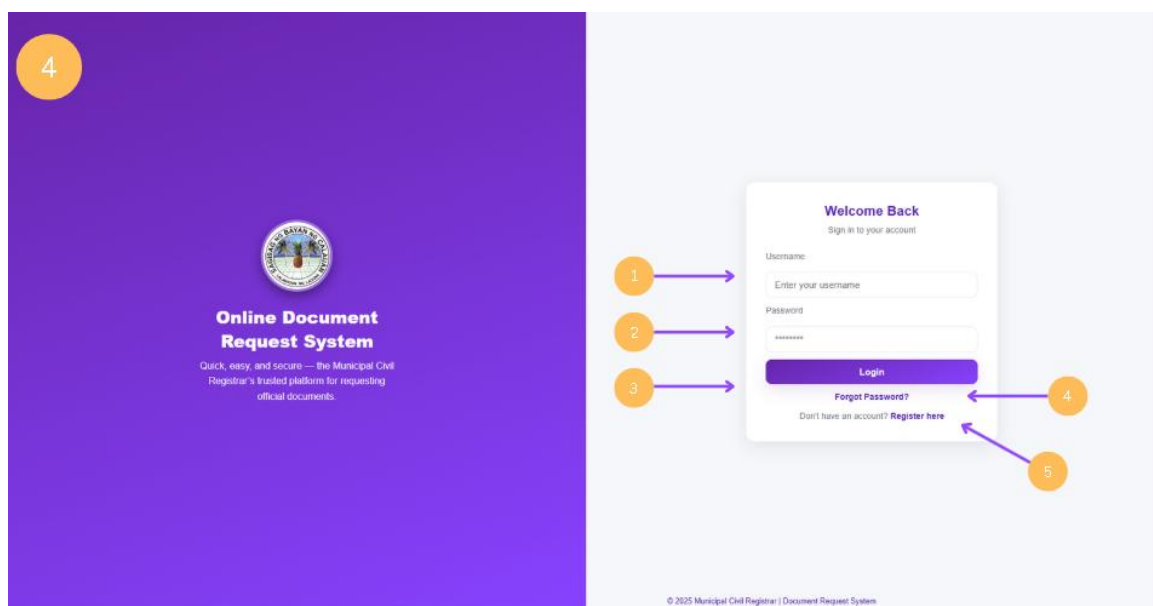
## 2. Create Account

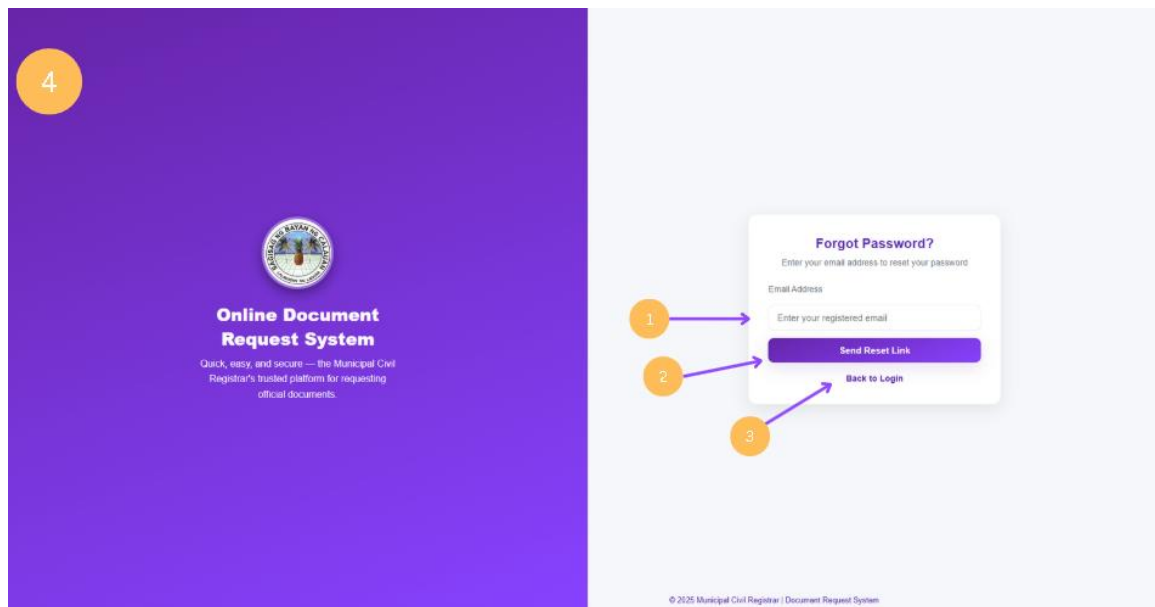
The section of the system where users can register by providing their personal information to create an account for requesting documents.

### Steps to Use

1. The user enters their **Last Name** in the first text field.
2. The user enters their **First Name** in the second text field.
3. The user enters their **Middle Name** to complete their personal information.
4. The user provides their **Email Address** for account verification and updates.
5. The user inputs their **Contact Number** for communication regarding document requests.

6. The user clicks the **Next** button to proceed to the next step of the registration process.
7. The user selects their **Address** from the dropdown menu.
8. The user inputs their desired **Username** in the text field.
9. The user clicks the **Choose File** button to upload a copy of their **Valid ID**.
10. The user inputs their desired **Password** in the text field.
11. The user re-enters the same password in the **Confirm Password** field.
12. The user clicks the **Back** button to return to the previous page (Landing Page).
13. The user clicks the **Register** button to finalize account creation.



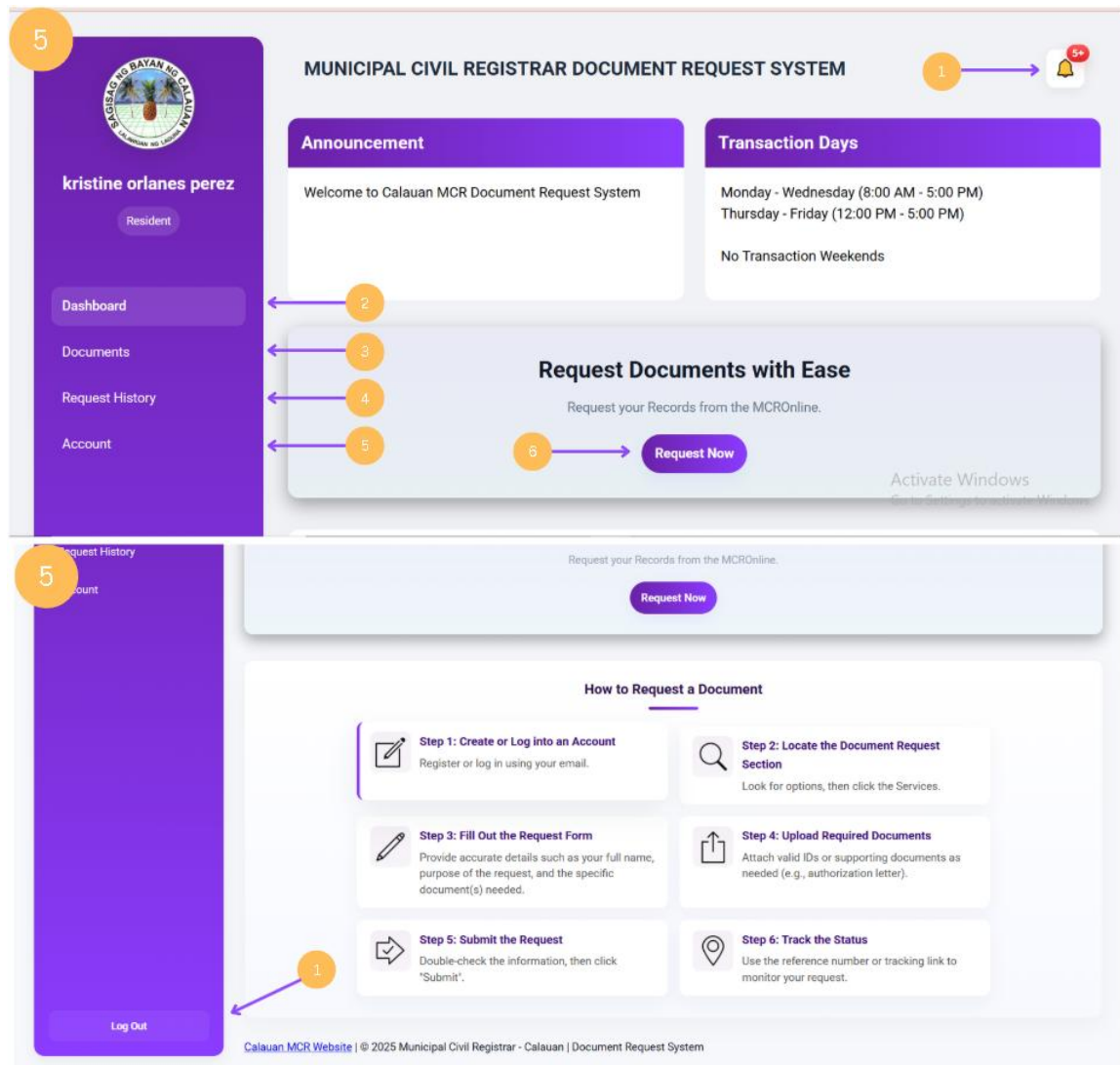


#### 4. Log-In

The entry point where verified users access their accounts to request documents and manage their transactions.

##### Steps to Use

1. The user inputs their **Username** in the first text field.
2. The user inputs their **Password** in the second text field.
3. The user clicks the **Login** button to access the account dashboard.
4. To reset a forgotten password, the user clicks **Forgot Password?**, which redirects to the password recovery process.
5. If the user does not have an existing account, the user clicks the **Register here** link to navigate to the **Sign-Up Page**.



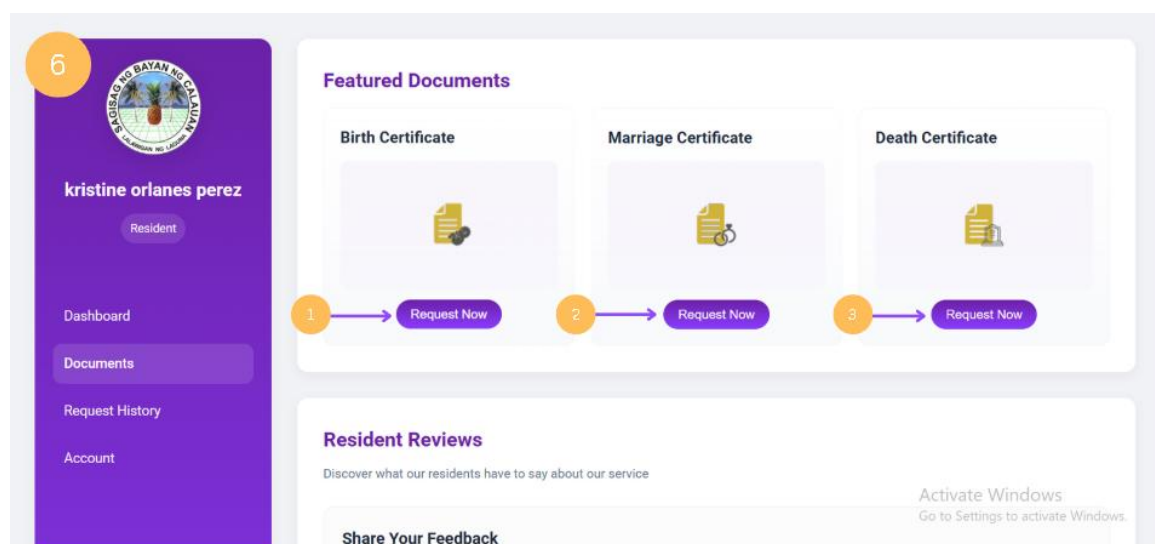
## 5. Dashboard

This is the user's main account page after logging in, providing an overview of important announcements, transaction hours, and navigation options.

### Steps to Use

1. Click the **Notification Bell** to view any system alerts or updates regarding your requests.

2. Click the **Dashboard** link to return to this main page.
3. Click the **Documents** link to navigate to the page where you can submit a new document request.
4. Click the **Request History** link to view the status and details of all your previous requests.
5. Click the **Account** link to view your profile details.
6. Click the **Request Now** button to start a new document request immediately.
7. The guide provides a step-by-step process covering how to **Locate the Document Request Section, Fill Out the Request Form, Upload Required Documents, Submit the Request**, and finally, **Track the Status** of your request.
8. Click the **Log Out** button to securely sign out of your account.



The screenshot shows a web interface for 'Resident Reviews'. On the left is a purple sidebar with a menu containing 'Test History' and 'Account'. The main content area is titled 'Resident Reviews' with the subtitle 'Discover what our residents have to say about our service'. Below this is a section titled 'Share Your Feedback'. It contains a 'Rating (1-5 Stars):' dropdown menu with the text 'Select Rating' and a downward arrow. Below the dropdown is a text area labeled 'Your Review:' with the placeholder text 'Type your honest review here...'. At the bottom right of the form is a purple button labeled 'Submit Review'. Three numbered callouts are present: callout 1 points to the 'Select Rating' dropdown, callout 2 points to the 'Your Review' text area, and callout 3 points to the 'Submit Review' button.

## 6. Documents

This is the main request page where users can select the type of civil document they need to request and share feedback on the service.

### Steps to Use

1. Click the **Request Now** button under **Birth Certificate** to initiate a request for that document.
2. Click the **Request Now** button under **Marriage Certificate** to initiate a request for that document.
3. Click the **Request Now** button under **Death Certificate** to initiate a request for that document.
4. In the **Resident Reviews** section, select a **Rating (1-5 Stars)** from the dropdown menu.
5. Type your honest feedback or review in the **Your Review** text area.

6. Click the **Submit Review** button to share your feedback with the system administrator.

The screenshot displays the 'BIRTH CERTIFICATE FORM' application page. On the left is a purple sidebar with the user's name 'kristine orlans perez' and a 'Resident' badge. The sidebar menu includes 'Dashboard', 'Documents', 'Request History', and 'Account'. The main content area is titled 'BIRTH CERTIFICATE FORM Request / Application' and contains several sections with numbered callouts:

- Request Details:**
  - 1: Request for (dropdown menu)
  - 2: Number of Copies
- Person's / Child's information:**
  - 3: Last Name
  - 4: First Name
  - 5: Middle Name
  - 6: Sex (radio buttons for Male and Female)
  - 7: Date of Birth (calendar icon)
  - 8: Place of Birth
  - 9: City/Municipality and Province
- Father's Name:**
  - 10: Last Name
  - 11: First Name
  - 12: Middle Name
- Mother's Name:**
  - 13: Maiden Name
  - 14: First Name
  - 15: Middle Name
- Upload Valid ID:**
  - 16: Choose File (No file chosen)
- Requester's Details:**
  - 17: Last Name (perez)
  - 18: First Name (kristine)
  - 19: Middle Name (o)
  - 20: Address (Balayhangin)
  - 21: City/Municipality, Province
  - 22: Contact Number (09982929388)
- Purpose of Request:** (Section header visible at the bottom)

The screenshot shows a web form for requesting a birth certificate. On the left is a purple sidebar with a 'Log Out' link. The main form area contains the following sections:

- Middle Name:** A text input field.
- Upload Valid ID:** A section with a 'Choose File' button and the text 'No file chosen'.
- Requester's Details:** A section with input fields for 'perez', 'kristine', a dropdown for 'o', 'Balayhangin', 'City/Municipality, Province', and '0992929399'.
- Purpose of Request:** A dropdown menu with 'Select Purpose' as the current selection.
- Email:** An input field containing 'perezkristine06@gmail.com'.
- Submit Request:** A purple button at the bottom right.

Numbered callouts are placed as follows:

- 7: Purple sidebar
- 20: Arrow pointing to the 'Purpose of Request' dropdown
- 21: Arrow pointing to the 'Email' input field
- 22: Arrow pointing to the 'Submit Request' button

## 7. Birth Certificate Form

This is the application form users must accurately fill out to request a copy of a Birth Certificate, including details about the person, the requestor, the purpose, and submission.

### Steps to Use

1. Under **Request Details**, select the type of request (e.g., **Copy Issuance**) from the dropdown menu.
2. Input the **Number of Copies** you wish to request.
3. Under **Person's / Child's Information**, input the **Last Name**.
4. Input the **First Name**.
5. Input the **Middle Name**.
6. Select the **Sex** (Male or Female) using the radio buttons.
7. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.

8. Input the **Place of Birth** (City/Municipality and Province).
9. Under **Father's Name**, input his **Last Name**.
10. Input his **First Name**.
11. Input his **Middle Name**.
12. Under **Mother's Name**, input her **Maiden Name**.
13. Input her **First Name**.
14. Input her **Middle Name**.
15. Under **Upload Valid ID**, click **Choose File** to upload the necessary identification document.
16. Under **Requester's Details**, input your **Last Name**.
17. Input your **First Name**.
18. Input your **Middle Initial**.
19. Input your **Address** (Barangay).
20. Input your **City/Municipality, Province** (and Country if abroad).
21. Input your **Mobile Number**.
22. (Continuation of Requester's Details) Input your **City/Municipality, Province** (and Country if abroad).
23. Input your **Mobile Number**.
24. Under **Purpose of Request**, specify the **purpose of your request** in the text area.
25. Verify your **Email** address (this field is usually pre-filled).
26. Click the **Submit Request** button to finalize and send your application.

The screenshot displays the 'MARRIAGE CERTIFICATE FORM' application interface. On the left is a purple sidebar with the user's name 'kristine orlanes perez' and a 'Resident' status. The sidebar menu includes 'Dashboard', 'Documents', 'Request History', and 'Account'. The main form area is divided into sections: 'Request Details' (1-2), 'Couple's Information' (3-11), 'Place of Marriage' (12), 'Upload Valid ID' (13), 'Requester's Details' (14-20), 'Purpose of Request' (21), and 'Email' (22). A 'Submit Request' button is at the bottom right (23). The form fields are as follows:

- Request Details:**
  - 1: Request for (Dropdown menu)
  - 2: Number of Copies
- Couple's Information:**
  - Husband's Name:**
    - 3: Last Name
    - 4: First Name
    - 5: Middle Name
    - 6: Date of Birth (dd/mm/yyyy)
  - Wife's Maiden Name:**
    - 7: Maiden Last Name
    - 8: First Name
    - 9: Middle Name
    - 10: Date of Birth (dd/mm/yyyy)
    - 11: Date of Marriage (dd/mm/yyyy)
- Place of Marriage:**
  - 12: City/Municipality and Province
- Upload Valid ID:**
  - 13: Choose File (No file chosen)
- Requester's Details:**
  - 14: Last Name (perez)
  - 15: First Name (kristine)
  - 16: Middle Name (o)
  - 17: Address (Balayhangin)
  - 18: City/Municipality, Province
  - 19: Phone Number (09992926399)
- Purpose of Request:**
  - 21: Select a purpose (Dropdown menu)
- Email:**
  - 22: Email Address (perezkristine86@gmail.com)
- Submit Request:** 23

## 8. Marriage Certificate Form

This is the application form users must accurately fill out to request a copy of a Marriage Certificate, covering details of the couple, the requestor, the purpose, and submission.

**Steps to Use**

1. Under **Request Details**, select the type of request (e.g., **Copy Issuance**) from the dropdown menu.
2. Input the **Number of Copies** you wish to request.
3. Under **Couple's Information** for **Husband's Name**, input his **Last Name**.
4. Input his **First Name**.
5. Input his **Middle Name**.
6. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
7. Under **Wife's Maiden Name**, input her **Maiden Last Name**.
8. Input her **First Name**.
9. Input her **Middle Name**.
10. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
11. Input the **Date of Marriage** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
12. Input the **Place of Marriage** (City/Municipality and Province).
13. Under **Upload Valid ID**, click **Choose File** to upload the necessary identification document.
14. Under **Requester's Details**, input your **Last Name**.
15. Input your **First Name**.
16. Input your **Middle Initial**.
17. Input your **Address** (House No., Street Name, Barangay).

18. Input your **City/Municipality, Province** (and Country if abroad).
19. Input your **Mobile Number**.
20. Under **Purpose of Request**, specify the **purpose of your request** in the text area.
21. Verify your **Email** address (this field is usually pre-filled).
22. Click the **Submit Request** button to finalize and send your application.

The screenshot displays the 'DEATH CERTIFICATE FORM' application interface. On the left is a purple sidebar with the user's name 'kristine orlanes perez' and a 'Resident' status. The sidebar menu includes 'Dashboard', 'Documents', 'Request History', and 'Account'. The main form area is divided into several sections:

- Request Details:** Includes 'Request for' (set to 'Copy Issuance') and 'Number of Copies'.
- Deceased Person's Information:** Fields for Last Name, First Name, Middle Name, Sex (dropdown), Date of Birth (dd/mm/yyyy), Place of Birth (City/Municipality and Province), Date of Death (dd/mm/yyyy), and Place of Death (City/Municipality and Province).
- Parents' Information:** Fields for Father's Name (Last, First, Middle) and Mother's Maiden Name (Maiden Last, First, Middle).
- Upload Valid ID:** A 'Choose File' button with the text 'No file chosen'.
- Requester's Details:** Fields for Last Name (perez), First Name (kristine), Middle Name (O), Address (Balayhangin), City/Municipality, Province (Country if abroad), and Mobile Number (09992629399).

Numbered callouts (1-23) point to specific fields and elements: 1 points to 'Request for', 2 to 'Number of Copies', 3 to 'Last Name', 4 to 'First Name', 5 to 'Middle Name', 6 to 'Select Sex', 7 to 'Date of Birth', 8 to 'Place of Birth', 9 to 'City/Municipality and Province', 10 to 'Date of Death', 11 to 'Place of Death', 12 to 'Father's Name', 13 to 'Mother's Maiden Name', 14 to 'Maiden Last Name', 15 to 'First Name', 16 to 'Middle Name', 17 to 'Choose File', 18 to 'perez', 19 to 'kristine', 20 to 'O', 21 to 'Balayhangin', 22 to 'City/Municipality, Province (Country if abroad)', and 23 to '09992629399'.

The screenshot displays a web form for requesting a Death Certificate. On the left, a purple sidebar contains a 'Log Out' button and a 'Submit Request' button. The main form area includes the following sections:

- Middle Name:** A text input field.
- Upload Valid ID:** A section with a 'Choose File' button and the text 'No file chosen'.
- Requester's Details:** A section with input fields for 'perez', 'kristine', 'o', 'Babayhagin', and 'City/Municipality, Province (Country if abroad)'. Below these is a phone number field containing '09992929399'.
- Purpose of Request:** A dropdown menu with 'Select Purpose' as the placeholder text.
- Email:** An input field containing 'persokristine88@gmail.com'.
- Submit Request:** A large purple button at the bottom right of the form.

Numbered callouts are placed as follows: 9 (top left sidebar), 24 (arrow to Purpose of Request dropdown), 25 (arrow to Email input), and 26 (arrow to Submit Request button).

## 9. Death Certificate Form

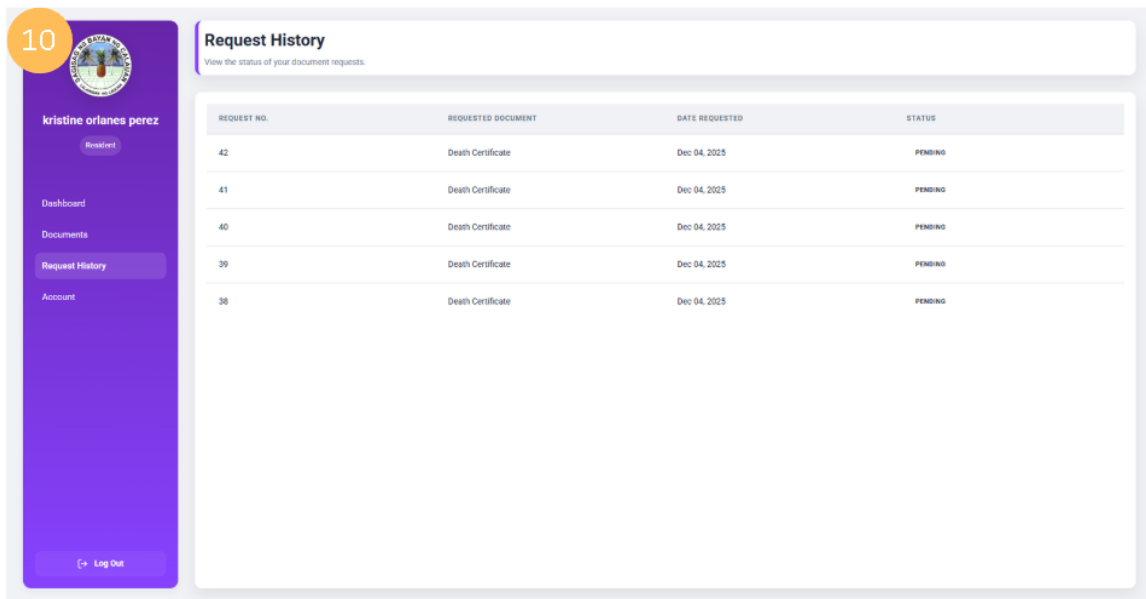
This is the application form users must accurately fill out to request a copy of a Death Certificate, covering details of the deceased, the requestor, the purpose, and submission.

### Steps to Use

1. Under **Request Details**, select the type of request (e.g., **Copy Issuance**) from the dropdown menu.
2. Input the **Number of Copies** you wish to request.
3. Under **Deceased Person's Information**, input the **Last Name**.
4. Input the **First Name**.
5. Input the **Middle Name**.
6. Select the **Sex** from the dropdown menu.

7. Input the **Date of Birth** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
8. Input the **Place of Birth** (City/Municipality and Province).
9. Input the **Date of Death** in the format \$dd/mm/yyyy\$ or use the calendar icon to select it.
10. Input the **Place of Death** (City/Municipality and Province).
11. Under **Parents' Information** for **Father's Name**, input his **Last Name**.
12. Input his **First Name**.
13. Input his **Middle Name**.
14. Under **Mother's Maiden Name**, input her **Maiden Last Name**.
15. Input her **First Name**.
16. Input her **Middle Name**.
17. Under **Upload Valid ID**, click **Choose File** to upload the necessary identification document.
18. Under **Requester's Details**, input your **Last Name**.
19. Input your **First Name**.
20. Input your **Middle Initial**.
21. Input your **Address** (House No., Street Name, Barangay).
22. Input your **City/Municipality, Province** (and Country if abroad).
23. Input your **Mobile Number**.
24. Under **Purpose of Request**, specify the **purpose of your request** in the text area.
25. Verify your **Email** address (this field is usually pre-filled).

26. Click the **Submit Request** button to finalize and send your application.



**10**

**kristine orlans perez**

**Request History**  
View the status of your document requests.

| REQUEST NO. | REQUESTED DOCUMENT | DATE REQUESTED | STATUS  |
|-------------|--------------------|----------------|---------|
| 42          | Death Certificate  | Dec 04, 2025   | PENDING |
| 41          | Death Certificate  | Dec 04, 2025   | PENDING |
| 40          | Death Certificate  | Dec 04, 2025   | PENDING |
| 39          | Death Certificate  | Dec 04, 2025   | PENDING |
| 38          | Death Certificate  | Dec 04, 2025   | PENDING |

## 10. Request History

This section allows users to view a list of all their previously submitted document requests and check the current status of each one.

### Steps to Use

1. View the **Request No.** column for the unique identifier of your request.
2. View the **Requested Document** column to see which type of certificate was requested.
3. View the **Date Requested** column to see the submission date.
4. View the **Status** column to monitor the progress of the request (e.g., PROCESSING, READY, FOR RELEASE).

11

kristine orlanes perez

Resident

Dashboard

Documents

Request History

Account

Log Out

Account

kristine orlanes perez

Resident No: N/A

Full Name

kristine orlanes perez

Address

Balayhangin

Contact No.

09962929399

Email

perezkristine860@gmail.com

## 11. Account

This section displays the user's registered personal and contact information.

### Steps to Use

1. Review the **Full Name** field to verify the name linked to the account.
2. Review the **Address** field to verify the registered location.
3. Review the **Contact No.** field to verify the mobile number.
4. Review the **Email** field to verify the registered email address.

*H. IT Practitioners Profile***DOROTHY MEI CABUYAO****Part-time College Instructor**

Tiaong, Quezon | dotiimc@gmail.com

**SUMMARY**

Tech savvy and passionate individual with good communication and creative skills. Well-organized and able to multitask while maintaining a high level of professionalism and attention to detail.

**PROFESSIONAL EXPERIENCE**

Graphics Intern, **Beppo Inc.** | Makati City Feb 2023 to June 2023

- Create advertising materials, social media content, and marketing collateral, while maintaining brand identity and assisting in designing marketing assets

Photo and Video Editor, **Diggity Studio & 360 Booth** | Candelaria, Quezon Feb 2024 - June 2024

- Managed multimedia content creation including video editing, photo retouching, and graphic designs.

Part-time College Instructor, **LSPU SPCC - CCS** | San Pablo City, Laguna Aug 2024 to Present

- Taught multiple undergraduate courses across Computer Studies, Engineering, and Industrial Technology, covering both theoretical and practical areas.
- Prepared and delivered engaging lectures and classroom activities that aligned with institutional learning outcomes.

**SKILLS**

|                            |                                    |                           |
|----------------------------|------------------------------------|---------------------------|
| Microsoft and Google Suite | Graphic Design and Video Editing   | Hardware Trouble Shooting |
| Adobe Creative Suite       | Basic Programming and UX/UI Design | Communication             |

**EDUCATION**

**Bachelor of Science in Computer Science Major in Graphics and Visualization** 2019-2023  
Laguna State Polytechnic University San Pablo City Campus

**Master of Science in Computer Science** Present  
University of the Philippines Los Baños (UPLB)

**ADDITIONAL INFORMATION**

- **Languages:** English, Filipino
- **Certifications:** Computer Systems Servicing NC II, Fundamentals in 2d Animation, Principles of Graphic Design, Google UX Design, Visual Graphic Design NC III
- **Awards/Activities:** Proficiency Awardee, Service Awardee, Best Research Paper Awardee, Special Awardee in Multimedia, Magna Cum Laude

**CHARACTER REFERENCES****RONNEL A. DELA CRUZ**

Associate Dean  
LSPU-SPCC  
ronnel.delacruz@lspu.edu.ph

**JOESUA R. MANZANERO**

Assistant Professor I  
LSPU-SPCC  
josesua.manzanero@lspu.edu.ph

**ELYZA PAULITE**

OJT Supervisor  
Beppo Inc. | Makati City  
+9696457799

RAPHAEL AFRICANO

Web Developer

+63 916 5467 285 | raph.dev.sp@gmail.com | Brgy. San Lucas 1, San Pablo City, Laguna

Professional Summary

A highly motivated web developer with a Bachelor's degree in Information Technology. Equipped with hands-on experience in backend development, database management, and API integration. Eager to contribute technical expertise and problem-solving skills in a dynamic and collaborative environment.

Work Experience

Developer | Bakawan Data Analytics Inc

September 2024 – Present

- Develop and maintain data-driven web applications and backend systems.
- Collaborate with cross-functional teams to analyze requirements and deliver scalable solutions.
- Optimize database performance and ensure secure data transactions.

IT Intern | Back-End Developer

Fortress Data Service Asya Philippines Inc, San Pablo City, Laguna | February 2024 – May 2024

- Developed and maintained APIs for seamless software component communication.
- Managed SQL and NoSQL databases for efficient data storage and retrieval.
- Wrote clean and scalable backend code using Go, PHP, and Node.js.

Academic Project

IT Student | Full Stack Web Developer

Exploring Bangkong Kahoy: A Virtual Tour and Booking System

Pamantasan ng Lungsod ng San Pablo | March 2023 – January 2024

- Developed using Laravel, JavaScript, and Bootstrap with MySQL database.
- Integrated Panolens.js to provide immersive 3D panoramic views of the destination.

### Technical Skills

Programming Languages: JavaScript, PHP, Dart, Go

Frameworks: Laravel, Flutter, React

Databases: MySQL, PostgreSQL

Tools & Platforms: Git, Postman, Visual Studio Code, Docker, NetBeans, Google Cloud SDK, Firebase

Core Competencies: Coding (Advanced), Debugging (Experienced), Software Design and Testing, Web Development

### Education

Bachelor of Science in Information Technology

Major in Web and Game Development

Pamantasan ng Lungsod ng San Pablo | 2020 – 2024

### Seminars & Trainings

- PHP Web Application Framework: CodeIgniter 4 (Intermediate) | November 2023
- Cybersecurity Fundamentals, Google Meet | December 2023
- All About Angular, Angular Philippines | June 2023
- Cybercrime Protection and Its Laws | April 2023
- Building a Strong Foundation: Exploring Databases, Flutter, Golang, and IT Infrastructure | March 2023

### Awards

First Placer – Back-End Developer

Project: Agri Shop Mobile Application

Fortress Data Service Asya Philippines Inc, San Pablo City, Laguna

### References

Available upon request



## JESSICA T. BIANES

### PRODUCT SPECIALIST 1

+639-263-894-847

jessica.bianes@fortress-asya.com

#49 Schetelig Avenue, San Pablo City, Laguna

#### PROFESSIONAL SUMMARY

Motivated and results-oriented Product Specialist 1/Account Executive with nearly 4 years of experience in managing client relationships and driving product sales. Proven track record in identifying customer needs, delivering tailored solutions, and achieving sales targets. Capable at working collaboratively with cross-functional teams to enhance product offerings and improve customer satisfaction.

#### ACADEMIC BACKGROUND

**PAMANTASAN NG LUNGSOD NG SAN PABLO**  
COMPUTER PROGRAMMING  
TECHNOLOGY 2017 - 2018

**PAMANTASAN NG LUNGSOD NG SAN PABLO**  
BS IN INFORMATION  
TECHNOLOGY  
Major in Software Development 2018 - 2022

#### CAREER HISTORY

**PRODUCT SPECIALIST/ACCOUNT EXECUTIVE** 2022 to Present  
FDS ASYA PHILIPPINES INC.

##### Key Responsibilities:

- Assisted in the rollout and management of company products, ensuring proper functionality and alignment with client needs.
- Conducted product demonstrations and presentations for clients, translating technical details into clear, user-friendly explanations.
- Provided first-level technical support and troubleshooting, addressing product issues promptly.
- Gathered and documented customer feedback, collaborating with the development team to enhance product features.
- Maintained accurate product documentation and internal reports for sales and support teams.
- Coordinated with sales and marketing departments to optimize product positioning and messaging.
- Assisted in product research and testing before new releases.
- Supported customer queries and escalated complex issues to management.
- Documented user feedback and identified recurring issues for internal review.
- Collaborated with cross-functional teams to ensure smooth product updates and releases.

#### CORE SKILLS

- Product expertise and feature understanding
- Ability to conduct product demonstrations
- Identifying product issues and recommending solutions
- Cross-functional teamwork with Sales, Marketing, QA, and Development
- Familiarity with product lifecycle concepts

#### WORK REFERENCES

##### RAYMOND R. VILLAPANDO

Deputy Director for BRM  
FDS ASYA PHILIPPINES INC.  
raymond.villapando@fortress-asya.com



# ALEXIS C. FANDIALAN

## EDUCATION

**Bachelor of Science in Business Administration Major in Marketing Management**

Lyceum of the Philippines University – Laguna 2015 – 2019  
Km 54 Nat'l Highway Makiling Calamba City Laguna

## WORK EXPERIENCE

**FDS ASYA PHILIPPINES INC. – Product Specialist**

**February 2024 – Up to Present**

*Build and maintain strong client relationships. Identify and pursue sales opportunities. Serve as the primary contact for assigned clients. Stay updated on company products and industry trends. Negotiate and close deals. Provide reports on sales performance and client feedback. Collaborate with marketing, customer service, and product teams.*

**A3 Phoenix Stone - PRULIFE UK - Admin Assistant**

**February 2022 – January 2024**

*Assisting Unit Manager and Agents concern, processing the applications and flows of New Business & Recruitment, make phone calls and answering phone calls, prepare presentation and documents for meetings, organizing and managing training, awarding event for the Branch.*

**China Banking Corporation – Customer Relations Assistant**

**April 2021 – August 2021**

*Process the applications of clients for banking facilities, interview customers to obtain information needed for opening accounts, Introduction of bank products & services to clients.*

**Bank of the Philippine Island - Customer Service Associate**

**December 2019 - June 2020**

*Serves customers by completing account transactions, provides account services to customer by receiving deposits, payments, cashing checks, issuing savings withdrawals, and recording night and mail deposit, prepares cashier checks, travelers check.*

## PROFESSIONAL OVERVIEW

Pursuing a position in a globally competitive company which provides challenging but rewarding role that will bring growth and development to oneself and at the same time support the company in achieving its organizational goals.

## CONTACT

PHONE: +63 917 129 2029

ADDRESS:

Pinaglabanan Road Purok 2,  
Brgy Sta Isabel, San Pablo City,  
Laguna 4000, Philippines

EMAIL:

alexisfandialan@gmail.com

## PERSONAL INFORMATION

Age: 27 years old

Date of Birth: 02/09/1998,

Civil Status: Single

Nationality: Filipino

Religion: Roman Catholic

Spoken: English, Filipino

CHRISALYN LAMIO

chrisalynlamio08@gmail.com  
09163907504

**SUMMARY** Information Systems graduate with foundational knowledge in team management and proficiency in Microsoft Excel. Known for being hardworking, dedicated, and persistent—committed to continuous learning and improvement. Eager to grow professionally and take on challenges with a resilient, goal-driven mindset.

**WORK EXPERIENCE** **Product Specialist** **July 9, 2024-Present**

- Plans and defines project scope, objectives, timelines, and resource allocation.
- Leads and coordinates team members to ensure timely and efficient task execution.
- Monitors project progress, manages risks, and resolves issues as they arise.
- Communicates regularly with stakeholders to provide updates and align expectations.
- Ensures project deliverables meet quality standards and conducts post-project evaluation.

**EDUCATION** **Bachelor of Science In Information Systems** **August 2023-June 2024**

CARD MRI Development Institute Inc.

- Final GPA: 1.48

**ACHIEVEMENTS** **College Achievements** **August 2023-June 2024**

CARD MRI Development Institute Inc.

- First Year-Third year College :**Deans Lister**
- Fourth Year College : **President Lister**
- Leadership Nominees

**KEY SKILLS**

- Office Suite software.
- Data entry.
- Organizational and time management skills.
- Corporate communications.
- Problem solving.
- Customer service.
- Attention to details.

## CHRISALYN LAMIO

chrisalynlamio08@gmail.com  
09163907504

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### TRAININGS ATTENDED

- |                                                         |                       |
|---------------------------------------------------------|-----------------------|
| <b>Certified Agile Scrum Master</b>                     | February 20-21, 2025  |
| • CARD BANK Maharlika Highway San Pablo City Laguna     |                       |
| <b>Jira Training</b>                                    |                       |
| • FDSAP San Pablo City Laguna                           | January 21, 2025      |
| • FDSAP San Pablo City Laguna                           |                       |
| <b>Business Analysis Fundamentals Training</b>          |                       |
| • CARD MRI Development Institute Inc. Tranca Bay Laguna | June 23-24, 2025      |
| <b>IT Risk Management Training</b>                      |                       |
| • CARD BANK Maharlika Highway San Pablo City Laguna     | September 18-19, 2025 |

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### CHARACTER REFERENCE

**Joy Escleto**

- BAKAWAN Data Analytics Inc. Product Specialist Head
- Phone Number:

**Pilar Fandiño**

- CARD MRI Development Institute Inc. BSIS College Dean
  - Phone Number:
-

**MARK ACEZEL L. MANLAPAZ** CORS MBA

manlapazmark555@gmail.com  
+63.917.5935533

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**Certification**

ISO 22301 Business Continuity Management Systems – Security and Resilience Lead Implementer  
The International Consortium for Organizational Resilience (ICOR)  
Certificate ID NO: 022217009BCM

**Employment Background****Head/AVP IT Risk Management**

Philippine Bank of Communications (PBCOM)  
IT Risk and Information Security Division  
April 2025 - Present

Responsibility: Lead the IT Risk Management Team in identifying, assessing, monitoring and reporting IT-related risks that may affect the Bank's operations, systems, and regulatory compliance.

**Risk Management**

- Identify and evaluate risks associated with IT systems, applications, and infrastructure
- Maintain an up-to-date IT risk register and assist in developing risk heat maps
- Conduct periodic IT risk assessments and threat modelling
- Monitor Key Risk Indicators (KRIS) and residual risks related to IT and cybersecurity
- Track the effectiveness of implemented controls and follow-up on remediation plans
- Recommend improvement on IT Risk Management System and its enterprise-wide implementation.
- Lead the periodic IT risk assessments, threat modelling and scenario analysis

**Policy and Framework Implementation**

- Ensure the effective implementation of IT risk management practices in alignment with BSP regulations, particularly Section 148 of MORB, industry frameworks, and the Bank's internal processes.
- Develop, update and/or recommend IT risk management framework, policies and procedures; must include among others, regulatory and best practice-prescribed standards and measures related to IT risk.

**Control and Evaluation Testing**

- Coordinate with ITG and Information Security Team to review technical and administrative controls
- Facilitate and/or perform risk control self-assessments (RCSAs) for IT functions and processes
- Perform application risk assessment (ARA) for the Bank's applications and systems
- Evaluate control effectiveness against emerging threats — IT risk, cyber risk, cyber fraud and digital risks

**Third Party and Project Risk**

- Conduct due diligence on third party service providers and vendors
- Facilitate and/or perform materiality assessment of third party service providers and vendors
- Participate and ensure IT risk coverage in project risk reviews -- for new systems, upgrades, IT initiatives and projects (e.g., digital transformation, cloud adoption, core system upgrades, etc.)

- Ensure that IT risk considerations are embedded in procurement, project/SDLC, deployment and post production.

#### Other Tasks

- Develop and implement a risk management tool including the validation and reporting of the result of the assessments, to reinforce a strong risk management culture and facilitate proactive management.
- Review and provide recommendations to the Information Security, Information Technology, and Bank policies, procedures, and guidelines in relation to IT Risk Management areas.
- Monitor and review Information Technology related incidents and Information Security events and ensure that risks from the threat and/or vulnerabilities are identified, and mitigating controls are performed.
- Assist the Information Security Head/CISO in monitoring IT risk by keeping abreast (when available) of the Bank's:
  - IT Risk Issues
  - I.T Risk Opportunities
  - Key IT Risk indicators and escalation triggers
  - Control gaps and policy exceptions
  - Risk aggregation data
  - Risk analysis exercises and corresponding results
- Serves as a secondary resource to the IT and IS Risk Division Head by researching and gathering materials on IT risk and information security matters, that includes:
  - Determining business criticality of IT resources
  - Updating IT risk scenario components
  - Development and maintenance of IT Key Risk Indicators (KRIs)
  - Support the conduct of information security programs
  - Work closely with the Operational Risk Division by gathering reports on risk exploited and corresponding implementation of mitigation solutions/strategies.
  - Monitor and review the conduct of the regular vulnerability assessment on critical assets of the Bank.
  - Stay informed about the latest development in the Information Security and Information Technology related field, including but not limited to virus updates and security tools, through on-line services, technical magazines, industry conferences, special training seminars and other methods.
  - Attend conferences, training, and seminars to remain familiar with the latest tools and techniques in the Information System Risk control areas.
- Keeps abreast and updates the IT and IS Risk Division Head/CISO on the following:
  - o Mapping of IT resources to business processes
  - o Maintaining the IT risk registers and risk maps
  - o Recommending policies and control requirements
- Prepare monthly report to Information Technology Steering Cyber Committee, and Risk Oversight Committee.

#### Lead/IT Risk Management

Union Digital Bank (UD)

Risk Management Tribe – Information Technology Risk Management

November 2024 – April 2025

**Lead/Information Security Risk Management**

SeaBank Philippines a Rural Bank Inc.

Risk Management

April 2024 – October 2024

**Lead/IT Risk Management**

Union Bank of the Philippines (UBP)

Enterprise Risk Management – Information Technology Risk Management

June 2021 – April 2024

**IT Risk Department Head/Deputy Senior Manager**

China Banking Corporation (Chinabank)

Operational Risk Division - Risk Management Group

Information Technology Risk Department

January 2020 – June 2021

**Manager**

Security Bank Corporation (SBC)

Information Security Compliance Information Security Division

Risk Management Group

September 2017 – December 2019

**Manager**

United Coconut Planters Bank (UCPB)

Information Technology Risk Section

Risk Management Division

September 2016 – September 2017

**Department Head/Assistant Manager**

Information Security Governance Division

Information Security Administration Department Rizal Commercial

Banking Corporation (RCBC)

Corporate Risk Management Services Group

September 2012 – August 2016

Operational Risk Analyst

Metropolitan Bank and Trust Company (METROBANK)

Risk Management Group-Operational Risk Division

Technology Risk Management Department

Nov 2008 – Sept 2012

**Training Attended**

May 2022 ITIL 4 Foundation

Mar 2022 Scrum Master

Sep 2020 World Cyber Security Summit

Sep 2020 World Cloud Show  
Aug 2020 World Artificial Intelligence (AI) Show Virtual Connect Manila  
Dec 2019 Microsoft Project  
Aug 2019 CyberSecurity Summit  
Feb 2017 Business Continuity Management Implementer  
Oct 2016 IT Risk Management, IT Risk Rating System and IT regulatory Updates  
Sep 2016 Information Security Officers Group (ISOG) Summit  
Nov 2015 Information Security Summit  
May 2015 Human Evocative Approach to Training (HEAT)  
Apr 2015 Effective Presentation Skills  
Dec 2014 NETWORK Vulnerability Assessment and Penetration Training  
Nov 2014 WEB APPS Vulnerability Assessment and Penetration Training  
Oct 2014 PCIDSS 3.0  
Jun 2014 BAIPhil - Foreign Account Tax Compliance Act – FATCA  
May 2014 ISACA - IT Governance Summit  
Mar 2014 Foreign Account Tax Compliance Act – FATCA  
Sep 2013 Documenting & Implementing ISO/IEC 27001:2005 ISMS  
Sep 2013 Auditing an Information Security Management System based on ISO 27001:2005  
Aug 2013 Intro to ISO/IEC 27001:2005 ISMS Standard

**Educational Background**

Master of Business Administration  
AMA University  
2020-2022

Teaching Professional Certificate Villagers Montessori  
Colleges Inc.  
2020

Bachelor of Science in Information Technology  
University of the East, Manila  
2004-2008

**Recognition and Award**

Heroes and Champions 2023

**Productivity Achievement Award**

Information Technology Risk Assessment for Changes and Enhancement (IT-RACE)  
Union Bank

*I. Project Team Assignment*

| Name and Signature        | Project Role                            | Email address/mobile#                                                                         | Signature |
|---------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------|-----------|
| PHOEBE<br>ISOBEL<br>PEREZ | Project Manager                         | <a href="mailto:phoebeisobelperez@gmail.com">phoebeisobelperez@gmail.com</a><br>0995 266 2409 |           |
| VINCE KYLE<br>CALAYAG     | Back End Designer/<br>System<br>Analyst | <a href="mailto:calayagvk@gmail.com">calayagvk@gmail.com</a> 0967 3623 375                    |           |
| KRISTINE<br>PEREZ         | Front End/ Database<br>Designer         | <a href="mailto:perezkristine86@gmail.com">perezkristine86@gmail.com</a><br>0998 2314 937     |           |
| LESTER YURIE<br>SALCEDO   | Network<br>Designer/Quality<br>Tester   | <a href="mailto:salcedoyurie2@gmail.com">salcedoyurie2@gmail.com</a><br>0938 4427 878         |           |
| ADONIS<br>BALAQUE         | Technical Writer                        | <a href="mailto:balaqueadonis06@gmail.com">balaqueadonis06@gmail.com</a><br>0948 4265 335     |           |
| RONNEL<br>GERVACIO        | Technical Writer                        | <a href="mailto:Ronnelgervacio15@gmail.com">Ronnelgervacio15@gmail.com</a><br>0915 2608 200   |           |

*J. Grammarian's Certificate*

**CERTIFICATION OF PROFESSIONAL PROOFREADING  
AND LANGUAGE EDITING**

TO WHOM IT MAY CONCERN:

This is to certify that the undergraduate research titled "**DEVELOPMENT OF A DOCUMENT REQUEST SYSTEM WITH K-MEANS CLUSTERING FOR THE MUNICIPAL REGISTRAR OF THE MUNICIPALITY OF CALAUAN**", a Capstone Project presented to the College of Computer Studies and Technology of **Pamantasan ng Lungsod ng San Pablo**, has been edited for proper English language, grammar, punctuation, spelling, and overall style by the undersigned.

Neither the research content nor the researchers' intention was altered in any way during the editing process.

Very truly yours,

**MAYRRIELL T. URSOLINO, LPT**

(SGD.)

*K. Developers' Profile*

Contact Number: 0948 426 5335  
Address: 22b Sitio Kapatiran, Brgy. Bukal Sur Candelaria, Quezon.  
Birthday: July 27, 2003  
Religion: Catholic  
Email Address: balaqueadonis06@gmail.com

**ADONIS A. BALAQUE****EDUCATION****2022 - present****College****Pamantasan ng Lungsod ng San Pablo**

Brgy. San Jose, San Pablo City, Laguna

**2020 - 2022****Senior High School****Recto Memorial National High School**

Marcial C. Punzalan Jr. Highway, Tiaong, Quezon

**2016 - 2020****Junior High School****Bukal Sur National High School**

Brgy. Buenavista West Candelaria, Quezon

**2014 - 2016****Primary****Bukal Sur Elementary School**

Brgy. Bukal Sur Candelaria, Quezon

**2010 - 2014****Lalig Ementary School**

Brgy. Lalig Tiaong, Quezon

**SKILLS**

- Computer Skills
- Teamwork Skills
- Communication Skills
- Emotional Intelligence
- Attention to Details

Contact Number: 0967 362 3375  
Address: Brgy, Bagong Bayan SPC Laguna  
Birthday: August 27, 2003  
Religion: Catholic  
Email Address: calayagvk@gmail.com

**VINCE KYLE C. CALAYAG****EDUCATION****2022 - present****College**

**Pamantasan ng Lungsod ng San Pablo**  
Brgy. San Jose, San Pablo City, Philippines 4000

**2020 - 2022****Senior High School**

**STI San Pablo**  
Lopez Jaena Bldg, San Pablo Laguna

**2016 - 2020****Junior High School**

**CREST Christian School**  
Brgy. San Francisco Calihan, San Pablo Laguna

**2013 - 2016****Primary**

**Don Enrique Baustista Elementary School**  
Brgy. San Francisco Calihan, San Pablo Laguna

**2010 - 2013**

**CREST Christian School**  
Brgy, San Francisco Calihan, San Pablo Laguna

**SKILLS**

- Computer Skills
- Teamwork Skills
- Critical Thinking
- Ability to Solve Problem

Contact Number: 09152608200  
Address: Brgy San Nicolas SPC  
Birthday: September 15, 2002  
Religion: Catholic  
Email Address: ronnelgervacio15@gmail.com

**GERVACIO RONNEL****EDUCATION****2022 - present****College**

Pamantasan ng Lungsod ng San Pablo

Brgy. San Jose, San Pablo City, Philippines 4000

**2021- 2022****Senior High School**

TCrecencia Drusila Lopez SHS

Brgy San Roque SPC

**2016 - 2020****Junior High School****Prudencia National High School**

Brgy San Nicolas SPC

**2010 - 2016****Primary****Prudencia D. Fulel Elementary School**

Brgy San Nicolas SPC

**2009 - 2010****Prudencia D. Fule**

Brgy San Nicolas SPC

**SKILLS**

- Computer Skills
- Leadership Skills
- Teamwork Skills
- Communication Skills
- Emotional Intelligence
- Attention to Details

Contact Number: 09982314937  
Address: Brgy. Lalog, Tiaong, Quezon  
Birthday: October 18, 2003  
Religion: Catholic  
Email Address: perezkristine860@yahoo.com



## PEREZ, KRISTINE O.

### EDUCATION

**2022 - present**

#### College

**Pamantasan ng Lungsod ng San Pablo**  
Brgy. San Jose, San Pablo City, Philippines 4000

**2020 - 2022**

#### Senior High School

**Recto Memorial National High School**  
Brgy. Quipot, Tiaong, Quezon

**2016 - 2020**

#### Junior High School

**Recto Memorial National High School**  
Brgy. Quipot, Tiaong, Quezon

**2012 - 2016**

#### Primary

**Lalog Elementary School**  
Brgy. Lalog, Tiaong, Quezon

**2010 - 2012**

**Lalog Elementary School**  
Brgy. Lalog, Tiaong, Quezon

### SKILLS

- Computer Skills
- Teamwork Skills
- Critical Thinking
- Attention to Details

Contact Number: 0995 266 2409  
Address: 204 Purok 2, Brgy. Lamot 1 Calauan, Laguna  
Birthday: May 29, 2004  
Religion: Catholic  
Email Address: phoebeisobelvperez@gmail.com

**PHOEBE ISOBEL V. PEREZ****EDUCATION****2022 - present****College**

Pamantasan ng Lungsod ng San Pablo  
Brgy. San Jose, San Pablo City, Philippines 4000

**2021 - 2022****Senior High School**

Trace College  
El Danda Street, Batong Malake, 4030  
Los Baños, Philippines

**2020 - 2021****STI College**

Lopez Jaena Street, San Pablo City, 4000 Laguna

**2016 - 2020****Junior High School****Liceo De Calauan**

M. Roxas vda de Soriano, Calauan, Laguna

**2010 - 2016****Primary****Calauan Central Elementary School**

Kanluran Calauan, Laguna

**2009 - 2010****Sister of the Child of Jesus**

Brgy. Lamot 2 Calauan, Laguna

**SKILLS**

- Computer Skills
- Leadership Skills
- Teamwork Skills
- Communication Skills
- Emotional Intelligence
- Attention to Details

Contact Number: 0938 442 7878  
Address: Brgy, Bulakin Tiaong, Quezon  
Birthday: Septermber 17, 2003  
Religion: Catholic  
Email Address: salcedoyurie2@gmail.com



## LESTER YURIE J. SALCEDO

### EDUCATION

**2022 - present**

#### College

**Pamantasan ng Lungsod ng San Pablo**  
Brgy. San Jose, San Pablo City, Philippines 4000

**2020 - 2022**

#### Senior High School

**Recto Memorial National Highschool**  
Brgy. Quipot Tiaong, Quezon

**2016 - 2020**

#### Junior High School

**Recto Memorial National Highschool**  
Brgy. Quipot Tiaong, Quezon

**2012 - 2016**

#### Primary

**Kiddie Corner School**  
Brgy. Villa Rosario, Tiaong Quezon

**2010 - 2012**

**St. Benedict Elementary School**  
Brgy, Lalog, Tiaong Quezon

### SKILLS

- Computer Skills
- Teamwork Skills
- Communication Skills
- Emotional Intelligence
- Attention to Details